

CONTENTS

- 01 Physical World, Units and Measurement
- 02 Kinematics
- 03 Laws of Motion
- 04 Work, Energy and Power
- 05 Rotational Motion (System of Particles and Rigid Body)
- 06 Gravitation
- 07 Properties of Bulk Matter (Solids and Fluids)
- 08 Thermodynamics
- 09 Kinetic Theory of Gases
- 10 Oscillations and Waves
- 11 Electrostatics
- 12 Current Electricity
- 13 Magnetic Effects of Current and Magnetism
- 14 Electromagnetic Induction and Alternating Current
- 15 Electromagnetic Waves
- 16 Optics
- 17 Dual Nature of Matter and Radiation
- 18 Atoms and Nuclei
- 19 Electronic Devices
- 20 Communication Systems
- FINAL REVISION — Formula, Relations, Tricks, Traps, Facts

CHAPTER 01 — PHYSICAL WORLD, UNITS AND MEASUREMENT**1. INTRODUCTION**

Physics starts with MEASUREMENT. Jab bhi koi quantity likhte hain — jaise "speed 20 m/s" — usme do cheezein hoti hain: ek NUMBER aur ek UNIT. Is chapter mein hum seekhte hain ki units kya hain, dimensions kya hain, aur dono ka use karke exam questions kaise turant solve hote hain. Ye chapter sabse zyada direct-scoring hai: dimensions aur unit conversion ke questions har paper mein aate hain.

2. CORE CONCEPTS

What is a physical quantity? Koi bhi cheez jo measure ki ja sake (length, mass, time, temperature, current) physical quantity hai. Ek UNIT (kg, metre, second) khud quantity NAHI hoti — kg mass ka unit hai, mass quantity hai. Yehi ek classic trap hai.

Fundamental vs derived. SI system mein 7 BASE units hain: length (m), mass (kg), time (s), electric current (A), temperature (K), amount of substance (mol), luminous intensity (cd). Baaki sab quantities DERIVED hain — force, work, power sab inhi 7 se banti hain.

What is a dimensional formula? Har quantity ko base dimensions $[M^a L^b T^c A^d]$ mein express karna. Mass (M), Length (L), Time (T), Current (A). Example: velocity = displacement/time, isliye $[\text{velocity}] = [L]/[T] = [M^0 L^1 T^{-1}]$. Homogeneity principle. Kisi bhi physical equation mein HAR term ka dimension same hona chahiye. Isi se hum unknown constant ka dimension nikalte hain — bas equation ke kisi bhi do same-dimension terms ko equate kar do.

3. KEY RELATIONS

- Dimension of a constant: us constant ko equation se alag karo, phir dimensions substitute karo. Example: $v = at + bt^2$ mein har term ka dimension velocity ka hai, isliye $[a] = [v/t] = LT^{-2}$ aur $[b] = [v/t^2] = LT^{-3}$.
- Same-dimension pairs (yaad rakhne wale twins): torque = work = energy; stress = pressure = Young's modulus;

momentum = impulse; angular momentum = Planck's constant.

- Dimensionless BUT with units: angle (radian), solid angle (steradian). Ye number hote hain phir bhi unit rakhte hain.
- Unitless AND dimensionless: strain, refractive index, specific gravity, relative permittivity, Poisson ratio, magnification, power factor — sab pure ratios hain.

4. FORMULAS

$[\text{velocity}] = [M^0 L^1 T^{-1}]$ $[\text{acceleration}] = [M^0 L^1 T^{-2}]$
 $[\text{force}] = [M^1 L^1 T^{-2}]$ $[\text{pressure}] = [\text{stress}] = [M^1 L^{-1} T^{-2}]$

$[\text{energy}] = [\text{work}] = [\text{torque}] = [M^1 L^2 T^{-2}]$ $[\text{power}] = [M^1 L^2 T^{-3}]$

$[\text{momentum}] = [\text{impulse}] = [M^1 L^1 T^{-1}]$ $[\text{angular momentum}] = [\text{Planck's } h] = [M^1 L^2 T^{-1}]$

$[\text{viscosity}] = [M^1 L^{-1} T^{-1}]$ $[\text{resistivity}] = [M^1 L^3 T^{-3} A^{-2}]$ $[G] = [M^{-1} L^3 T^{-2}]$

1 kWh = 3.6×10^6 J ; 1 N = 10^5 dyne ; 1 T = 10^4 gauss ; 1 pm = 10^{-12} m = 10^{-6} micron

$^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5/9$; $^{\circ}\text{F} = (9/5)^{\circ}\text{C} + 32$; 1 cal = 4.18 J

σ (Stefan) = $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$; pole strength unit = A·m

5. CONDITIONS / SPECIAL CASES

- 1 kWh = 3.6×10^6 J (kilo-watt-hour → joules).
- $[\text{potential}] = [M^1 L^2 T^{-3} A^{-1}]$; [magnetic field B] = $[M^1 T^{-2} A^{-1}]$ ($F = qvB$ se).
- Coefficient of viscosity η ka SI unit Pa·s (poiseuille), CGS unit poise.
- Time (second) hi ek aisi quantity hai jiska unit CGS aur SI dono mein SAME hai.
- km/h → m/s: $\times 5/18$ (54 km/h = 15 m/s).
- Light-year = light dwara vacuum mein 1 saal mein tay ki gayi doori = unit of DISTANCE (time nahi!). 1 ly $\approx 9.46 \times 10^{15}$ m.
- 'nit' = unit of luminance (photometric quantity).
- Density unit-conversion: 128 kg/m³ ko aise system mein badlo jahan mass unit 50 g aur length unit 25 cm ho → $128 \times (1000/50) \times (0.25/1)^3 = 40$.
- $^{\circ}\text{F} \leftrightarrow ^{\circ}\text{C}$: $^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5/9$; $^{\circ}\text{F} = (9/5)^{\circ}\text{C} + 32$. 200 $^{\circ}\text{F} \rightarrow (200 - 32) \times 5/9 = 93.3^{\circ}\text{C}$.
- Stefan-Boltzmann constant $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ (radiated energy $\propto T^4$).
- Pole strength ka unit A·m (magnetic dipole moment ka unit A·m²).
- Mechanical equivalent of heat: 1 cal = 4.18 J.
- CGS unit of electric dipole moment = statC·cm.

6. HOW TO RECOGNIZE THE QUESTION

- "Dimension of X find karo" → defining formula lo, dimensions substitute karo.
- "Kaunsa pair same dimension rakhta hai?" → dono ko formula se likho, compare karo.
- "Kaun dimensionless/unitless hai?" → strain, refractive index, specific gravity, angle.
- "Unit conversion" (kWh→J, N→dyne, T→gauss, pm→micron) → conversion ladder yaad rakho.
- "Fundamental vs derived?" → 7 base SI units se match karo.

7. SOLVING METHODS

- Dimension of constant: equation se constant ko symbolically alag karo, phir substitute karo.
- Energy/Volume ka dimension PRESSURE ke barabar hota hai — ye equivalence exam mein directly aata hai.

- Density/unit conversions ke liye ratio formula: new value = old $\times$ (old mass unit/new mass unit) $\times$ (new length unit/old length unit)³.

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — If $v = at + bt^2$, v is velocity and t is in seconds, then the dimension of b is:

What this tests: Homogeneity principle — har term ka dimension velocity (LT^{-1}) ke barabar hota hai.

Solution: $[b] \times [t^2] = [v] \rightarrow [b] = [v]/[t^2] = LT^{-1} / T^2 = LT^{-3}$.

Key takeaway: Kaunse power mein time aata hai, usi se exponent decide hota hai. b ka dimension LT^{-3} .

REAL PYQ EXAMPLE — The dimensional formula for specific gravity is:

What this tests: Specific gravity = density/density — ek pure ratio.

Solution: Ratio mein sab kuch cancel ho jaata hai $\rightarrow [M^0L^0T^0]$.

Key takeaway: Koi bhi pure ratio (strain, refractive index, specific gravity) dimensionless hota hai.

REAL PYQ EXAMPLE — A non-SI unit called 'nit' is the unit of which photometric quantity?

What this tests: Units ke facts — 'nit' luminance ka unit hai.

Solution: Luminance = light intensity per area; iska unit nit hai.

Key takeaway: Direct fact. Nit = luminance. Lumen = luminous flux (don't mix).

REAL PYQ EXAMPLE — A measured temperature is 200 °F. Its value in Celsius is:

What this tests: $^{\circ}C = (^{\circ}F - 32) \times 5/9$.

Solution: $^{\circ}C = (200 - 32) \times 5/9 = 168 \times 5/9 = 93.3^{\circ}C$.

Key takeaway: Pehle 32 ghat, phir 5/9 se multiply. $^{\circ}C \rightarrow ^{\circ}F$ iska ulta ($\times 9/5 + 32$).

9. 10-SECOND TRICKS

- Jo cheez sirf number hai (strain, refractive index, specific gravity, angle) $\rightarrow$ dimensionless. MCQ mein turant eliminate karo.
- Planck's constant aur angular momentum hamesha twins hain.
- $^{\circ}F \rightarrow ^{\circ}C: (F - 32) \times 5/9$; $200^{\circ}F = 93.3^{\circ}C$.
- 1 cal = 4.18 J (heat $\leftrightarrow$ mechanical work ka bridge).
- "kg" ek unit hai (quantity nahi); "current (A)" base quantity hai; "frequency (Hz)" derived hai — classic elimination points.

10. IMPORTANT FACTS

- 7 base SI units: metre, kilogram, second, ampere, kelvin, mole, candela.
- Frequency ka unit Hz = s^{-1} (derived).
- Permittivity ka unit: $C^2N^{-1}m^{-2}$. Permeability $\mu_0 = 4\pi \times 10^{-7} T m/A$.

11. COMMON TRAPS

TRAP: "kg" ek physical quantity hai.

CORRECT: kg ek UNIT hai; mass quantity hai. Ye trap bar-bar aata hai.

TRAP: Specific gravity/strain ka "koi unit nahi hai to koi dimension nahi" — galat way mein.

CORRECT: Ye dimensionless hain: dimension $[M^0L^0T^0]$ hota hai (zero nahi hota, zero-POWER hota hai).

TRAP: Angle ka koi unit nahi hota.

CORRECT: Angle ka unit RADIAN hota hai, lekin dimension nahi hota.

12. GRAPHS / DIAGRAMS

- Dimensions ka table hi is chapter ka "diagram" hai — FORMULAS section ki table yaad karo.
- Conversion ladder: 1 kWh = 3.6×10^6 J; 1 N = 10^5 dyne; 1 T = 10^4 gauss; 1 pm = 10^{-6} micron.

13. LAST-MINUTE REVISION

LAST-MINUTE REVISION

- [Energy] = [Work] = [Torque] = $[ML^2T^{-2}]$; [Power] = $[ML^2T^{-3}]$
- [Pressure] = [Stress] = [Young's modulus] = $[ML^{-1}T^{-2}]$ = [Energy/Volume]
- [Momentum] = [Impulse] = $[MLT^{-1}]$; [Angular momentum] = $[h] = [ML^2T^{-1}]$
- [Viscosity] = $[ML^{-1}T^{-1}]$; [Resistivity] = $[ML^3T^{-3}A^{-2}]$
- 1 kWh = 3.6×10^6 J; 1 N = 10^5 dyne; 1 T = 10^4 gauss; 1 pm = 10^{-6} micron

CHAPTER 02 — KINEMATICS

1. INTRODUCTION

Kinematics = motion ki study bina ye pooche ki motion kisne kiya. Yahan sirf position, velocity, acceleration aur time ka relation padhte hain. Ye chapter har exam ka backbone hai kyunki yahi se equations of motion, graphs aur projectile ke questions aate hain.

2. CORE CONCEPTS

What is displacement vs distance? Displacement ek VECTOR hai (direction ke saath), distance ek SCALAR hai. Closed loop mein ghoom kar wapas aane par displacement = 0, lekin distance > 0. Isliye average velocity = NET displacement/time (closed path mein zero), jabki average speed = total distance/time.

What is acceleration? Acceleration batata hai ki velocity kitni jaldi badal rahi hai. Agar har second velocity same amount se badle, acceleration constant hota hai, aur tab hum uniform-acceleration equations use kar sakte hain. $a = dv/dt$.

What is a projectile? Ek body jise kisi angle par upar phenka gaya ho. Uski horizontal velocity constant rehti hai (koi horizontal force nahi), vertical motion free fall hai (g neeche). In dono ko alag-alag treat karo.

Motion graphs. x-t graph ka SLOPE = velocity; v-t graph ka SLOPE = acceleration; v-t graph ka AREA = displacement; a-t graph ka AREA = velocity change; F-t graph ka AREA = impulse.

3. KEY RELATIONS

- Three equations (constant acceleration): $v = u + at$; $s = ut + (1/2)at^2$; $v^2 = u^2 + 2as$.
- Distance in nth second: $S_n = u + (a/2)(2n - 1)$.
- Free fall ($u = 0$, $a = g$): $h = (1/2)gt^2$; $v = \sqrt{2gh}$; $t = \sqrt{2h/g}$.
- Per-second distance ratio in free fall = 1 : 3 : 5 : 7 (odd numbers).
- Total distance ratio in 1,2,3,4 seconds = 1 : 4 : 9 : 16 (squares).
- Projectile: $H = u^2 \sin^2 \theta / 2g$; $R = u^2 \sin^2 \theta / g$; $T = 2u \sin \theta / g$; max range at $\theta = 45^{\circ}$.
- Circular motion: $a = v^2/r = \omega^2 r$; $v = \omega r$; $\omega = 2\pi/T$.

4. FORMULAS

$$v = u + at; s = ut + (1/2)at^2; v^2 = u^2 + 2as; s = ((u+v)/2)t$$

$$S_n = u + (a/2)(2n - 1)$$

$$a = dv/dt = v(dv/dx); a(\text{centripetal}) = v^2/r = \omega^2 r$$

$$H = u^2 \sin^2 \theta / 2g; R = u^2 \sin^2 \theta / g; T = 2u \sin \theta / g$$

Symbol meanings: u = initial velocity, v = final velocity, a = acceleration, s = displacement, t = time, θ = projection angle, $g = 9.8 \text{ m/s}^2$, r = radius, ω = angular speed.

5. CONDITIONS / SPECIAL CASES

- Body starts from rest $\rightarrow u = 0$, to $s = (1/2)at^2$. "Car from rest to 20 m/s in 10 s": $a = 2 \text{ m/s}^2$, $s = (1/2)(2)(100) = 100 \text{ m}$.
- Projectile: max range 45° par; complementary angles ($30^\circ + 60^\circ$, $20^\circ + 70^\circ$) par equal range.
- Highest point par projectile ki velocity sirf horizontal ($u \cos \theta$) hoti hai; acceleration g hi rehti hai (kabhi zero nahi).
- $s = xt^2 - yt + z$ (polynomial in t): $a = d^2s/dt^2 = 2x$ — CONSTANT, isliye $t = 4 \text{ s}$ par bhi $2x$ hi hoga.
- Average speed (distance segments): $v_{\text{avg}} = \text{total distance} / \text{total time}$. $1/3$ distance at 60 km/h , $2/3$ at $30 \text{ km/h} \rightarrow v_{\text{avg}} = 36 \text{ km/h}$.
- Escalator/moving walkway: combined time $t = t_1 t_2 / (t_1 + t_2)$ (walk speed + walkway speed).
- Coefficient of restitution $e = \text{separation speed} / \text{approach speed}$; bounce ke baad height = $e^2 h$ (perfectly elastic $e = 1$).
- Circular motion mein velocity double $\rightarrow$ acceleration $4 \times$ ($a \propto v^2$). Ratio after: before = $4:1$.

6. HOW TO RECOGNIZE THE QUESTION

- "nth second / per-second ratio" $\rightarrow S_n = u + (a/2)(2n-1)$; odd ratio $1:3:5:7$.
- "Slope/area of a graph" $\rightarrow$ slope $x-t = v$, slope $v-t = a$, area $v-t = s$.
- "Projectile range/height/time" $\rightarrow H, R, T$ formulas.
- "Uniform circular motion" $\rightarrow a = v^2/r = \omega^2 r$.
- "s as polynomial in t" $\rightarrow$ differentiate TWICE for acceleration.

7. SOLVING METHODS

- Constant-acceleration question: list karo u, v, a, s, t ; wahi equation lo jisme knowns + unknown hon.
- Ratio problems (heights $h, 2h$): time $\propto \sqrt{h} \rightarrow$ ratio $\sqrt{1} : \sqrt{2}$.
- Graph questions: slope ya area se quantity identify karo (curve ki shape se nahi).

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — The ratio of the distances travelled by a freely falling body in the 1st, 2nd, 3rd and 4th second:

What this tests: Distance in nth second, $u = 0$ (free fall).

Solution: $S_n = u + (a/2)(2n-1) = 0 + (g/2)(2n-1) \rightarrow S_n \propto (2n-1)$.

$n = 1, 2, 3, 4 \rightarrow 2n-1 = 1, 3, 5, 7$.

Key takeaway: Free-fall per-second distances hamesha odd numbers $1:3:5:7$.

REAL PYQ EXAMPLE — A car accelerates uniformly from rest to 20 m/s in 10 seconds. How far does it travel?

What this tests: Uniform acceleration, $u = 0, v = 20, t = 10$.

Solution: $s = ((u+v)/2) \times t = (0+20)/2 \times 10 = 100 \text{ m}$.

Fast method: Average-velocity method: $v_{\text{avg}} = (u+v)/2 = 10 \text{ m/s} \rightarrow s = 10 \times 10 = 100 \text{ m}$.

Key takeaway: Average velocity $\times$ time is the fastest route for uniform acceleration.

REAL PYQ EXAMPLE — Two objects are dropped from heights h and $2h$. Ratio of times to reach the ground:

What this tests: Free fall: $t = \sqrt{(2h/g)}$, so $t \propto \sqrt{h}$.

Solution: $t_1:t_2 = \sqrt{h} : \sqrt{(2h)} = 1 : \sqrt{2}$.

Key takeaway: Time $\propto \sqrt{\text{height}}$. Double height $\rightarrow$ time $\times \sqrt{2}$ (not double).

9. 10-SECOND TRICKS

- Odd-number trick: $1:3:5:7$ ALWAYS free-fall per-second ratio.
- Double velocity in circular motion $\rightarrow$ acceleration $\times 4$.
- Ball up then caught $\rightarrow$ total time = $2u/g$.
- Slope = derivative, area = integral — har motion graph par apply karo.

10. IMPORTANT FACTS

- Retardation = negative acceleration (deceleration).

11. COMMON TRAPS

TRAP: Average speed aur average velocity same hoti hai.

CORRECT: Average velocity = NET displacement/time (closed loop $\rightarrow$ zero); average speed = total distance/time.

TRAP: $s = xt^2 - yt + z$ mein $t = 4 \text{ s}$ par alag acceleration hota hai.

CORRECT: $a = 2x$ (constant) — t par depend nahi karta. Ye trap baar-baar aata hai.

TRAP: Retardation nikalne ke liye direct $a = dv/dt$ lagao bina $v = 0$ wala time nikale.

CORRECT: Pehle $v = 0$ se time nikalo, phir $a = dv/dt$. ($x = 8 + 12t - t^3$: $v = 12 - 3t^2 = 0 \rightarrow t = 2 \text{ s} \rightarrow a = -6t = -12 \rightarrow$ retardation 12 m/s^2 .)

12. GRAPHS / DIAGRAMS

- $x-t$ graph: slope = velocity; straight line = uniform velocity.
- $v-t$ graph: slope = acceleration; AREA = displacement.
- $a-t$ graph: area = change in velocity.
- Projectile path ek PARABOLA hai; range 45° par maximum.

13. LAST-MINUTE REVISION

LAST-MINUTE REVISION

- $v = u + at$; $s = ut + (1/2)at^2$; $v^2 = u^2 + 2as$; $S_n = u + (a/2)(2n-1)$
- Free fall: per-second $1:3:5:7$; total $1:4:9:16$; $t \propto \sqrt{h}$
- Circular: $a = v^2/r = \omega^2 r$; double $v \rightarrow 4 \times a$
- Projectile: $H = u^2 \sin^2 \theta / 2g$; $R = u^2 \sin 2\theta / g$; $T = 2u \sin \theta / g$

CHAPTER 03 — LAWS OF MOTION

1. INTRODUCTION

Ye chapter force aur motion ke beech ka connection samjhata hai. Newton ke teen laws, momentum, impulse aur friction — sab yahin aate hain. Inka use hi baad mein work-energy aur rotational motion ka base banta hai.

2. CORE CONCEPTS

Newton's 1st law (inertia). Body apni state (rest ya uniform motion) tab tak nahi badalti jab tak net external force na lage. Mass hi inertia ka measure hai. Bus suddenly start ho to passenger peeche girta hai — ye INERTIA hai (1st law), action-reaction nahi.

Newton's 2nd law. $F = ma = dp/dt$. Force = momentum ke change ki rate. Yahi quantitative law hai: force se acceleration, ya mass, ya momentum change sab nikalte hain.

Newton's 3rd law. Har action ka equal aur opposite reaction. Boat se jump karo to boat peeche jaati hai — ye action-reaction hai. Ye law SAB interacting bodies par lagta hai.

What is momentum? $p = mv$ (vector). Isolated system (no external force) mein total momentum conserved rehta hai — isi se recoil problems solve hoti hain.

What is impulse? $J = F \times \Delta t$ = change in momentum (Δp). Same momentum change ke liye: jitna LAMBA time, utna CHOTA force. Isi liye airbag, cushioned bed — stopping time badha kar force kam karte hain.

What is equilibrium? Body par net force ZERO (translational) aur net torque ZERO (rotational) ho to body equilibrium mein hai. Zero resultant banane ke liye MINIMUM 2 forces chahiye — equal aur opposite (1 force kabhi zero nahi de sakta, 3-4 zaroori nahi).

What is a resultant? Do vectors A aur B ka resultant $R = \sqrt{A^2 + B^2 + 2AB \cos\theta}$. Perpendicular ($\theta = 90^\circ$) $\rightarrow R = \sqrt{A^2 + B^2}$; same direction $\rightarrow A+B$; opposite $\rightarrow |A-B|$.

What is an Atwood machine? Pulley par do masses m_1, m_2 ($m_2 > m_1$) hang ho $\rightarrow$ acceleration $a = (m_2 - m_1)g / (m_1 + m_2)$. Massless string hai to tension dono sides par SAME.

What is apparent weight in a lift? Lift upar accelerate $\rightarrow R = m(g+a)$ (feel heavy). Neeche accelerate $\rightarrow R = m(g-a)$. Free fall ($a = g$) $\rightarrow R = 0$ (weightless).

Which forces are NOT electromagnetic? Tension, friction, normal, spring — sab electromagnetic hain (atom-atom charge forces). GRAVITY aur NUCLEAR force electromagnetic NAHI hain.

3. KEY RELATIONS

- $F = ma$; $p = mv$; $J = F\Delta t = \Delta p$.
- Kinetic energy vs momentum: $K = p^2/2m$. $K \propto v^2$ ho jaye $\rightarrow p \propto \sqrt{K}$, NOT $4 \times$.
- Momentum conserve: $m_1v_1 = m_2v_2 \rightarrow$ recoil $v = (m \times v)/M$.
- Static friction max: $f_{\text{max}} = \mu_s N$; bina slip ke max acceleration $a = \mu_s g$.
- Resultant: $R = \sqrt{A^2 + B^2 + 2AB \cos\theta}$; $\perp \rightarrow \sqrt{A^2 + B^2} \rightarrow 3 \text{ kN} \ \& \ 4 \text{ kN} \rightarrow 5 \text{ kN}$.
- Atwood: $a = (m_2 - m_1)g / (m_1 + m_2)$; lift: $R = m(g \pm a)$.
- Angle between vectors: $\cos\theta = (A \cdot B) / (|A||B|)$.
- Minimum forces for zero resultant = 2 (equal, opposite).
- Force from a, mass: $F = ma$ (bus 8 tonnes, $a = 0.5 \rightarrow F = 4000 \text{ N}$).

4. FORMULAS

$F = ma = dp/dt$; $p = mv$; $J = F\Delta t = \Delta p$
 $f_{\text{max}} = \mu_s N$; $a = \mu_s g$
 $K = p^2/2m \rightarrow p = \sqrt{2mK}$
 $R = \sqrt{A^2 + B^2 + 2AB \cos\theta}$; $\cos\theta = (A \cdot B) / (|A||B|)$
 $a_{\text{atwood}} = (m_2 - m_1)g / (m_1 + m_2)$; $R_{\text{lift}} = m(g \pm a)$
 Symbol meanings: F = force (N), m = mass (kg), a = acceleration, p = momentum, μ_s = coefficient of static friction, N = normal force, J = impulse.

5. CONDITIONS / SPECIAL CASES

- Momentum double (mass fixed) $\rightarrow$ velocity double. 3000 units $\rightarrow$ 6000 units.
- $4 \times KE \rightarrow 2 \times$ momentum. ($K = p^2/2m$ ka $\sqrt{\text{lete hain.}}$)
- Airbag/cushion $\rightarrow$ time badhao $\rightarrow$ force kam. Hamesha "increase time" wala option.
- Recoil speed = (chhota mass $\times$ uski velocity)/bada mass. Bullet 100 g at 100 m/s, gun 20 kg $\rightarrow v = (0.1 \times 100)/20 = 0.5 \text{ m/s}$.
- 3 kN aur 4 kN $\perp \rightarrow$ resultant 5 kN (3-4-5 triangle).

- Momentum 25% badhao ($p \rightarrow 1.25p$) $\rightarrow K = p^2/2m \rightarrow K \times (1.25)^2 = 1.5625 \rightarrow 56.25\%$ increment.
- Atwood: heavier side neeche, $a = (m_2 - m_1)g / (m_1 + m_2)$.
- Lift upar $\rightarrow R = m(g+a)$; neeche $\rightarrow R = m(g-a)$.

6. HOW TO RECOGNIZE THE QUESTION

- "Force / mass / acceleration" $\rightarrow F = ma$.
- "Momentum double / velocity double" $\rightarrow p = mv$.
- "Recoil of a gun" $\rightarrow$ momentum before = after = 0.
- "Impulse / impact force" $\rightarrow J = F\Delta t = \Delta p$ (longer time $\rightarrow$ smaller force).
- "Max acceleration without slipping" $\rightarrow a = \mu_s g$.
- "Resultant of two forces" $\rightarrow R = \sqrt{A^2 + B^2 + 2AB \cos\theta}$.
- "Zero resultant minimum forces" $\rightarrow 2$.
- "Pulley / two masses" $\rightarrow$ Atwood $a = (m_2 - m_1)g / (m_1 + m_2)$.
- "Weight in a lift" $\rightarrow m(g \pm a)$.

7. SOLVING METHODS

- Recoil: total momentum before = after (usually 0).
- $F = ma$ directly for force/mass.
- Impulse: $J = Ft = \Delta p$.
- Friction threshold: $F \leq \mu_s N$ for rest; slipping starts at $a = \mu_s g$.

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — A bullet of mass 100 g is fired from a gun of mass 20 kg with a velocity of 100 m/s. The recoil velocity of the gun is:

What this tests: Conservation of linear momentum (initially at rest).

Solution: $0 = (0.1)(100) + (20)(v) \rightarrow v = -10/20 = -0.5 \text{ m/s}$ (minus = opposite direction).

Key takeaway: Recoil = $(m \times v)/M$ — chhota mass $\times$ velocity, bade mass se divide.

REAL PYQ EXAMPLE — An unbalanced force of 90 N produces an acceleration of 15 m/s². The mass is:

What this tests: Newton's 2nd law directly.

Solution: $m = F/a = 90/15 = 6 \text{ kg}$.

Key takeaway: $m = F/a$ — force aur acceleration se mass ek step mein.

REAL PYQ EXAMPLE — A bus of mass 8 tonnes rolls down a hill with acceleration 0.5 m/s². The force is:

What this tests: $F = ma$ with unit conversion (8 tonnes = 8000 kg).

Solution: $F = 8000 \times 0.5 = 4000 \text{ N}$.

Key takeaway: Tonnes $\rightarrow$ kg pehle convert karo ($\times 1000$).

REAL PYQ EXAMPLE — A ball has 3000 units of momentum. If velocity is doubled, new momentum is:

What this tests: $p \propto v$ (mass fixed).

Solution: Double $v \rightarrow$ double $p \rightarrow 6000$ units.

Key takeaway: Momentum $\propto$ velocity (mass constant).

$KE \propto v^2$ — dono confuse mat karo.

REAL PYQ EXAMPLE — A particle's linear momentum is increased by 25%. The percentage increase in kinetic energy is:

What this tests: $K = p^2/2m$ — momentum ka factor square hota hai.

Solution: $p \rightarrow 1.25p \rightarrow K \propto (1.25)^2 = 1.5625 \rightarrow 56.25\%$ increase.

Key takeaway: $p \propto n \rightarrow K \propto n^2$. $1.25^2 = 1.5625$ (56.25% up).

REAL PYQ EXAMPLE — Two forces of 3 kN and 4 kN act perpendicular to each other. Their resultant is:

What this tests: Perpendicular vectors → Pythagoras.

Solution: $R = \sqrt{3^2 + 4^2} = 5 \text{ kN}$.

Key takeaway: 3-4-5 right triangle → resultant 5 kN.

REAL PYQ EXAMPLE — Two masses m_1 and m_2 ($m_2 > m_1$) hang from a frictionless pulley. The acceleration is:

What this tests: Atwood machine formula.

Solution: $a = (m_2 - m_1)g / (m_1 + m_2)$.

Key takeaway: Net force $(m_2 - m_1)g$ ko total mass $(m_1 + m_2)$ se divide karo.

9. 10-SECOND TRICKS

- Momentum double → velocity double.
- $K \propto n$ -times → $p \propto \sqrt{n}$ -times.
- Longer stopping time = smaller force (airbag). Always pick "increase time".
- Recoil speed = (small mass × velocity)/big mass.
- $3 \text{ kN} \perp 4 \text{ kN} \rightarrow 5 \text{ kN}$; $p \times 1.25 \rightarrow K \times 1.5625$.
- Zero resultant ke liye minimum 2 forces (equal-opposite).

10. IMPORTANT FACTS

- Force ek vector hai; friction hamesha relative motion oppose karti hai.
- Newton's 3rd law SAB interacting bodies par lagta hai (boat-jump, recoil, rocket).

11. COMMON TRAPS

TRAP: Bus suddenly start → passenger peeche girta hai = Newton's 3rd law.

CORRECT: Ye INERTIA hai (1st law). Boat se jump → boat peeche = 3rd law (action-reaction).

TRAP: $4 \times KE \rightarrow 4 \times$ momentum.

CORRECT: $K = p^2/2m$ se $p = \sqrt{2mK} \rightarrow 4 \times KE = 2 \times$ momentum.

TRAP: Airbag force badhata hai.

CORRECT: Airbag stopping TIME badhata hai → force GHATTI hai ($J = F\Delta t$).

TRAP: Zero resultant ke liye 1 ya 3 forces kaafi hain.

CORRECT: Minimum 2 forces (equal aur opposite). 1 force kabhi zero resultant nahi deta.

TRAP: Momentum 25% badhao to KE bhi 25% badhti hai.

CORRECT: $K \propto p^2 \rightarrow (1.25)^2 = 1.5625 \rightarrow 56.25\%$ increase.

12. GRAPHS / DIAGRAMS

- Free-body diagram: pehle weight, normal, friction, applied force draw karo.
- Force-time graph: area = impulse = momentum change.
- Momentum conserved with changing mass → $v \propto 1/m$ (hyperbola).
- Friction vs applied force: static friction $\mu_s N$ tak badhta hai, phir kinetic $\mu_k N$ par gir jaata hai.

13. LAST-MINUTE REVISION**LAST-MINUTE REVISION**

- $F = ma = dp/dt$; $p = mv$; $J = F\Delta t = \Delta p$
- $f_{\text{max}} = \mu_s N$; slipping threshold $a = \mu_s g$
- Recoil: $v = (m \times v)/M$; $K = p^2/2m \rightarrow K \times 4 \Rightarrow p \times 2$

CHAPTER 04 — WORK, ENERGY AND POWER**1. INTRODUCTION**

Force ka motion par effect "work" ke through measure hota hai. Work-energy theorem hi aage ke saare chapters (potential, kinetic, springs, power) ki neev hai. Ye chapter short hai lekin highly scoring — dot product, spring energy aur power ke fixed question types hain.

2. CORE CONCEPTS

What is work? $W = F \cdot s = F s \cos\theta$. Force ka displacement ki direction mein component hi kaam karta hai. Work ek SCALAR hai aur NEGATIVE bhi ho sakta hai — jab force motion oppose kare (retarding force → negative work).

What is energy? Kaam karne ki capacity. Kinetic energy $K = (1/2)mv^2$ (motion ki energy), potential energy $U = mgh$ (height ki) ya $(1/2)kx^2$ (spring ki).

What is power? Kaam karne ki RATE. $P = W/t = Fv$. Jitna jaldi kaam, utni zyada power.

What is a conservative force? Jis force ke liye closed path par work = 0 (gravity, spring). Tab total mechanical energy conserve hoti hai: gain in KE = loss in PE.

Work-energy theorem. $W_{\text{total}} = \Delta K$ — net work = kinetic energy ka change. Ye theorem hi har "force → speed" problem ka shortcut hai.

What is elastic energy in a wire? Wire ko stretch karne par stored energy $U = (1/2)F \times \Delta L$. Energy density (per volume) = $(1/2) \times \text{stress} \times \text{strain} = (1/2)Y(\text{strain})^2$ — bilkul spring ke $(1/2)kx^2$ jaisa, bas wire ke terms mein.

Work done pulling a hanging chain. Uniform chain ka kuch hissa table se latka ho to use upar kheenchne ka work = (hanging mass) $\times g \times$ (uske centre of mass ki raise).

Ek-third length latki ho to $W = (m/3) \times g \times (L/6) = mgL/18$.

3. KEY RELATIONS

- $W = F \cdot s = F_x s_x + F_y s_y + F_z s_z$ (dot product). $F = 2i + 3j + k$, $s = i + 2j - 4k \rightarrow W = 2 + 6 - 4 = 4 \text{ J}$.
- Spring: $U = (1/2)kx^2$; work from x_1 to $x_2 = (1/2)k(x_2^2 - x_1^2)$.
- Spring 2 cm → 100 J; further 2 cm (total 4 cm) → energy $4 \times = 400 \text{ J} \rightarrow$ increase 300 J. ($U \propto x^2$.)
- Power: $P = W/t = Fv$.
- Variable force: $W = \int F dx$. $F = 3ax^2 - 3$ from 0 to 2 → $8a - 6$.
- Wire energy: $U = (1/2)F\Delta L = (1/2)Y \times (\Delta L/L)^2 \times \text{volume}$. Energy density = $(1/2) \times \text{stress} \times \text{strain}$.
- Chain (1/3 length L hanging): $W = (m/3) \times g \times (L/6) = mgL/18$.

4. FORMULAS

$$W = F s \cos\theta = F \cdot s ; \Delta K = W$$

$$K = (1/2)mv^2 = p^2/2m ; U = mgh ; U_{\text{spring}} = (1/2)kx^2$$

$$P = W/t = F v \cos\theta$$

$$W_{\text{spring}}(x_1 \rightarrow x_2) = (1/2)k(x_2^2 - x_1^2)$$

$$U_{\text{wire}} = (1/2)F\Delta L ; \text{energy density} = (1/2) \times \text{stress} \times \text{strain} = (1/2)Y(\text{strain})^2$$

$$W_{\text{chain}} = (\text{hanging mass}) \times g \times (\text{height of its centre of mass})$$

Symbol meanings: F = force, s = displacement, θ = angle between them, k = spring constant, v = velocity, m = mass, $g = 9.8 \text{ m/s}^2$.

5. CONDITIONS / SPECIAL CASES

- $\theta = 0 \rightarrow W = Fs$ (max). $\theta = 90^\circ \rightarrow W = 0$ (pushing wall, bag horizontal le jaana). $\theta = 180^\circ \rightarrow W = -Fs$ (retarding force).
- Conservative field: (p) ME conserved, (q) closed-path work = 0, (r) gain KE = loss PE — TEENO correct.
- Spring 1000 N/m, 10 cm → 20 cm: $W = 0.5 \times 1000 \times (0.04 - 0.01) = 15 \text{ J}$.

- Athlete 60 kg, 100 m in 10 s $\rightarrow v \approx 10 \text{ m/s} \rightarrow K = 0.5 \times 60 \times 100 = 3000 \text{ J}$ (range 2000–5000 J).

6. HOW TO RECOGNIZE THE QUESTION

- "Work by force over displacement (vectors)" $\rightarrow$ dot product $W = F \cdot s$.
- "Retarding force ka work" $\rightarrow$ negative.
- "Spring energy / work between two stretches" $\rightarrow W = (1/2)k(x_2^2 - x_1^2)$.
- "Variable force $F(x)$ " $\rightarrow$ integrate $F dx$.
- "Power / kinetic-energy estimate" $\rightarrow P = W/t$, $K = (1/2)mv^2$.

7. SOLVING METHODS

- Dot-product work: matching components multiply karke add karo (signs dhyan se).
- $F(x)$ force: $\int F dx$ from x_1 to x_2 .
- Estimate problems: rough speed le kar $K = (1/2)mv^2$.

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — Calculate the work done by $F = 2\mathbf{i} + 3\mathbf{j} + k$ on a particle displaced by $s = \mathbf{i} + 2\mathbf{j} - 4\mathbf{k}$.

What this tests: Work = dot product of force and displacement.

Solution: $W = (2)(1) + (3)(2) + (1)(-4) = 2 + 6 - 4 = 4 \text{ J}$.

Key takeaway: Matching components multiply karo; $-4k$ wale sign ka dhyan rakho.

REAL PYQ EXAMPLE — A spring ($k = 1000 \text{ N/m}$) is stretched from 10 cm to 20 cm. Work done:

What this tests: Spring work between two positions — difference of squares.

Solution: $W = (1/2)k(x_2^2 - x_1^2) = 0.5 \times 1000 \times (0.04 - 0.01) = 15 \text{ J}$.

Key takeaway: $W = (1/2)k(x_2^2 - x_1^2)$, NOT $(1/2)k(x_2 - x_1)^2$.

REAL PYQ EXAMPLE — When a spring is stretched by 2 cm, stored energy is 100 J. Stretched further by 2 cm, energy increases by:

What this tests: $U \propto x^2$ — doubling stretch quadruples energy.

Solution: Total 4 cm $\rightarrow U = 4 \times 100 = 400 \text{ J}$. Increase = $400 - 100 = 300 \text{ J}$.

Key takeaway: $U \propto x^2$: stretch $2\times \rightarrow$ energy $4\times$.

"Further" $\rightarrow$ naya minus purana.

REAL PYQ EXAMPLE — When a force retards the motion of a body, the work done is:

What this tests: Sign of work — force opposes displacement.

Solution: $\theta = 180^\circ \rightarrow W = Fs \cos 180^\circ = -Fs \rightarrow$ negative.

Key takeaway: Motion oppose karne wala force hamesha negative work karta hai.

REAL PYQ EXAMPLE — Energy stored in a stretched wire is ($Y =$ Young's modulus, $S =$ strain):

What this tests: Elastic energy density in a wire.

Solution: Energy density = $(1/2) \times \text{stress} \times \text{strain} = (1/2)YS^2$.

Key takeaway: Wire energy density = $(1/2)Y(\text{strain})^2$ — spring ke $(1/2)kx^2$ ka wire version.

REAL PYQ EXAMPLE — A uniform chain of length L and mass m lies on a table with one-third of its length hanging. Work to pull it up:

What this tests: Work = mgh for the hanging part's centre of mass.

Solution: $W = (m/3) \times g \times (L/6) = mgL/18$.

Key takeaway: Hanging part ka CM $L/6$ neeche hai — sirf hanging mass par mgh lagao.

9. 10-SECOND TRICKS

- Force opposing motion $\rightarrow$ work negative (instant).
- Spring stretch $2\times \rightarrow$ energy $4\times$.
- Dot product: $(2)(1) + (3)(2) + (1)(-4) = 4$ — one line.
- Wire energy density = $(1/2) \times \text{stress} \times \text{strain}$.
- Chain pull-up: $W = (\text{hanging mass}) \times g \times (\text{CM height})$.

10. IMPORTANT FACTS

- Work aur energy scalars hain, same dimension $[ML^2T^{-2}]$.
- Friction hamesha negative work karti hai.

11. COMMON TRAPS

TRAP: Spring work = $(1/2)k(x_2 - x_1)^2$.

CORRECT: $W = (1/2)k(x_2^2 - x_1^2)$ — difference of squares.

TRAP: Dot product mein $-4k$ component ka sign chhodna.

CORRECT: Saare components ke signs rakho: $2+6-4 = 4$.

TRAP: Retarding force positive work karti hai.

CORRECT: Retarding force ka work NEGATIVE hota hai.

12. GRAPHS / DIAGRAMS

- Force-displacement graph: AREA = work done.
- Spring F vs x : origin se straight line; area = $(1/2)kx^2$.
- Power-time graph: area = energy.

13. LAST-MINUTE REVISION

LAST-MINUTE REVISION

- $W = F s \cos \theta$; $K = (1/2)mv^2$; $U = (1/2)kx^2$; $P = W/t = Fv$
- Retarding force $\rightarrow$ negative work; $\theta = 90^\circ \rightarrow$ zero work
- Spring: $W = (1/2)k(x_2^2 - x_1^2)$; $U \propto x^2$

CHAPTER 05 — ROTATIONAL MOTION (SYSTEM OF PARTICLES AND RIGID BODY)

1. INTRODUCTION

Jab body ghoomti hai (rotation), tab linear quantities ki jagah angular quantities aati hain: ω (angular velocity), α (angular acceleration), I (moment of inertia), τ (torque), L (angular momentum). Exam mein sabse zyada marks moment-of-inertia table aur parallel/perpendicular axis theorem se aate hain.

2. CORE CONCEPTS

What is moment of inertia? $I =$ rotational inertia = Σmr^2 .

Jaise mass linear motion ka "resistance" hai, waise I rotation ka resistance hai. Ye body ke mass aur axis se mass ki doori (distribution) par depend karta hai — shape aur axis badle to I badalta hai.

What is torque? $\tau = r \times F = rF \sin \theta = I\alpha$. Torque force aur MOMENT ARM (perpendicular distance) par depend karta hai — mass par nahi. Yahi exam ka favourite trap hai.

What is angular momentum? $L = I\omega = mvr$ (particle ke liye). Jab tak external torque na ho, L conserved rehta hai. Isi se Kepler ka 2nd law (equal areas) aata hai.

What is centre of mass? System ka wo point jahan poora mass concentrated maan sakte hain. External force na ho to CM ka velocity constant rehta hai.

Radius of gyration. $k =$ wo distance jahan saara mass rakhne par same I mile: $I = Mk^2$.

CM of uniform bodies. Uniform cube/sphere/rod ka CM GEOMETRIC CENTRE par hota hai — cube ka CM body ke centre par (face ya edge par nahi).

CM formula. $x_{cm} = \Sigma(mx)/\Sigma m$; $V_{cm} = \Sigma(mv)/\Sigma m$. Do bodies ka system: $V_{cm} = (m_1v_1 + m_2v_2)/(m_1 + m_2)$.

KE of centre of mass. System ki CM motion ki KE = $(1/2)(m_1 + m_2)V_{cm}^2$. (Total KE = CM ki KE + relative motion ki KE.)

Angular acceleration in UCM. Uniform circular motion mein ω CONSTANT $\rightarrow$ angular acceleration $\alpha = 0$.

3. KEY RELATIONS

- $L = I\omega$; $\tau = I\alpha = rF \sin\theta$; $KE_{rot} = (1/2)I\omega^2$.
- Parallel axis: $I = I_{cm} + Md^2$. Perpendicular axis (lamina): $I_z = I_x + I_y$.
- Ring: MR^2 ($\perp$ centre), $MR^2/2$ (diameter), $3MR^2/2$ (tangent).
- Disc: $MR^2/2$ ($\perp$ centre), $MR^2/4$ (diameter), $5MR^2/4$ (tangent in plane).
- Sphere: $2MR^2/5$ (diameter). Rod: $ML^2/12$ (centre $\perp$), $ML^2/3$ (end $\perp$).
- Disc diameter = $I \rightarrow \perp$ axis through rim = $6I$ ($I = MR^2/4 \rightarrow MR^2 = 4I$; rim = $2I + 4I$).
- CM: $x_{cm} = \Sigma(mx)/\Sigma m$; $V_{cm} = \Sigma(mv)/\Sigma m$; $KE_{cm} = (1/2)(\Sigma m)V_{cm}^2$.
- $\tau = r \times F$ (cross product); magnitude $rF \sin\theta$; direction right-hand rule se.
- UCM: ω constant $\rightarrow \alpha = 0$.

4. FORMULAS

Body / Axis	Moment of inertia
Ring, axis $\perp$ plane through centre	MR^2
Ring, about a diameter	$MR^2/2$
Ring, about a tangent (in plane)	$3MR^2/2$
Disc, axis $\perp$ plane through centre	$MR^2/2$
Disc, about a diameter	$MR^2/4$
Solid sphere, about diameter	$2MR^2/5$
Solid cylinder, about its own axis	$MR^2/2$
Rod, about centre ($\perp$)	$ML^2/12$
Rod, about end ($\perp$)	$ML^2/3$

5. CONDITIONS / SPECIAL CASES

- Torque $\propto$ force $\times$ moment arm (NOT mass, NOT radius alone).
- L conserved $\rightarrow I$ double ho to ω half ($L = I\omega$). Rotating boy hands chest mein kheenche $\rightarrow I$ ghatta $\rightarrow \omega$ badhta.
- Do bodies mutual gravity se ek dusre ki taraf $\rightarrow V_{cm} = 0$ (no external force). 2 kg at 20 m/s aur 4 kg at 10 m/s opposite $\rightarrow V_{cm} = 0$.
- Ring tangent = $3MR^2/2$ (parallel axis se: $MR^2 + MR^2/2$). $5MR^2/2$ galat hai.

6. HOW TO RECOGNIZE THE QUESTION

- "MI about a given axis" $\rightarrow$ table + parallel/perpendicular theorem.
- "Torque kis par depend" $\rightarrow$ force $\times$ moment arm.
- " L conserved (I badla)" $\rightarrow L = I\omega$.
- "CM velocity of two bodies" $\rightarrow$ external force zero $\rightarrow V_{cm}$ constant.

7. SOLVING METHODS

- MI axis ke hisaab se: parallel/perpendicular theorem lagao.
- Radius of gyration: $I = Mk^2$ se k nikalo.
- Angular displacement track par: sweep hua angle padho.

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — What is the moment of inertia of a ring of mass M and radius R about a tangent to the circle?

What this tests: Parallel-axis theorem from a diameter.

Solution: $I_{tangent} = I_{diameter} + MR^2 = MR^2/2 + MR^2 = 3MR^2/2$.

Key takeaway: Ring tangent = $3MR^2/2$ ($5MR^2/2$ galat hai).

REAL PYQ EXAMPLE — Moment of inertia of a disc about a diameter is I . Its MI about an axis $\perp$ to its plane through a point on the rim is:

What this tests: Perpendicular-axis + parallel-axis theorem.

Solution: $I_{\perp center} = I_{diameter} + I_{diameter} = 2I$. Rim: $I = 2I + MR^2$. Disc ke liye $MR^2 = 4I \rightarrow 2I + 4I = 6I$.

Key takeaway: Disc: $I_{diameter} = I \rightarrow \perp$ axis through centre = $2I$, rim = $6I$.

REAL PYQ EXAMPLE — Torque depends on:

What this tests: Definition of torque.

Solution: $\tau = \text{force} \times \text{moment arm (perpendicular distance)}$.

Key takeaway: Torque = force $\times$ moment arm — "mass" wala option galat.

REAL PYQ EXAMPLE — Two bodies of 2 kg (20 m/s) and 4 kg (10 m/s) move toward each other under mutual gravity. Velocity of centre of mass:

What this tests: No external force $\rightarrow V_{cm}$ constant (initially zero).

Solution: $V_{cm} = (2 \times 20 + 4 \times (-10))/6 = 0$.

Key takeaway: Mutual attraction = internal force $\rightarrow V_{cm}$ nahi badalta.

REAL PYQ EXAMPLE — The centre of mass of a uniform cube is at:

What this tests: Uniform body $\rightarrow$ geometric centre.

Solution: Cube ka CM uske geometrical centre par hota hai (face/edge/diagonal par nahi).

Key takeaway: Uniform (symmetric) body ka CM = geometric centre.

REAL PYQ EXAMPLE — A ball of mass M (2 m/s) hits a 1 kg ball (1 m/s). KE of centre of mass is $4/3$ J. Find M :

What this tests: $KE_{cm} = (1/2)(M+1)V_{cm}^2$; $V_{cm} = (2M + 1 \times 1)/(M+1)$.

Solution: $(1/2)(M+1)[(2M+1)/(M+1)]^2 = 4/3 \rightarrow (2M+1)^2 = (8/3)(M+1)$.

$3(4M^2 + 4M + 1) = 8M + 8 \rightarrow 12M^2 + 4M - 5 = 0 \rightarrow M = 0.5$ kg.

Key takeaway: $KE_{cm} = (1/2)(\Sigma m)V_{cm}^2$ — pehle V_{cm} nikalo, phir KE.

9. 10-SECOND TRICKS

- Ring tangent = $3MR^2/2$; disc tangent = $5MR^2/4$; sphere = $2MR^2/5$.
- I double $\rightarrow \omega$ half (L constant).
- Kepler equal areas = angular momentum conservation.
- Uniform body $\rightarrow$ CM at geometric centre.
- UCM mein $\alpha = 0$ (ω constant).

10. IMPORTANT FACTS

- Rigid body = fixed shape (particles ke beech distance constant).
- Angular momentum ka dimension = Planck's constant ka dimension = $[ML^2T^{-1}]$.

11. COMMON TRAPS

TRAP: Torque "mass" ya "radius" par depend karta hai.

CORRECT: Torque = force $\times$ moment arm.

TRAP: Ring tangent = $5MR^2/2$.

CORRECT: Ring tangent = $3MR^2/2$ (parallel axis se: $MR^2/2 + MR^2$).

TRAP: Solid sphere ka $k = r$ wale axis ka distance $\sqrt{0.4r}$ hota hai (printed answer).

CORRECT: METHOD yaad karo: $I = Mk^2 = Mr^2 = (2/5)Mr^2 + Md^2 \rightarrow d = r\sqrt{(3/5)}$. Numbers ki rat mat maro.

12. GRAPHS / DIAGRAMS

- MI axes diagram: ring, disc, sphere, rod — centre/diameter/tangent/end sab draw karke yaad karo.
- ω -t graph (constant α): straight line, slope = α .

13. LAST-MINUTE REVISION

LAST-MINUTE REVISION

- $\tau = I\alpha = rF \sin\theta$; $L = I\omega$; $KE = (1/2)I\omega^2$; $I = Mk^2$
- Ring: MR^2 , $MR^2/2$ (diam), $3MR^2/2$ (tangent)
- Disc: $MR^2/2$, $MR^2/4$ (diam), rim- $\perp$ -axis $6I$; Sphere: $2MR^2/5$; Rod: $ML^2/12$, $ML^2/3$
- Parallel axis: $I = I_{cm} + Md^2$

CHAPTER 06 — GRAVITATION

1. INTRODUCTION

Gravitation batata hai ki masses ek dusre ko kyun kheenchti hain. Is chapter ke teen pillar: Newton's law, acceleration due to gravity (g), aur escape velocity. Satellite energy aur Kepler's laws bhi isi se judi hain.

2. CORE CONCEPTS

What is Newton's law of gravitation? $F = GMm/r^2$ — attractive, dono bodies ko jodne wali line ke along. $G = 6.67 \times 10^{-11}$ (universal constant).

What is g ? Planet ki surface par gravity ka acceleration: $g = GM/R^2$. Ye planet ke mass aur radius par depend karta hai — aapke mass par nahi.

What is escape velocity? Wo minimum speed jisse body planet ka gravity chhod kar hamesha ke liye nikal jaye: $v_e = \sqrt{2GM/R} = \sqrt{2gR}$. Earth ke liye 11.2 km/s.

What is weightlessness in a satellite? Astronaut aur satellite dono SATH mein free fall kar rahe hote hain $\rightarrow$ relative weight zero. Ye FREE FALL hai, "gravity zero" nahi — gravity wahi lagi rehti hai.

Kepler's laws. (1) elliptical orbit, Sun at focus; (2) equal areas in equal time = angular momentum conservation; (3) $T^2 \propto r^3$.

3. KEY RELATIONS

- $g = GM/R^2$; g' at height h ($h \ll R$): $g' = g(1 - 2h/R)$.
- g at depth d : $g' = g(1 - d/R)$ — Earth ke centre par zero.
- $g = (4/3)\pi G\rho R$ (density form): same radius $\rightarrow g \propto \rho$; same density $\rightarrow g \propto R$.
- $v_e = \sqrt{2gR} = \sqrt{2GM/R} = 11.2$ km/s.
- Planet $4M$ aur $R/2 \rightarrow g = G(4M)/(R/2)^2 = 16g$.
- Satellite: $v = \sqrt{GM/r}$; $T = 2\pi\sqrt{r^3/GM}$; $E = -GMm/2r$; $PE = 2E$; $KE = -E$.
- Equal escape velocity $\rightarrow M/R$ equal (NOT $M \times R$).

4. FORMULAS

$F = GMm/r^2$; $g = GM/R^2$; $v_e = \sqrt{2GM/R} = \sqrt{2gR}$
 Satellite: $PE = 2 \times E_{total}$; $KE = -E_{total}$; $T = 2\pi\sqrt{r^3/GM}$

Kepler: $T^2 \propto r^3$; areal velocity constant

Symbol meanings: G = universal gravitational constant, M = planet mass, R = planet radius, r = orbit radius, m = body mass.

5. CONDITIONS / SPECIAL CASES

- g pole se equator tak DECREASE hota hai (rotation + bulge).
- Earth ka radius 1% kam (mass fixed) $\rightarrow g \sim 2\%$ badhta hai ($g = GM/R^2$).
- $2M$ aur $R/2$ wale planet par v_e double ($11.2 \rightarrow 22.4$ km/s).
- Satellite ka time period central mass M aur radius r par depend karta hai, satellite ke mass par NAHI.
- Satellite: $PE = 2 \times$ total energy. Total energy $E_0 \rightarrow PE = 2E_0$, $KE = -E_0$.
- Radii 1:4, densities 1:2 $\rightarrow g$ ratio = $(\rho_1 R_1)/(\rho_2 R_2) = (1 \times 1)/(2 \times 4) = 1:8$.
- Height par g $1/4$ rah jaye $\rightarrow (1 - 2h/R) = 1/4 \rightarrow h = 3R/8$.

6. HOW TO RECOGNIZE THE QUESTION

- "Planet ka g (M , R diye)" $\rightarrow g \propto M/R^2$.
- "Escape velocity" $\rightarrow \sqrt{2GM/R}$.
- "Satellite energy / time period" $\rightarrow PE = 2E_{total}$; T formula.
- "Kepler" $\rightarrow$ equal areas = angular momentum.

7. SOLVING METHODS

- "Mass kx , radius $1/mx$ " $\rightarrow g' = k m^2 \times g$. Radius factor ko SQUARE karo.
- Equal escape velocity $\rightarrow M/R$ equal ($M \times R$ nahi).

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — If the mass of a planet is four times that of earth and radius is half, gravitational acceleration is:

What this tests: $g \propto M/R^2$.

Solution: $g' = 4/(1/2)^2 = 4 \times 4 = 16g$.

Key takeaway: Radius factor square karo: $(1/2)^2 = 1/4$, to $4/(1/4) = 16$.

REAL PYQ EXAMPLE — The escape velocity on earth is 11.2 km/s. On a planet with twice the mass and half the radius, it is:

What this tests: $v_e \propto \sqrt{M/R}$.

Solution: $v_e' = \sqrt{2/(1/2)} = \sqrt{4} = 2 \rightarrow 2 \times 11.2 = 22.4$ km/s.

Key takeaway: $v_e \propto \sqrt{M/R}$: mass ratio direct, radius ratio ulta, dono ka square root.

REAL PYQ EXAMPLE — A satellite in a circular orbit has total energy E_0 . Its potential energy is:

What this tests: Satellite energy relation.

Solution: $PE = 2 \times E_{total} = 2E_0$ ($KE = -E_0$).

Key takeaway: Satellite: $PE = 2E_{total}$, $KE = -E_{total}$ — fixed relation.

REAL PYQ EXAMPLE — The escape velocities of planets A and B are equal. Which statement is correct?

What this tests: Equal v_e condition.

Solution: $v_e = \sqrt{2GM/R}$ equal $\rightarrow M/R$ equal ($M \propto R$), NOT $M \times R$ equal.

Key takeaway: Equal escape velocity $\rightarrow M_1/R_1 = M_2/R_2$ (product nahi).

9. 10-SECOND TRICKS

- $4M + R/2 \rightarrow 16g$; $2M + R/2 \rightarrow v_e \times 2$.

- Weightless astronaut = free fall.
- Equal escape velocity → equal M/R .

10. IMPORTANT FACTS

- $g \approx 9.8 \text{ m/s}^2$; Moon $\approx 1.63 \text{ m/s}^2 (\approx g/6)$. 10 kg: Earth 98 N, Moon 16.3 N.
- Kepler 2nd law = angular momentum conservation.

11. COMMON TRAPS

TRAP: Equal escape velocity → $M_1 R_1 = M_2 R_2$.

CORRECT: $M_1/R_1 = M_2/R_2$ hota hai ($M \times R$ galat).

TRAP: Weightless = gravity nahi hai.

CORRECT: Weightless = free fall; gravity maujood hai.

TRAP: Pole → equator g badhta hai.

CORRECT: g DECREASE hota hai.

12. GRAPHS / DIAGRAMS

- g vs r : Earth ke andar $g \propto r$ (linear rise); bahar $g \propto 1/r^2$ (fall).
- g vs depth: centre par zero tak linear decrease.
- Orbit: orbital velocity chakkar mein rakhti hai; escape velocity hamesha ke liye chhod deti hai.

13. LAST-MINUTE REVISION

LAST-MINUTE REVISION

- $F = GMm/r^2$; $g = GM/R^2$; $v_e = \sqrt{2gR}$; $T^2 \propto r^3$
- $4M$ & $R/2 \rightarrow 16g$; 1% smaller $R \rightarrow g$ up $\sim 2\%$; $2M$ & $R/2 \rightarrow v_e \times 2$
- Satellite: $PE = 2E_{\text{total}}$; $T = 2\pi r \sqrt{r^3/GM}$ (satellite mass se independent)

CHAPTER 07 — PROPERTIES OF BULK MATTER (SOLIDS AND FLUIDS)

1. INTRODUCTION

Ye chapter solids (elasticity) aur fluids (liquids/gases: pressure, surface tension, viscosity) dono cover karta hai. Har sub-topic se 1-2 fixed questions aate hain: Young's modulus, excess pressure, terminal velocity, gauge pressure — sab short aur formula-based hain.

2. CORE CONCEPTS

What is elasticity? Body par force lagao to wo deform hoti hai. Stress = F/A (force per area), strain = $\Delta L/L$ (change per original). Young's modulus Y = stress/strain = stress-strain curve ke LINEAR part ka slope.

What is pressure? Force per area. Gauge pressure = absolute — atmospheric = $P - P_a$. (Jo gauge padhta hai wo atmospheric ke upar ka extra pressure hai.)

What is surface tension? Liquid surface ka wo tendency jisse wo apna area minimum karti hai — isliye drop spherical hota hai. Ye force/length hai aur temperature badhne par DECREASE hoti hai. Soap bubble ki 2 surfaces hoti hain ($\Delta P = 4T/r$), drop ki 1 ($\Delta P = 2T/r$).

What is viscosity? Liquid ki internal friction — layers ke beech ka drag. Terminal velocity $v_t \propto r^2$ (radius square).

Pascal vs Archimedes vs Bernoulli. Pascal: enclosed fluid ka pressure sab taraf equally transmit hota hai (hydraulic lift). Archimedes: buoyant force = displaced fluid ka weight. Bernoulli = energy conservation; continuity = mass conservation.

What is calorimetry? Heat transfer $Q = mc\Delta T$. m = mass, c = specific heat, ΔT = temperature change. Mixture: heat gained = heat lost. Water ki specific heat sabse zyada: $1 \text{ cal/g}^\circ\text{C} = 4186 \text{ J/kg}^\circ\text{C}$.

What is thermal expansion? Garam karne par solid expand hota hai. Linear $\Delta L = \alpha L \Delta T$; area $\Delta A = \beta A \Delta T$ ($\beta = 2\alpha$);

volume $\Delta V = \gamma V \Delta T$ ($\gamma = 3\alpha$). Isliye $\alpha : \beta : \gamma = 1 : 2 : 3$. Thermometer isi expansion par chalta hai.

Why does hot glass crack on sudden cooling? Bahar ki layer pehle sikud jaati hai, andar nahi → thermal STRESS → crack (glass heat ka poor conductor hai).

What is capillary rise? Patli tube mein liquid upar chadhta hai: $h = 2T \cos\theta/(r\rho g)$. Tube patli (r chhota) → rise zyada. Water-glass $\theta = 0^\circ \rightarrow h = 2T/(r\rho g)$.

Float fraction. Floating body ka submerged fraction = $\rho_{\text{body}}/\rho_{\text{liquid}}$. Cube density 800 kg/m^3 , water $1000 \rightarrow 80\%$ andar, 20% bahar.

Breaking stress. Wo maximum stress jis par material toot jaata hai (strength). Steel ki elasticity sabse zyada (steel > copper > rubber).

Moduli. Bulk modulus $K = -\Delta P/(\Delta V/V)$ (compress karne ka resistance); modulus of rigidity G = shear stress/shear strain.

3. KEY RELATIONS

- $Y = (F/A)/(\Delta L/L) = FL/(A\Delta L)$ = stress-strain curve ka linear slope.
- Excess pressure: drop $2T/r$; bubble $4T/r$; ratio $\propto 1/r$ (radii 5:3 → pressure 3:5).
- $v_t \propto r^2 \rightarrow$ radius double $\rightarrow v_t \times 4$.
- $P = \rho gh$; gauge = $P - P_a$.
- Poisson ratio σ = lateral strain/longitudinal strain (unitless).
- Work to stretch wire = $(1/2)F \times \Delta L$ (elastic PE).
- $Q = mc\Delta T$; heat lost = heat gained.
- $\alpha : \beta : \gamma = 1 : 2 : 3$; $\Delta L = \alpha L \Delta T$; $\Delta V = \gamma V \Delta T$.
- Capillary: $h = 2T \cos\theta/(r\rho g)$.
- Float: submerged fraction = $\rho_{\text{body}}/\rho_{\text{liq}}$; n drops combine $\rightarrow$ radius $\propto n^{1/3}$.
- $K = -\Delta P/(\Delta V/V)$; G = shear stress/shear strain.

4. FORMULAS

Stress = F/A ; Strain = $\Delta L/L$; $Y = FL/(A\Delta L)$

$\Delta P(\text{drop}) = 2T/r$; $\Delta P(\text{bubble}) = 4T/r$; Surface energy = $T \times \Delta A$

$v_t = 2r^2(\rho - \sigma)g / 9\eta$

Gauge pressure = $P - P_a$; $P = \rho gh$

$Q = mc\Delta T$; $\Delta L = \alpha L \Delta T$; $\Delta A = 2\alpha A \Delta T$; $\Delta V = 3\alpha V \Delta T$

$h = 2T \cos\theta/(r\rho g)$; submerged fraction = $\rho_{\text{body}}/\rho_{\text{liq}}$

Bulk modulus $K = -\Delta P/(\Delta V/V)$; rigidity G = shear stress/shear strain

Symbol meanings: Y = Young's modulus, F = force, A = area, L = original length, ΔL = elongation, T = surface tension, r = radius, ρ = density, η = viscosity coefficient.

5. CONDITIONS / SPECIAL CASES

- Young's modulus = stress-strain curve ke LINEAR part ka SLOPE.
- Convex meniscus $\rightarrow$ contact angle obtuse $\rightarrow$ mercury (140°). Water/alcohol glass par 0° (concave).
- Temperature badhao $\rightarrow$ surface tension ghatta hai.
- Elasticity = wo ability jisse body permanent change ko RESIST karti hai (stress hatane par original shape wapas aa jaati hai).
- Brim-full cup mein floating ice pighle $\rightarrow$ water level SAME (displaced water = ice ka weight).
- Water drops diameter 1 cm & 1.5 cm $\rightarrow$ radii 1:1.5 $\rightarrow \Delta P$ ratio = $1.5:1 = 3:2$.
- 8 equal drops combine $\rightarrow R = n^{1/3} r = 2r \rightarrow \Delta P$ becomes $1/2$ ($\Delta P \propto 1/r$).
- Water drop 27 droplets mein toota $\rightarrow r' = r/3 \rightarrow \Delta P' = 3\Delta P$.

- Capillary $d = 2$ mm, rise 15 mm, $\theta = 0^\circ$, SG 0.8 $\rightarrow T = \frac{r\rho gh}{2} = \frac{(1 \times 10^{-3})(800)(9.8)(0.015)}{2} = 0.0588$ N/m.
- Gas at constant T: $PV = \text{constant}$ (Boyle's law).

6. HOW TO RECOGNIZE THE QUESTION

- "Young's modulus / stress-strain slope" $\rightarrow Y = \frac{\text{stress}}{\text{strain}}$.
- "Excess pressure drop/bubble" $\rightarrow \frac{2T}{r}$ ya $\frac{4T}{r}$; ratio = radii ka inverse.
- "Terminal velocity vs radius" $\rightarrow \propto r^2$.
- "Gauge pressure" $\rightarrow P - P_a$. "Poisson ratio" $\rightarrow$ lateral/longitudinal.
- "Meniscus/contact angle" $\rightarrow$ convex = mercury (140°).
- "Heat / temperature change / specific heat" $\rightarrow Q = mc\Delta T$.
- "Length/volume on heating" $\rightarrow \alpha \Delta T / \gamma \Delta T$.
- "Capillary rise" $\rightarrow h = \frac{2T \cos\theta}{r\rho g}$.
- "Fraction floating" $\rightarrow \rho_{\text{body}}/\rho_{\text{liq}}$.

7. SOLVING METHODS

- Excess pressure ratio $\rightarrow$ radius ratio invert karo.
- Terminal velocity: factor ko square karo ($2r \rightarrow 4\times$).

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — The soap bubble has radii in the ratio 5 : 3. The ratio of excess pressure inside them is:

What this tests: $\Delta P \propto 1/r$.

Solution: $\Delta P_1/\Delta P_2 = r_2/r_1 = 3/5$.

Key takeaway: Excess pressure ratio = radius ratio ka ULTA (3:5).

REAL PYQ EXAMPLE — A body is falling freely in a viscous liquid. If its radius is doubled, its terminal velocity becomes:

What this tests: $v_t \propto r^2$.

Solution: $2^2 = 4 \rightarrow 4$ times.

Key takeaway: Radius square karo: $2\times$ radius $\rightarrow 4\times$ terminal velocity.

REAL PYQ EXAMPLE — If the pressure at a point is P and atmospheric pressure is P_a , the gauge pressure is:

What this tests: Definition of gauge pressure.

Solution: Gauge = absolute – atmospheric = $P - P_a$.

Key takeaway: Gauge = $P - P_a$ (na P, na $P+P_a$).

REAL PYQ EXAMPLE — A glass capillary shows a convex meniscus. The liquid is:

What this tests: Convex meniscus $\rightarrow$ contact angle $> 90^\circ$.

Solution: Mercury-glass = 140° (obtuse) $\rightarrow$ mercury.

Key takeaway: Convex = mercury (140°); water/alcohol = 0° (concave).

REAL PYQ EXAMPLE — Which material property is obtained from the slope of the linear portion of the stress-strain curve?

What this tests: Stress-strain slope.

Solution: Slope = stress/strain = Young's modulus.

Key takeaway: Linear region ka slope hi Y hai.

REAL PYQ EXAMPLE — A gas of 240 ml is heated from 27°C to 227°C at constant pressure. New volume:

What this tests: Charles' law — $V \propto T$ (kelvin mein).

Solution: $V_2 = V_1 \times \frac{T_2}{T_1} = 240 \times \frac{500}{300} = 400$ ml.

Key takeaway: $V \propto T$ (kelvin): $300 \rightarrow 500$ K, to $240 \rightarrow 400$ ml.

REAL PYQ EXAMPLE — A capillary of diameter 2 mm is dipped in liquid (specific gravity 0.8). Rise 15 mm, contact angle 0° . Surface tension:

What this tests: $h = \frac{2T \cos\theta}{r\rho g} \rightarrow T = \frac{r\rho gh}{2}$.

Solution: $T = \frac{(1 \times 10^{-3})(800)(9.8)(15 \times 10^{-3})}{2} = 0.0588$ N/m.

Key takeaway: $T = \frac{r\rho gh}{2}$ — r metre mein, $\rho = 800$ kg/m³, h metre mein.

REAL PYQ EXAMPLE — A metal ball just passes through a ring. Only the ball is heated. It will:

What this tests: Thermal expansion of the ball.

Solution: Ball expand hota hai $\rightarrow$ ab ring mein pass nahi karega.

Key takeaway: Heat ball $\rightarrow$ expand $\rightarrow$ won't pass through the ring.

REAL PYQ EXAMPLE — A water drop is divided into 27 equal droplets. Pressure difference inside each droplet becomes:

What this tests: $r' = r/3 \rightarrow \Delta P \propto 1/r$.

Solution: $\Delta P' = 3\Delta P$ (radius ek-third $\rightarrow$ pressure $3\times$).

Key takeaway: $27 = 3^3 \rightarrow$ radius $\div 3 \rightarrow \Delta P \times 3$.

REAL PYQ EXAMPLE — A cube floats in water with some part outside. The fraction submerged depends on:

What this tests: Float condition: submerged fraction = $\rho_{\text{body}}/\rho_{\text{liq}}$.

Solution: Density ratio decide karta hai kitna hissa andar hai.

Key takeaway: Submerged fraction = $\rho_{\text{body}}/\rho_{\text{water}}$.

9. 10-SECOND TRICKS

- Bubble $\frac{4T}{r}$, drop $\frac{2T}{r}$ (bubble ki 2 surfaces).
- Radius double $\rightarrow v_t \times 4$.
- Stress-strain slope = Young's modulus.
- $Q = mc\Delta T$ (heat = mass $\times c \times \Delta T$).
- $\alpha:\beta:\gamma = 1:2:3$; radius $\div n \rightarrow \Delta P \times n$ (drops).
- Capillary: $h = \frac{2T \cos\theta}{r\rho g}$.

10. IMPORTANT FACTS

- Water ki surface tension temperature ke saath ghatti hai.
- Venturi-meter flow speed; manometer pressure difference; Pascal = pressure; Archimedes = buoyancy.
- Water: $c = 1 \text{ cal/g}^\circ\text{C} = 4186 \text{ J/kg}^\circ\text{C}$ (highest); $1 \text{ cal} = 4.18 \text{ J}$.
- Steel sabse elastic; thermometer thermal expansion par chalta hai.
- Liquid ka η temperature ke saath GHATTA hai; gas ka η BADHTA hai.

11. COMMON TRAPS

TRAP: Gauge pressure = P ya $P + P_a$.

CORRECT: Gauge = $P - P_a$ (absolute minus atmospheric).

TRAP: Convex meniscus water ka hota hai.

CORRECT: Convex = mercury (contact angle $140^\circ > 90^\circ$).

TRAP: Stress-strain slope = ultimate strength.

CORRECT: Slope = Young's modulus.

TRAP: Heat dene par body ka temperature hamesha badhta hai.

CORRECT: Phase change (ice $\rightarrow$ water) mein temperature CONSTANT rehta hai (latent heat).

TRAP: Area expansion coefficient = 3α .

CORRECT: $\beta = 2\alpha$ (area), $\gamma = 3\alpha$ (volume) — 1:2:3 ka order.

TRAP: Ice melts in brim-full cup → water overflows.

CORRECT: Level SAME rehta hai (displaced water = ice ka weight).

12. GRAPHS / DIAGRAMS

- Stress-strain curve: linear slope = Y , phir yield point, phir fracture.
- Meniscus: $<90^\circ$ concave (water), $>90^\circ$ convex (mercury).
- Soap bubble 2 surfaces, drop 1 surface → $4T/r$ vs $2T/r$.

13. LAST-MINUTE REVISION

LAST-MINUTE REVISION

- $Y = FL/(A\Delta L)$ = stress-strain ka linear slope
- $\Delta P(\text{drop}) = 2T/r$; $\Delta P(\text{bubble}) = 4T/r$; $v_t \propto r^2$
- Gauge = $P - P_a$; Bernoulli = energy; Continuity = mass

CHAPTER 08 — THERMODYNAMICS

1. INTRODUCTION

Thermodynamics = heat, work aur internal energy ka relation. Sirf teen cheezein solidly yaad karo: first law ($\Delta Q = \Delta U + \Delta W$), isothermal/adiabatic processes, aur Carnot efficiency. Ye teeno hi har paper mein aate hain.

2. CORE CONCEPTS

What is the first law? Energy conservation: $\Delta Q = \Delta U + \Delta W$. Heat ka kuch hissa internal energy badhata hai (ΔU), kuch hissa kaam karta hai (ΔW). Work "on" gas ho to ΔU badhta hai, "by" gas ho to ΔU ghat-ta hai.

What is internal energy? Ideal gas ki U sirf TEMPERATURE par depend karti hai. Isi liye isothermal (T constant) mein $\Delta U = 0$.

Isothermal process. T constant → $\Delta U = 0$ → saari heat work ban jaati hai: $Q = W$.

Adiabatic process. Heat exchange nahi ($Q = 0$) → $\Delta U = -W$. Expansion mein gas kaam karti hai → ΔU negative (temperature girta hai). Compression mein ulta.

What is the second law + entropy? Second law ENTROPY define karta hai (natural processes ki direction). Full cycle par $\oint(\delta Q - \delta W) = 0$.

Carnot engine. Do temperatures ke beech maximum efficiency wala ideal engine: $\eta = 1 - T_2/T_1$ (temperatures KELVIN mein!).

Isochoric (constant volume). V constant → koi expansion nahi → work $W = 0$. Tab $\Delta Q = \Delta U$ (saari heat internal energy ban jaati hai).

Isobaric (constant pressure). $W = P\Delta V$ (pressure × volume change).

Refrigerator. Heat engine ka ULTA — sink se heat nikal kar source ko deta hai. Coefficient of performance $COP = Q_2/W = T_2/(T_1 - T_2)$ (kelvin mein). Ye engine ki efficiency ka reciprocal NAHI hai.

Polytropic process $PV^n = \text{constant}$. $n = 1$ → isothermal, $n = \gamma$ → adiabatic. Work done $W = (P_1V_1 - P_2V_2)/(n - 1)$.

3. KEY RELATIONS

- $\Delta Q = \Delta U + \Delta W$; isothermal: $Q = W$, $\Delta U = 0$; adiabatic: $Q = 0$, $\Delta U = -W$.
- $\eta = 1 - T_2/T_1$; $\eta = W/Q_1$ ($W = \eta Q_1$). 75% of 300 J → 225 J.
- 127 °C–327 °C → 400 K–600 K → $\eta = 1 - 400/600 = 33.3\%$.
- 200 J work ON gas (adiabatic) → $\Delta U = +200$ J.
- Isochoric: $W = 0$, $\Delta Q = \Delta U$; Isobaric: $W = P\Delta V$.
- Refrigerator $COP = T_2/(T_1 - T_2)$. 0 °C–100 °C → $273/100 = 2.73$.

- $PV^2 = \text{constant} \rightarrow W = (P_1V_1 - P_2V_2)/(2 - 1)$.

4. FORMULAS

$$\Delta Q = \Delta U + \Delta W ; \eta_{\text{carnot}} = 1 - T_2/T_1 ; \eta = W/Q_1$$

$$T(K) = t(^{\circ}\text{C}) + 273$$

$$\text{Isochoric: } W = 0 ; \text{Isobaric: } W = P\Delta V ; COP_{\text{ref}} = T_2/(T_1 - T_2)$$

$$PV^n = \text{const} \rightarrow W = (P_1V_1 - P_2V_2)/(n - 1)$$

Symbol meanings: ΔQ = heat supplied, ΔU = internal energy change, ΔW = work by system, T_1 = source temp, T_2 = sink temp (both kelvin), Q_1 = heat absorbed.

5. CONDITIONS / SPECIAL CASES

- Carnot $\eta = 1/6$, sink 62 °C kam → η double: $T_1 = 372$ K (99 °C), $T_2 = 310$ K (37 °C).
- Isothermal $Q = 100$ J → $W = 100$ J, $\Delta U = 0$.
- Work 333 cal, ΔU 167 cal → $Q = 333 + 167 = 500$ cal.
- Same 20 K gap: sabse LOW cold/hot ratio jita: 40 K/20 K → $\eta = 50\%$.
- Source temperature badhao → Carnot efficiency badhti hai.
- Heat aur work PATH functions hain (process par depend karte hain); internal energy STATE function hai (process par nahi).
- P-V diagram: area under curve = work; adiabat isotherm se STEEPER (PV^γ).
- Isochoric (V constant) → $W = 0$ → $\Delta Q = \Delta U$.
- Refrigerator 0 °C & 100 °C → $COP = 273/(373 - 273) = 2.73$.
- Gas law names: $PV = \text{constant}$ (Boyle, T const); $V \propto T$ (Charles, P const); $P \propto T$ (Gay-Lussac, V const).

6. HOW TO RECOGNIZE THE QUESTION

- " ΔQ , ΔU , W " → first law.
- "Isothermal / adiabatic" → $\Delta U = 0$ / $Q = 0$.
- "Carnot efficiency" → $\eta = 1 - T_2/T_1$ (kelvin!).
- " $\oint(\delta Q - \delta W)$ " → 0.

7. SOLVING METHODS

- Carnot ke liye hamesha °C → K (add 273).
- "Work ON gas" → ΔU up; "BY gas" → ΔU down.
- η double wale tricks: do equations bana ke solve karo.

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — The efficiency of a Carnot engine operating between 127 °C and 327 °C is:

What this tests: Carnot formula — pehle kelvin mein convert karo.

Solution: 400 K aur 600 K → $\eta = 1 - 400/600 = 1/3 = 33.3\%$.

Key takeaway: Carnot hamesha kelvin mein: °C + 273 pehle.

REAL PYQ EXAMPLE — An adiabatic process occurs where 200 J of work is done on the gas. Change in internal energy:

What this tests: Adiabatic: $Q = 0$.

Solution: $\Delta U = -W = -(-200) = +200$ J (work ON gas → ΔU badhta hai).

Key takeaway: Work ON gas → ΔU +ve; work BY gas → ΔU -ve.

REAL PYQ EXAMPLE — In an isothermal process, if heat given is 100 J, change in internal energy and work done are:

What this tests: Isothermal: T constant → $\Delta U = 0$.

Solution: $\Delta Q = \Delta U + W \rightarrow 100 = 0 + W \rightarrow W = 100$ J.

Key takeaway: Isothermal: $\Delta U = 0$, $Q = W$.

REAL PYQ EXAMPLE — Second law of thermodynamics defines:

What this tests: What the second law introduces.

Solution: Entropy (internal energy nahi, heat nahi).

Key takeaway: Second law = entropy; zeroth law = temperature.

REAL PYQ EXAMPLE — A Carnot engine has 75% efficiency and absorbs 300 J. Work done:

What this tests: $\eta = W/Q_1$.

Solution: $W = 0.75 \times 300 = 225 \text{ J}$.

Key takeaway: $W = \eta \times Q_1$ — efficiency ko heat se multiply karo.

REAL PYQ EXAMPLE — Work done by a gas in an isochoric process is:

What this tests: Isochoric = constant volume.

Solution: $\Delta V = 0 \rightarrow W = P\Delta V = 0$.

Key takeaway: Volume constant $\rightarrow$ koi work nahi ($W = 0$).

REAL PYQ EXAMPLE — A Carnot refrigerator operates between 0 °C and 100 °C. Its coefficient of performance is:

What this tests: $COP = T_2/(T_1 - T_2)$.

Solution: $COP = 273/(373 - 273) = 273/100 = 2.73$.

Key takeaway: $COP = T_{\text{cold}}/(T_{\text{hot}} - T_{\text{cold}})$ — kelvin mein.

REAL PYQ EXAMPLE — A gas expands with $PV^2 = \text{constant}$ from $P_1 = 320 \text{ kPa}$, $V_1 = 0.04 \text{ m}^3$ to $P_2 = 80 \text{ kPa}$. Work done:

What this tests: Polytropic work formula ($n = 2$).

Solution: $V_2 = V_1\sqrt{P_1/P_2} = 0.04 \times 2 = 0.08 \text{ m}^3$.

$W = (P_1V_1 - P_2V_2)/(n-1) = (12800 - 6400)/1 = 6400 \text{ J} = 6.4 \text{ kJ}$.

Key takeaway: $W = (P_1V_1 - P_2V_2)/(n-1)$; pehle V_2 nikalo ($PV^n = \text{const}$ se).

9. 10-SECOND TRICKS

- Isothermal $\rightarrow \Delta U = 0$, $Q = W$ (instant).
- Adiabatic $\rightarrow Q = 0$, $\Delta U = -W$.
- $\eta = 1 - T_{\text{cold}}/T_{\text{hot}}$ (hamesha kelvin).
- Isochoric $\rightarrow W = 0$ (instant).
- Refrigerator $COP = T_{\text{cold}}/(T_{\text{hot}} - T_{\text{cold}})$.

10. IMPORTANT FACTS

- Second law entropy define karta hai; zeroth law temperature.
- Full cycle pe $\oint(\delta Q - \delta W) = 0$.

11. COMMON TRAPS

TRAP: Carnot mein °C direct use karna.

CORRECT: Pehle 273 add karo ($127^\circ\text{C} = 400 \text{ K}$).

TRAP: Adiabatic expansion mein ΔU zero.

CORRECT: Expansion mein gas kaam karti hai $\rightarrow \Delta U$ NEGATIVE.

TRAP: Second law internal energy define karta hai.

CORRECT: Second law ENTROPY define karta hai.

TRAP: Refrigerator ka COP = engine ki efficiency.

CORRECT: $COP = T_2/(T_1 - T_2)$ — engine $\eta = 1 - T_2/T_1$ se bilkul alag.

TRAP: Isochoric mein gas kaam karti hai.

CORRECT: V constant $\rightarrow W = P\Delta V = 0$.

12. GRAPHS / DIAGRAMS

- P-V: isotherm = hyperbola ($PV = \text{const}$); adiabat STEEPER ($PV^\gamma = \text{const}$).
- Carnot cycle: 2 isotherms + 2 adiabats; enclosed area = net work per cycle.

13. LAST-MINUTE REVISION**LAST-MINUTE REVISION**

- $\Delta Q = \Delta U + \Delta W$; isothermal $\Delta U = 0$; adiabatic $Q = 0$
- $\eta_{\text{carnot}} = 1 - T_2/T_1$ (kelvin); $\eta = W/Q_1$
- $127-327^\circ\text{C} \rightarrow 33.3\%$; work ON gas $\rightarrow \Delta U$ up

CHAPTER 09 — KINETIC THEORY OF GASES**1. INTRODUCTION**

Ye chapter gas ko molecules ke level par samajhta hai.

Idhar 4-5 fixed facts hain: ideal gas assumptions, KE-temperature relation, $\gamma = C_p/C_v$, sound speed in gas, aur van der Waals constant. Sab compact aur direct hain.

2. CORE CONCEPTS

What is an ideal gas? Bahut saare molecules, negligible volume, elastic collisions, koi intermolecular force nahi (sirf collision ke waqt). Real gas mein ye assumptions toot-ti hain, isliye van der Waals correction aata hai.

Pressure kahan se aata hai? Gas ke molecules walls se takra kar pressure banate hain.

What is KE-temperature relation? Average translational KE per molecule = $(3/2)k_{\text{BT}}$. Poore gas ki $E = (3/2)Nk_{\text{BT}} \rightarrow$ yaad rakho $2E = 3k_{\text{BNT}}$ (N factor k_{B} ke saath hai).

What are C_p and C_v ? C_p = constant pressure par specific heat, C_v = constant volume par. Inka ratio $\gamma = C_p/C_v = 1 + 2/f$, jahan f = degrees of freedom. Diatomic ($f=5$) $\rightarrow \gamma = 1.4$; monatomic ($f=3$) $\rightarrow \gamma = 1.66$.

What is sound speed in a gas? $v = \sqrt{\gamma P/\rho}$. Constant temperature par pressure double karo to density bhi double $\rightarrow v$ UNCHANGED.

What is rms speed? Molecules ki speeds ka root-mean-square: $v_{\text{rms}} = \sqrt{3RT/M} = \sqrt{3P/\rho} = \sqrt{(3k_{\text{BT}}/m)}$. Matlab: $v_{\text{rms}} \propto \sqrt{T}$ (temperature ka square root) aur $v_{\text{rms}} \propto 1/\sqrt{M}$ (molar mass ka ulta square root). Constant temperature par pressure se INDEPENDENT hai. Gas law names. Boyle: $PV = \text{constant}$ (T const). Charles: $V \propto T$ (P const). Gay-Lussac: $P \propto T$ (V const). Teeno milkar ideal gas law $PV = nRT$ banate hain.

3. KEY RELATIONS

- $PV = nRT = Nk_{\text{BT}}$; KE/molecule = $(3/2)k_{\text{BT}}$.
- $\gamma = C_p/C_v = 1 + 2/f$; diatomic 1.4; monatomic 1.66.
- $v = \sqrt{\gamma P/\rho}$ — constant T par pressure se independent.
- van der Waals: $(P + a/V^2)(V - b) = RT$; a ka dimension $[\text{ML}^5\text{T}^{-2}]$.
- $v_{\text{rms}} = \sqrt{3RT/M} = \sqrt{3P/\rho}$. O_2 ($M = 32$) at $300 \text{ K} \rightarrow 483 \text{ m/s}$; at $1200 \text{ K} \rightarrow 483 \times 2 = 966 \text{ m/s}$.
- Pressure double + T half $\rightarrow v_{\text{rms}} \times (1/\sqrt{2})$ (P se independent, sirf $\sqrt{T}$ se).
- Boyle: PV const; Charles: $V \propto T$; Gay-Lussac: $P \propto T$.

4. FORMULAS

$PV = nRT$; $E = (3/2)Nk_{\text{BT}}$; $\gamma = C_p/C_v = 1 + 2/f$
 $v = \sqrt{\gamma P/\rho}$; van der Waals 'a': $[\text{ML}^5\text{T}^{-2}]$
 $v_{\text{rms}} = \sqrt{3RT/M} = \sqrt{3P/\rho}$; $v_{\text{rms}} \propto \sqrt{T}$, $\propto 1/\sqrt{M}$
 Boyle: $PV = \text{const}$; Charles: $V \propto T$; Gay-Lussac: $P \propto T$

Symbol meanings: P = pressure, V = volume, T = temperature (K), n = moles, N = number of molecules, k_{B} = Boltzmann constant, f = degrees of freedom.

5. CONDITIONS / SPECIAL CASES

- Diatomic $\gamma = 1.4$; monatomic $\gamma = 1.66$.
- Pressure double (constant T) $\rightarrow$ sound speed SAME (ρ bhi double).

- 27 °C ka rise = 27 K ka rise (1 °C interval = 1 K).
- van der Waals 'a' = attraction correction, 'b' = molecular volume.

6. HOW TO RECOGNIZE THE QUESTION

- "KE vs temperature" → $E = (3/2)Nk_{BT}$.
- "γ of a gas" → diatomic 1.4, monatomic 1.66.
- "Sound speed vs pressure" → unchanged.
- "van der Waals a" → $[ML^5T^{-2}]$.

7. SOLVING METHODS

- KE-T relation: N factor sahi jagah — $2E = 3k_{BNT}$.
- $\gamma = 1 + 2/f$ (f = degrees of freedom).

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — In gases of diatomic molecules, the ratio of the two specific heats C_p/C_v is:

What this tests: $\gamma = 1 + 2/f$ with f = 5 (diatomic).

Solution: $\gamma = 1 + 2/5 = 7/5 = 1.4$.

Key takeaway: Diatomic = 1.4 ; monatomic = 1.66.

REAL PYQ EXAMPLE — The velocity of sound in air is 332 m/s. If the pressure is doubled, the velocity becomes:

What this tests: $v = \sqrt{\gamma P/\rho}$; constant T par ρ bhi double.

Solution: P/ρ ratio same → v unchanged = 332 m/s.

Key takeaway: Constant T par sound speed pressure se independent.

REAL PYQ EXAMPLE — The relationship between translational KE (E) of gas molecules and temperature (T) is:

What this tests: KE-temperature relation with N molecules.

Solution: $E = (3/2)Nk_{BT} \rightarrow 2E = 3k_{BNT}$.

Key takeaway: $2E = 3k_{BNT}$ — N factor kB ke saath.

REAL PYQ EXAMPLE — The dimensional formula of constant 'a' in van der Waals equation $(P + a/V^2)(V - b) = RT$:

What this tests: a/V^2 ka dimension pressure ke barabar.

Solution: $[a] = [P][V^2] = ML^{-1}T^{-2} \times L^6 = ML^5T^{-2}$.

Key takeaway: a/V^2 pressure jaisa → a ka dimension $[ML^5T^{-2}]$.

REAL PYQ EXAMPLE — The rms speed of oxygen molecules at 300 K is about 483 m/s. The rms speed at 1200 K is:

What this tests: $v_{rms} \propto \sqrt{T}$.

Solution: $v_2 = v_1 \times \sqrt{(1200/300)} = 483 \times 2 = 966$ m/s.

Key takeaway: Temperature 4x → rms speed 2x ($\sqrt{4}$).

REAL PYQ EXAMPLE — The rms speed of a gas is v. If pressure is doubled and temperature is halved, the new rms speed is:

What this tests: $v_{rms} \propto \sqrt{T}$ (constant T par pressure se independent).

Solution: $v' = v \times \sqrt{(1/2)} = v/\sqrt{2}$.

Key takeaway: rms speed sirf $\sqrt{T}$ aur $1/\sqrt{M}$ par depend — pressure nahi.

REAL PYQ EXAMPLE — PV = constant at constant temperature is which law?

What this tests: Gas law names.

Solution: Boyle's law (T constant → PV constant).

Key takeaway: Boyle = PV; Charles = V/T; Gay-Lussac = P/T.

9. 10-SECOND TRICKS

- Diatomic → 1.4 ; monatomic → 1.66 (one-word answers).

- Sound speed pressure se independent (constant T).
- van der Waals 'a' → $[ML^5T^{-2}]$.
- rms speed: T 4x → v 2x ; M 4x → v half.
- Boyle PV, Charles V/T, Gay-Lussac P/T.

10. IMPORTANT FACTS

- Kinetic theory gas ko point-like, random motion wale particles maanti hai.
- Temperature 27 °C badhana = 27 K badhana.

11. COMMON TRAPS

TRAP: Diatomic $\gamma = 1.66$.

CORRECT: 1.66 monatomic hai; diatomic = 1.4.

TRAP: Pressure double → sound speed double.

CORRECT: Constant T par ρ bhi double → v unchanged.

TRAP: $E = (3/2)Nk_{BT}$ mein N bahar hota hai.

CORRECT: $2E = 3k_{BNT}$ — N, kB ke saath hi hota hai.

12. GRAPHS / DIAGRAMS

- KE vs T: origin se straight line ($E \propto T$).
- Sound speed vs pressure (constant T): flat line.
- P-V: isotherm hyperbola vs steeper adiabat.

13. LAST-MINUTE REVISION

LAST-MINUTE REVISION

- $PV = nRT$; $2E = 3k_{BNT}$
- $\gamma = 1 + 2/f$; diatomic 1.4 ; monatomic 1.66
- $v = \sqrt{\gamma P/\rho}$; van der Waals a: $[ML^5T^{-2}]$

CHAPTER 10 — OSCILLATIONS AND WAVES

1. INTRODUCTION

Ye chapter do cheezein cover karta hai: SHM (simple harmonic motion) aur waves (sound/mechanical vs EM). SHM ke questions fixed pattern ke hain — velocity at $x = A/2$, a_{max}/v_{max} ratio, energy $\propto A^2$.

2. CORE CONCEPTS

What is SHM? Wo motion jisme restoring force displacement ke $\propto$ aur opposite hota hai: $F = -kx$. Mean position ke around back-and-forth oscillation.

Mean vs extreme. Mean position par velocity MAXIMUM aur acceleration ZERO. Extreme par velocity zero aur acceleration MAXIMUM. Ye hi sabse common question hai. SHM velocity. $v = \omega\sqrt{(A^2 - x^2)}$. Mean ($x=0$) par $v = v_{max} = \omega A$; extreme ($x=A$) par $v = 0$.

What is a wave? Energy ka ek jagah se dusri jagah transfer, bina matter transfer ke. Mechanical waves (sound) ko medium chahiye; EM waves ko nahi.

Sound. Mechanical wave; speed solids > liquids > gases. Bats ultrasonic waves (reflection) se obstacles detect karte hain.

What is a wave number? $k = 2\pi/\lambda$ — 2π length mein kitni waves (propagation constant bhi kehte hain). $y = A \sin(\omega t - kx)$ mein x ka coefficient hi k hai.

What is a stationary wave? Do SAME waves opposite directions mein multi hain to stationary wave banti hai. Isme ENERGY TRANSFER NAHI hota (energy nodes ke beech trapped). Nodes = zero displacement, antinodes = maximum.

What is intensity of a wave? Intensity $\propto$ (amplitude)² — amplitude double → intensity 4x.

SHM phase relations. $x = A \sin \omega t$. Velocity $v = dx/dt$ leads x by 90°; acceleration $a = -\omega^2 x$ is 180° out of phase with x (a leads v by 90°). Mean par v max; extreme par a max.

Phase difference of two waves. $y = A \sin(\omega t + \phi_1)$ aur $y = A \sin(\omega t + \phi_2) \rightarrow$ phase difference $= |\phi_1 - \phi_2|$.

3. KEY RELATIONS

- $v = \omega \sqrt{A^2 - x^2}$; $v_{\max} = \omega A = 2\pi A/T$.
- $x = A/2$ par: $v = v_{\max} \sqrt{3/4} = (\sqrt{3}/2)v_{\max}$.
- Ratio $a_{\max}/v_{\max} = \omega = 2\pi f$.
- $a = \omega^2 x \rightarrow \omega = \sqrt{a/x} = \sqrt{2/0.02} = 10 \text{ rad/s}$.
- SHM energy $E \propto A^2$ (A double $\rightarrow$ KE $\times 4$).
- $k = 2\pi/\lambda$; $v = \omega/k$; particle velocity in wave: $v_{\max} = A\omega$.
- $I \propto A^2$; stationary wave $\rightarrow$ energy transfer ZERO.
- SHM: v leads x by 90° ; a leads v by 90° ($a = -\omega^2 x$).
- Time to reach x from mean: $t = (T/2\pi) \times \sin^{-1}(x/A)$.
Mean $\rightarrow A/2$: $t = T/12$ (fixed value).
- Two waves: $\Delta\phi = |\phi_1 - \phi_2|$ ($\pi/3 - \pi/4 = \pi/12$).

4. FORMULAS

$T = 2\pi\sqrt{m/k}$ (spring) ; $T = 2\pi\sqrt{L/g}$ (pendulum)
 $v_{\max} = \omega A$; $a_{\max} = \omega^2 A$; $v = \omega \sqrt{A^2 - x^2}$; $E = (1/2)m\omega^2 A^2$
 $k = 2\pi/\lambda$; $v = \omega/k$; $I \propto A^2$
 Particle velocity in wave: $v_{\max} = A\omega$
 Symbol meanings: A = amplitude, ω = angular frequency,
 x = displacement from mean, T = time period, f = frequency, m = mass, k = spring constant, L = pendulum length.

5. CONDITIONS / SPECIAL CASES

- Mean: v max, $a = 0$. Extreme: $v = 0$, a max.
- A double $\rightarrow$ max KE $\times 4$ ($KE \propto A^2$).
- Sound: solids > liquids > gases speed.
- Pendulum length badhao $\rightarrow$ T badhta hai ($T \propto \sqrt{L}$).

6. HOW TO RECOGNIZE THE QUESTION

- "Velocity at $x = A/2$ " $\rightarrow (\sqrt{3}/2)v_{\max}$.
- " $a_{\max}/v_{\max}$ ratio" $\rightarrow \omega = 2\pi f$.
- " ω from a and x " $\rightarrow \sqrt{a/x}$.
- "Sound vs EM / medium" $\rightarrow$ sound mechanical.

7. SOLVING METHODS

- $x = A/2 \rightarrow v_{\max}$ ko $\sqrt{3}/2$ se multiply.
- Angular speed: $\omega = \sqrt{a/x}$.
- Energy: $E \propto A^2$.

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — Velocity at mean position of a particle executing SHM is v . Velocity at a distance equal to half the amplitude is:

What this tests: $v = \omega \sqrt{A^2 - x^2}$ at $x = A/2$.
 Solution: $v = v_{\max} \sqrt{1 - 1/4} = v_{\max} \times \sqrt{3}/2 = (\sqrt{3}/2)v$.
 Key takeaway: $x = A/2 \rightarrow v = (\sqrt{3}/2)v_{\max}$ — standard value yaad rakho.

REAL PYQ EXAMPLE — A body executing SHM has amplitude 0.01 m and frequency 50 Hz. The ratio of max acceleration to max velocity is:

What this tests: $a_{\max}/v_{\max} = \omega = 2\pi f$.
 Solution: $a_{\max}/v_{\max} = 2\pi \times 50 = 100\pi$.
 Key takeaway: Ye ratio hamesha $\omega = 2\pi f$ hota hai.

REAL PYQ EXAMPLE — The displacement and acceleration of a harmonic oscillator are 0.02 m and 2 m/s². Its angular speed is:

What this tests: $a = \omega^2 x$.
 Solution: $\omega = \sqrt{a/x} = \sqrt{2/0.02} = \sqrt{100} = 10 \text{ rad/s}$.
 Key takeaway: $\omega = \sqrt{a/x}$ — instantaneous a aur x se direct.

REAL PYQ EXAMPLE — Which of the following is NOT an electromagnetic wave?

What this tests: EM vs mechanical distinction.
 Solution: Radio, IR, UV EM hain; SOUND mechanical hai.
 Key takeaway: Sound = mechanical (medium chahiye); EM ko nahi.

REAL PYQ EXAMPLE — If the amplitude of SHM is doubled, maximum kinetic energy increases by:

What this tests: $KE \propto A^2$.
 Solution: $2^2 = 4 \rightarrow 4$ times.
 Key takeaway: Energy $\propto A^2$ — amplitude double $\rightarrow$ energy $\times 4$.

REAL PYQ EXAMPLE — In $y = a \sin(\omega t + kx)$, the quantity k is called:

What this tests: Wave parameters.
 Solution: k = wave number / propagation constant = $2\pi/\lambda$.
 Key takeaway: $k = 2\pi/\lambda$ — x ka coefficient.

REAL PYQ EXAMPLE — In which type of wave is energy not transferred?

What this tests: Stationary wave property.
 Solution: Stationary wave (standing wave) — energy nodes ke beech trapped.
 Key takeaway: Stationary = no energy transfer; progressive = energy travels.

REAL PYQ EXAMPLE — The phase difference between velocity and acceleration in SHM is:

What this tests: SHM phase relations.
 Solution: v leads x by 90° ; $a = -\omega^2 x$ (180° from x) $\rightarrow v$ aur a mein 90° .
 Key takeaway: SHM: v leads x by 90° ; a leads v by 90° .

REAL PYQ EXAMPLE — Two waves $y_1 = 20 \sin(4\pi t + \pi/4)$ and $y_2 = 20 \sin(4\pi t + \pi/3)$. Phase difference:

What this tests: $\Delta\phi = |\phi_1 - \phi_2|$.
 Solution: $\Delta\phi = \pi/3 - \pi/4 = \pi/12$.
 Key takeaway: Same $\omega \rightarrow$ phase difference = phases ka antar.

REAL PYQ EXAMPLE — Intensity of light depends on:

What this tests: $I \propto A^2$.
 Solution: Intensity $\propto$ (amplitude)² — frequency/wavelength se nahi.
 Key takeaway: Amplitude double $\rightarrow$ intensity $4\times$.

9. 10-SECOND TRICKS

- Mean $\rightarrow v$ max ; extreme $\rightarrow a$ max.
- $x = A/2 \rightarrow (\sqrt{3}/2)v_{\max}$; A double $\rightarrow$ energy $\times 4$.
- $k = 2\pi/\lambda$; stationary $\rightarrow$ no energy transfer.
- Intensity $\propto A^2$; v leads x by 90° , a leads v by 90° .

10. IMPORTANT FACTS

- Ultrasonic > 20 kHz; sound speed in air $\approx 332 \text{ m/s}$.
- Echo sunne ke liye minimum distance $\approx 17 \text{ m}$ (sound 340 m/s, 0.1 s ka gap).
- SHM examples: pendulum (chhota angle), spring-block.
 NOT SHM: uniform circular motion, earth ka rotation.

11. COMMON TRAPS

- TRAP: Sound ek EM wave hai.
 CORRECT: Sound mechanical wave hai; radio/IR/UV EM hain.
- TRAP: Mean position par velocity zero.
 CORRECT: Mean par velocity MAXIMUM hoti hai.
- TRAP: Solids mein sound sabse dheemi hoti hai.
 CORRECT: Solids > liquids > gases (solids mein sabse tez).

12. GRAPHS / DIAGRAMS

- SHM x-t sinusoidal; v x se 90° aage, a x se 180° aage.
- Energy vs position: PE parabola ($\propto x^2$), total energy horizontal line.
- Transverse (crest-trough) vs longitudinal (compression-rarefaction).

13. LAST-MINUTE REVISION**LAST-MINUTE REVISION**

- $v = \omega\sqrt{A^2 - x^2}$; $v_{\max} = \omega A$; $a_{\max} = \omega^2 A$; $a_{\max}/v_{\max} = 2\pi f$
- $x = A/2 \rightarrow (\sqrt{3}/2)v_{\max}$; A doubled $\rightarrow KE \times 4$
- Sound: mechanical; solids > liquids > gases; bats = ultrasonic

CHAPTER 11 — ELECTROSTATICS**1. INTRODUCTION**

Electrostatics = rest par charges ka study. Idhar chaar fixed areas hain: Coulomb's force, electric field/potential, dipole, aur capacitors. Sab formula-based hain aur ratios se turant solve hote hain.

2. CORE CONCEPTS

What does a charge at rest produce? Sirf ELECTRIC field (magnetic field nahi). Magnetic field sirf MOVING charge banata hai. Ye direct fact-question hai.

What is Coulomb's law? $F = k q_1 q_2 / r^2$ ($k = 9 \times 10^9$). Medium (dielectric) mein force ghat jaati hai: $F' = F/K = F/\epsilon_r$.

What is electric field? $E = F/q = kq/r^2$ (point charge). Parallel plates ke beech uniform field $E = V/d$.

What is potential? $V = kq/r$. Conductors ke ANDAR field zero aur potential constant hota hai.

What is a dipole? +q aur -q ka pair, separation d. Dipole moment $p = qd$. Field mein dipole par torque $\tau = pE \sin\theta$.

What is a capacitor? Charge store karne wala device. $C = Q/V = \epsilon_0 A/d$. Energy $U = (1/2)CV^2 = Q^2/2C$.

What is potential difference from work? $V = W/q$ — charge q ko move karne mein jo work W karna pade, wo potential difference se aata hai. Equipotential surface (same V) par move karne mein work = 0.

What is surface charge density? $\sigma = Q/A$ — charge per area. Spherical shell: $\sigma = Q/(4\pi R^2)$. Thin shell ke just bahar $E = \sigma/\epsilon_0$.

What is charge quantisation? Charge discrete hota hai: $q = ne$ ($n = \text{integer}$, $e = 1.6 \times 10^{-19} \text{ C}$). e ka koi fraction possible nahi.

How does a dipole's field vary with r? Axial aur equatorial dono par $E \propto 1/r^3$; potential $V \propto 1/r^2$. (Point charge ke $E \propto 1/r^2$ se zyada jaldi girta hai.)

PE of a charge system. Do charges ki PE = $kq_1 q_2 / r$. Teen charges equilateral triangle (side x) par system ki $U = 3kQ^2/x$ — aur yehi work hai system arrange karne mein.

3. KEY RELATIONS

- $F = kq_1 q_2 / r^2$; $E = F/q = kq/r^2$; $E(\text{plates}) = V/d$.
- Field ratio: $E_2/E_1 = (q_2/q_1)(r_1/r_2)^2$. 2 μC at r vs 8 μC at 2r $\rightarrow (8/2)(1/2)^2 = 1$.
- $C = \epsilon_0 A/d$; $U = (1/2)CV^2 = Q^2/2C$.
- Series: $1/C = 1/C_1 + 1/C_2$; Parallel: $C = C_1 + C_2$.
- Dipole SHM: $\omega = \sqrt{2qE/md}$.
- Gauss: $\oint E \cdot dA = q/\epsilon_0$ (electricity); $\oint B \cdot dA = 0$ (magnetism).
- $V = W/q$; equipotential par $W = 0$.
- $\sigma = Q/A = Q/4\pi R^2$ (shell); $E(\text{shell outside}) = \sigma/\epsilon_0$.
- $q = ne$ (quantisation); $e = 1.6 \times 10^{-19} \text{ C}$.

- Conductor ka saara charge OUTER surface par rehta hai ($E \text{ inside} = 0$) — external field mein bhi redistribution surface par hi hoti hai.
- Do charges ke beech CONDUCTOR plate rakho $\rightarrow$ net force ZERO ho jaata hai (conductor ke andar $E = 0$).
- Dipole: $E \propto 1/r^3$, $V \propto 1/r^2$. System PE: $U = kq_1 q_2 / r$.

4. FORMULAS

$F = (1/4\pi\epsilon_0) q_1 q_2 / r^2$; $E = F/q = kq/r^2$; $E = V/d$
 $\tau = pE \sin\theta$ ($p = qd$); dipole SHM $\omega = \sqrt{2qE/md}$
 $C = Q/V = \epsilon_0 A/d$; $U = (1/2)CV^2 = Q^2/2C$
 $V = W/q$; $\sigma = Q/A$; $E = \sigma/\epsilon_0$ (just outside conductor)

$U_{\text{system}} = k q_1 q_2 / r$; dipole: $E \propto 1/r^3$, $V \propto 1/r^2$
 Symbol meanings: q = charge, r = distance, $k = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$, E = field, V = potential, p = dipole moment, C = capacitance, ϵ_0 = permittivity, A = plate area, d = separation.

5. CONDITIONS / SPECIAL CASES

- Dielectric $K=2 \rightarrow$ force $F/2$.
- Charged spherical conductor ke andar: $E = 0$, V constant.
- Plates -10 V aur +30 V, 2 cm $\rightarrow E = 40/0.02 = 2000 \text{ V/m}$.
- $3 \times 4 \mu\text{F} \rightarrow 6 \mu\text{F}$: do series (2 μF) + ek parallel.
- Distance badhao (charge constant) $\rightarrow C$ ghathe, $U = Q^2/2C$ badhe. d double $\rightarrow C$ half (10 $\mu\text{F} \rightarrow 5 \mu\text{F}$).
- -5 C ko 100 J mein move karo $\rightarrow V = W/q = 100/5 = 20 \text{ V}$.
- Equipotential (5 V) par 10 C move $\rightarrow W = q\Delta V = 10 \times 0 = 0$.
- Shell $\sigma = 88.54 \text{ C/m}^2$, 12 mm bahar $\rightarrow E = \sigma/\epsilon_0 = 88.54/8.85 \times 10^{-12} \approx 10^{13} \text{ N/C}$.
- 3 charges +Q triangle side x $\rightarrow U = 3kQ^2/x$.

6. HOW TO RECOGNIZE THE QUESTION

- " $E_2:E_1$ (q, r diye)" $\rightarrow (q_2/q_1)(r_1/r_2)^2$.
- "Force in medium" $\rightarrow F/K$.
- "Capacitor combination / energy" $\rightarrow$ series/parallel, $U = Q^2/2C$.
- "Dipole oscillation" $\rightarrow \omega = \sqrt{2qE/md}$.
- "Field inside conductor" $\rightarrow 0$.
- "Work to move charge" $\rightarrow V = W/q$.
- "Surface charge density / shell" $\rightarrow \sigma = Q/A$.
- "Dipole field vs distance" $\rightarrow \propto 1/r^3$.

7. SOLVING METHODS

- Field ratio: charge aur distance alag-alag scale karo.
- Capacitor network: step-by-step reduce.
- Energy vs distance (Q const): $U \propto d$.

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — A point charge 2 μC produces E_1 at distance r; another 8 μC at 2r produces E_2 .

Ratio $E_2 : E_1$ is:

What this tests: $E \propto q/r^2$.

Solution: $E_2/E_1 = (8/2) \times (r/2r)^2 = 4 \times 1/4 = 1$.

Key takeaway: Distance ratio ko SQUARE karo: $(1/2)^2 = 1/4$.

REAL PYQ EXAMPLE — Two plates are at potentials -10 V and +30 V, separation 2 cm. The electric field is:

What this tests: $E = \Delta V/d$.

Solution: $E = (30 - (-10))/0.02 = 40/0.02 = 2000 \text{ V/m}$.

Key takeaway: Total ΔV lo (-10 se +30 = 40 V).

REAL PYQ EXAMPLE — The force between two charges is F . If a medium of relative permittivity 2 is introduced, the force becomes:

What this tests: $F' = F/\epsilon_r$.

Solution: $F' = F/2$.

Key takeaway: Dielectric $K \rightarrow$ force F/K .

REAL PYQ EXAMPLE — Three capacitors of $4 \mu\text{F}$ each are connected to give $6 \mu\text{F}$. This is done by:

What this tests: Capacitor combination.

Solution: Do series ($2 \mu\text{F}$) + ek parallel ($4 \mu\text{F}$) $\rightarrow 2 + 4 = 6 \mu\text{F}$.

Key takeaway: 2 series ($2 \mu\text{F}$) + 1 parallel = $6 \mu\text{F}$.

REAL PYQ EXAMPLE — If the distance between capacitor plates is increased keeping charge constant, stored energy:

What this tests: $U = Q^2/2C$, $C = \epsilon_0 A/d$.

Solution: $C \propto 1/d \rightarrow C$ ghatta $\rightarrow U$ badhta hai.

Key takeaway: Charge constant + $d \uparrow \rightarrow U \uparrow$ ($Q^2/2C$).

REAL PYQ EXAMPLE — A charge at rest produces:

What this tests: Field of a static charge.

Solution: Electric field only (no magnetic field).

Key takeaway: Rest charge $\rightarrow E$ only; moving charge $\rightarrow E + B$.

REAL PYQ EXAMPLE — The work done on moving a charge of -5 C in an electric field is 100 J . The potential difference is:

What this tests: $V = W/q$.

Solution: $V = 100/5 = 20 \text{ V}$ (magnitude — sign sirf direction batata hai).

Key takeaway: $V = W/q$ — work ko charge se divide.

REAL PYQ EXAMPLE — The work done in moving a charge of 10 C between two points on a 5 V equipotential surface is:

What this tests: Equipotential $\rightarrow$ same $V \rightarrow \Delta V = 0$.

Solution: $W = q\Delta V = 10 \times 0 = 0$.

Key takeaway: Equipotential surface par move $\rightarrow$ work ZERO.

REAL PYQ EXAMPLE — Three charges $+Q$ are at the vertices of an equilateral triangle of side x . Work to assemble them:

What this tests: System PE = sum of pair energies.

Solution: $U = 3 \times kQ^2/x$ (teen pair, har pair ki energy kQ^2/x).

Key takeaway: Triangle: 3 pair $\rightarrow 3kQ^2/x$.

REAL PYQ EXAMPLE — The electric field strength and potential due to a dipole vary with distance r as:

What this tests: Dipole distance dependence.

Solution: $E \propto 1/r^3$; $V \propto 1/r^2$.

Key takeaway: Dipole $E \propto r^{-3}$ (point charge ke r^{-2} se tez girta hai).

9. 10-SECOND TRICKS

- Rest charge $\rightarrow$ electric field only.
- Dielectric $K \rightarrow$ force F/K .
- E inside conductor = 0 (hamesha).
- Equipotential move $\rightarrow W = 0$.
- Dipole: $E \propto 1/r^3$, $V \propto 1/r^2$.

10. IMPORTANT FACTS

- ϵ_0 unit: $\text{C}^2\text{N}^{-1}\text{m}^{-2}$.
- Gauss magnetism: $\oint \mathbf{B} \cdot d\mathbf{A} = 0$ (no monopoles).

11. COMMON TRAPS

TRAP: $E_2:E_1$ mein distance ratio square karna bhoolna.

CORRECT: $(r_1/r_2)^2$ hamesha likho.

TRAP: Charge constant, distance badhao $\rightarrow U$ ghatte.

CORRECT: $C = \epsilon_0 A/d$ ghata hai $\rightarrow U = Q^2/2C$ BADHTA hai.

TRAP: Rest charge magnetic field bhi banata hai.

CORRECT: Rest charge sirf electric field banata hai.

12. GRAPHS / DIAGRAMS

- Field lines: + se nikalti, — par khatam; kabhi cross nahi.
- E vs r (point charge): $1/r^2$ curve.
- Equipotential surfaces field lines ke $\perp$.

13. LAST-MINUTE REVISION**LAST-MINUTE REVISION**

- $F = kq_1q_2/r^2$ (medium $/K$); $E = kq/r^2$; $E = V/d$
- $\tau = pE \sin\theta$; dipole SHM $\omega = \sqrt{2qE/md}$
- $C = \epsilon_0 A/d$; $U = (1/2)CV^2 = Q^2/2C$
- E inside conductor = 0; V constant

CHAPTER 12 — CURRENT ELECTRICITY**1. INTRODUCTION**

Current electricity = moving charges. Ohm's law, resistance, series/parallel, Kirchhoff's laws aur power — ye sab hi exam ke core hain. Fixed numericals hain jo ratio method se turant solve hote hain.

2. CORE CONCEPTS

What is current? $I = Q/t$ — charge ke flow ki rate.

Conventional current positive charge ki direction mein maana jaata hai. Wire current flow ke waqt NEUTRAL rehti hai (electrons aate-jaate hain, charge balance rehta hai).

What is Ohm's law? $V = IR$. Resistance $R = \rho L/A$ — material (ρ), length aur area par depend karta hai.

What are Kirchhoff's laws? KCL (current law) = CHARGE conservation (junction par current in = out). KVL (voltage law) = ENERGY conservation (loop mein voltage sum = 0).

What is power? $P = VI = I^2R = V^2/R$. AC mein $P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos\phi$.

What is internal resistance? Battery ke andar ki khud ki resistance r . Circuit mein current $I = E/(R + r)$. Terminal voltage $V = E - Ir$ (current draw karne par EMF se kam).

What is drift velocity? Electrons ka average slow net motion (electric field ke against): $I = neAv_d$. n = electrons/ m^3 , e = charge, A = area, v_d = drift velocity. I aur A dono double $\rightarrow v_d$ SAME ($2/2 = 1$).

What is conductance? Resistance ka reciprocal: $G = 1/R$, unit siemens (S).

What is a temperature coefficient? $R = R_0(1 + \alpha\Delta T)$. Metals ka α positive (R badhta hai), semiconductors ka α NEGATIVE (R ghatta hai).

3. KEY RELATIONS

- $R = \rho L/A$: length aur area DONO double $\rightarrow R$ SAME ($2/2 = 1$).
- $\rho_1 = 2 \times 10^{-8}$, $\rho_2 = 6 \times 10^{-8}$ (same L, A) $\rightarrow R_1/R_2 = 1:3$.
- Parallel (2 resistors): product/sum $\rightarrow 10 \text{ k}\Omega \parallel 30 \text{ k}\Omega = 7.5 \text{ k}\Omega$.
- 3 equal R parallel + R series $\rightarrow 4R/3$.
- Same battery: $E = I_1R_1 = I_2R_2$.
- $p(t) = v(t) \cdot i(t)$; $w(t) = \int p(t)dt$.
- $I = E/(R+r)$; $V_{\text{terminal}} = E - Ir$. ($E = 4 \text{ V}$, $r = 2 \Omega$, $R = 7 \Omega \rightarrow I = 4/9 \text{ A}$.)
- $I = neAv_d \rightarrow v_d \propto I/A$. I aur A dono double $\rightarrow v_d$ unchanged.

- EMF cell ke ELECTRODES/electrolyte par depend karta hai — circuit (R, I) par nahi.
- R aur V se length: $R = \rho L/A$, $V = AL \rightarrow L = \sqrt{(RV/\rho)}$.

4. FORMULAS

$$V = IR ; R = \rho L/A ; R_s = R_1 + R_2 ; 1/R_p = 1/R_1 + 1/R_2$$

$$P = VI = I^2 R = V^2/R ; P_{avg} = V_{rms} I_{rms} \cos\phi$$

$$I = E/(R+r) ; V = E - Ir ; G = 1/R$$

$$I = neAv_d ; R = R_0(1 + \alpha\Delta T)$$

Symbol meanings: I = current, V = voltage, R = resistance, ρ = resistivity, L = length, A = cross-sectional area, P = power.

5. CONDITIONS / SPECIAL CASES

- L double $\rightarrow$ R double; A double $\rightarrow$ R half; dono $\rightarrow$ no change.
- Semiconductor (germanium) mein temperature badhne par R GHATTA hai.
- $V = 100 \sin 100t$, $I = 100 \sin(100t + \pi/3)$ mA $\rightarrow P_{avg} = (100/\sqrt{2})(0.1/\sqrt{2})\cos 60^\circ = 2.5$ W.
- Resistivity material property hai; resistance shape par depend karta hai.
- Battery 4 V, $r = 2 \Omega$, $R = 7 \Omega \rightarrow I = 4/(7+2) = 4/9$ A.
- I aur A dono double $\rightarrow v_d$ SAME (I/A ratio constant).
- Heater wire: nichrome (high resistivity + high melting point).
- n cells parallel (har cell r) $\rightarrow$ total internal resistance r/n .

6. HOW TO RECOGNIZE THE QUESTION

- "R = $\rho L/A$ (L, A badle)" $\rightarrow$ ratio method.
- "Series/parallel" $\rightarrow$ product/sum (do resistors).
- "KCL vs KVL" $\rightarrow$ charge vs energy.
- "Same battery, two currents" $\rightarrow$ E equate karo.
- "Battery + internal resistance" $\rightarrow I = E/(R+r)$.
- "Drift velocity vs I, A" $\rightarrow I = neAv_d$.

7. SOLVING METHODS

- Do resistors parallel: product/sum.
- Same-battery problems: $E = I_1 R_1 = I_2 R_2$.
- Resonance par $\cos\phi = 1 \rightarrow P = E_{rms} I_{rms}$.

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — If two resistors of 10 k Ω and 30 k Ω are connected in parallel, the effective resistance is:

What this tests: Two-resistor parallel shortcut.

$$\text{Solution: } R = 10 \times 30 / (10 + 30) = 300/40 = 7.5 \text{ k}\Omega.$$

Key takeaway: Product/sum — do resistors ke liye fastest.

REAL PYQ EXAMPLE — When the length and area of cross-section both are doubled, the resistance:

What this tests: $R = \rho L/A$.

$$\text{Solution: } R' = \rho(2L)/(2A) = \rho L/A = R \rightarrow \text{unchanged.}$$

Key takeaway: L aur A dono double $\rightarrow$ R SAME ($2/2 = 1$).

REAL PYQ EXAMPLE — The current in a circuit is 5 A. When 2 Ω is added, it drops to 4 A. The original resistance is:

What this tests: Same battery $\rightarrow$ same emf.

$$\text{Solution: } 5R = 4(R+2) \rightarrow 5R = 4R+8 \rightarrow R = 8 \Omega.$$

Key takeaway: $E = I_1 R_1 = I_2 R_2$ equate karo.

REAL PYQ EXAMPLE — Kirchhoff's current law is based on conservation of ____ while voltage law is based on conservation of ____:

What this tests: KCL vs KVL meaning.

Solution: KCL = charge ; KVL = energy.

Key takeaway: KCL charge, KVL energy — order yaad rakho.

REAL PYQ EXAMPLE — In an AC circuit $V = 100 \sin(100t)$ V and $I = 100 \sin(100t + \pi/3)$ mA. Power dissipated:

What this tests: $P_{avg} = V_{rms} I_{rms} \cos\phi$.

$$\text{Solution: } P = (100/\sqrt{2})(0.1/\sqrt{2})(\cos 60^\circ) = 5 \times 0.5 = 2.5 \text{ W.}$$

Key takeaway: Peak $\rightarrow$ rms ($/\sqrt{2}$), phir $\cos\phi$ multiply karo.

REAL PYQ EXAMPLE — Two wires have resistivities 2×10^{-8} and $6 \times 10^{-8} \Omega\text{m}$ with same length and area. Ratio R_1/R_2 :

What this tests: $R \propto \rho$ (same L, A).

$$\text{Solution: } R_1/R_2 = 2/6 = 1:3.$$

Key takeaway: Same shape $\rightarrow R \propto \rho$ directly.

REAL PYQ EXAMPLE — A battery of EMF 4 V and internal resistance 2 Ω is connected to a 7 Ω resistor. The current is:

What this tests: $I = E/(R+r)$.

$$\text{Solution: } I = 4/(7+2) = 4/9 \text{ A.}$$

Key takeaway: Total resistance = $R + r$ (battery ka r bhi series mein).

REAL PYQ EXAMPLE — The drift velocity of electrons is v for current I and area A. If I and A both double, new drift velocity is:

What this tests: $I = neAv_d \rightarrow v_d = I/(neA)$.

$$\text{Solution: } v_d' = (2I)/(ne \cdot 2A) = v_d \rightarrow \text{same.}$$

Key takeaway: I aur A dono $\times 2 \rightarrow$ ratio same $\rightarrow v_d$ unchanged.

REAL PYQ EXAMPLE — Which equation shows the relation between current i and drift velocity v_d ?

What this tests: $I = neAv_d$.

Solution: $i = neAv_d$ (n = free electrons/ m^3 , e = charge, A = area).

Key takeaway: $i = neAv_d$ — drift velocity ka defining relation.

REAL PYQ EXAMPLE — On increasing temperature, the resistance of semiconductors:

What this tests: Semiconductor temperature behaviour.

Solution: Semiconductor: R GHATTA hai (negative temperature coefficient).

Key takeaway: Metals $R \uparrow$, semiconductors $R \downarrow$ with temperature.

9. 10-SECOND TRICKS

- L aur A dono double $\rightarrow$ R unchanged.
- 10 k Ω || 30 k $\Omega \rightarrow$ 7.5 k Ω .
- KCL = charge, KVL = energy.
- $I = E/(R+r)$; $V = E - Ir$.
- I aur A dono double $\rightarrow$ drift velocity same.

10. IMPORTANT FACTS

- Conventional current = positive charge ka flow.
- Conductor: $R \uparrow$ with T; semiconductor: $R \downarrow$ with T.

11. COMMON TRAPS

TRAP: KCL energy ka conservation hai.

CORRECT: KCL = charge, KVL = energy (order matter karta hai).

TRAP: Wire current flow mein charged ho jaati hai.

CORRECT: Wire neutral rehti hai.

TRAP: L aur A dono double $\rightarrow$ R double ya half.

CORRECT: $2/2 = 1 \rightarrow$ R unchanged.

TRAP: Terminal voltage hamesha EMF ke barabar hoti hai.

CORRECT: $V = E - Ir$ — current draw karne par EMF se kam.

TRAP: Drift velocity electrons ki itni hi hoti hai jitni current.

CORRECT: Drift velocity bahut slow ($\sim$ mm/s); electric field tez travel karta hai.

12. GRAPHS / DIAGRAMS

- Ohm's law: V-I straight line through origin, slope = R.
- Non-ohmic (diode, filament): curved V-I.
- Series: ek path, same current; parallel: same voltage, split current.

13. LAST-MINUTE REVISION

LAST-MINUTE REVISION

- $R = \rho L/A$; series/parallel rules
- $P = VI = I^2 R = V^2/R$; $P_{avg} = V_{rms} I_{rms} \cos\phi$
- KCL = charge; KVL = energy; wire neutral

CHAPTER 13 — MAGNETIC EFFECTS OF CURRENT AND MAGNETISM

1. INTRODUCTION

Ye chapter current se magnetic field banane aur magnetic field mein charge par force padhne dono cover karta hai. Fixed patterns: wire/solenoid ka field, circular motion ($r = mv/qB$), magnetic moment, aur hysteresis.

2. CORE CONCEPTS

What creates a magnetic field? Moving charge/current. Long wire ka field $B = \mu_0 I / 2\pi r$ ($\propto 1/r$). Solenoid ke andar $B = \mu_0 n I$ (uniform).

What is Lorentz force? $F = qvB \sin\theta$. $v \perp B$ ho to particle CIRCLE mein ghoomta hai: $r = mv/qB$.

What is magnetic dipole moment? Coil ka $m = NIA$ (N turns, area A). Direction right-hand rule se: clockwise current (upar se dekho) $\rightarrow$ moment DOWNWARD.

What is the time period of circular motion? $T = 2\pi m/qB$ — VELOCITY par depend NAHI karta (sirf m, q, B).

What does the hysteresis loop area give? Energy loss per magnetisation cycle.

What is the field of a circular coil? Centre par $B = \mu_0 I / 2R$ (N turns $\rightarrow$ $\times N$). Axis par distance x par $B = \mu_0 I R^2 / 2(R^2 + x^2)^{3/2}$. (Solenoid ka $B = \mu_0 n I$ ALAG hai.)

What is Biot-Savart law? Current element Idl ka field $dB = (\mu_0 / 4\pi) Idl \sin\theta / r^2$. Wire aur coil ke B formulas isi se aate hain.

What is a cyclotron? Charged particle ko circular paths mein accelerate karta hai. Frequency $f = qB / 2\pi m$ (velocity se independent — sirf q, B, m).

Force between parallel currents. Same direction $\rightarrow$ ATTRACTION; opposite $\rightarrow$ repulsion. $F/L = \mu_0 I_1 I_2 / 2\pi d$.

Magnetic torque. Field mein magnet/loop par $\tau = MB \sin\theta$ (M = magnetic moment, B = field). Maximum $\theta = 90^\circ$ par.

What is susceptibility? $\chi = \mu_r - 1$, jahan $\mu_r = \mu/\mu_0$ relative permeability. Diamagnetic χ negative (temperature se independent); paramagnetic $\chi \propto 1/T$.

Ferromagnetic examples. Iron, nickel, cobalt (aluminium paramagnetic, copper diamagnetic). Freely suspended magnet N-S align karta hai.

3. KEY RELATIONS

- $B(\text{wire}) = \mu_0 I / 2\pi r$; $B(\text{solenoid}) = \mu_0 n I = \mu_0 (N/L)I$.
- $r = mv/qB$; $T = 2\pi m/qB$ (velocity independent).

- $m = NIA$; $\chi = \mu_r - 1$; $W = MH(1 - \cos\alpha)$.

- Solenoid 50 cm, 100 turns, 2.5 A $\rightarrow B = 4\pi \times 10^{-7} \times (100/0.5) \times 2.5 = 6.28 \times 10^{-4}$ T.

- $B(\text{coil centre}) = \mu_0 I / 2R$; axis ($x = R$): $B/\sqrt{8}$.

- Radius half (current same) $\rightarrow B \times 2$ ($B \propto 1/R$).

- Cyclotron: $f = qB / 2\pi m$; $F/L = \mu_0 I_1 I_2 / 2\pi d$; $\tau = MB \sin\theta$.

- Magnetic force hamesha velocity ke $\perp$ hoti hai $\rightarrow$ koi WORK nahi karti $\rightarrow$ particle ki KE CONSTANT rehti hai (sirf direction badalti hai).

- $\chi = \mu_r - 1$; $\mu = \mu_0 \mu_r$.

4. FORMULAS

$F = qvB \sin\theta$; $F = BIl \sin\theta$; $B(\text{wire}) = \mu_0 I / 2\pi r$; $B(\text{sol}) = \mu_0 n I$

$r = mv/qB$; $T = 2\pi m/qB$; $m = NIA$; $\chi = \mu_r - 1$

$B(\text{coil}) = \mu_0 I / 2R$; $B_{\text{axis}} = \mu_0 I R^2 / 2(R^2 + x^2)^{3/2}$

$dB = (\mu_0 / 4\pi) Idl \sin\theta / r^2$; $f_{\text{cyclotron}} = qB / 2\pi m$

$F/L = \mu_0 I_1 I_2 / 2\pi d$; $\tau = MB \sin\theta$; $\mu = \mu_0 \mu_r$

Symbol meanings: B = magnetic field (T), I = current, r = distance/radius, n = turns per metre, N = turns, A = area, m = dipole moment, q = charge, v = velocity, M = magnetic moment, H = field intensity.

5. CONDITIONS / SPECIAL CASES

- v ke $\perp$ aur $\parallel$ dono components $\rightarrow$ HELICAL path (sirf $\perp \rightarrow$ circle).

- B badhao $\rightarrow$ radius ghatta hai ($r = mv/qB$).

- T sirf m , q , B par depend; velocity par nahi.

- Ferromagnetic above Curie temperature $\rightarrow$ PARAMAGNETIC.

- 1 T = 10^4 gauss. Angle of dip: poles 90° , equator 0° .

- Coil radius half (I same) $\rightarrow$ centre $B \times 2$.

- Coil axis $x = R \rightarrow B_{\text{axis}} = B_{\text{centre}} / (\sqrt{2}) = B/\sqrt{8}$.

- Parallel currents same direction $\rightarrow$ attract.

- Diamagnetic: temperature se independent; paramagnetic: $\chi \propto 1/T$.

6. HOW TO RECOGNIZE THE QUESTION

- "Wire/solenoid field" $\rightarrow \mu_0 I / 2\pi r$; $\mu_0 n I$.

- "Radius vs B " $\rightarrow \propto 1/B$.

- " T of circular motion" $\rightarrow 2\pi m/qB$ (v nahi).

- "Moment direction" $\rightarrow$ right-hand rule.

- "Coil ke centre ka field" $\rightarrow \mu_0 I / 2R$ ($\times N$ turns).

- "Cyclotron frequency" $\rightarrow qB / 2\pi m$.

- "Parallel wires" $\rightarrow \mu_0 I_1 I_2 / 2\pi d$ (same $\rightarrow$ attract).

7. SOLVING METHODS

- Solenoid: $n = N/L$ pehle, phir $\mu_0 n I$.

- Radius vs field: inverse.

- Magnet rotate: $W = MH(1 - \cos\alpha)$.

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — A long solenoid of 50 cm length having 100 turns carries a current of 2.5 A. The field at the centre is ($\mu_0 = 4\pi \times 10^{-7}$ T m/A):

What this tests: $B = \mu_0 n I$ with $n = N/L$.

Solution: $B = 4\pi \times 10^{-7} \times (100/0.5) \times 2.5 = 6.28 \times 10^{-4}$ T.

Key takeaway: Pehle $n = N/L$ nikalo, phir $\mu_0 n I$.

REAL PYQ EXAMPLE — A charged particle moves perpendicular to a magnetic field. If the field is increased, the radius of its path:

What this tests: $r = mv/qB \rightarrow r \propto 1/B$.

Solution: B badhao $\rightarrow r$ ghatta hai.

Key takeaway: $r \propto 1/B$ — increasing B shrinks the circle.

REAL PYQ EXAMPLE — A charge moves in a circle perpendicular to a magnetic field. Its time period does NOT depend on:

What this tests: $T = 2\pi m/qB$.

Solution: T mein velocity nahi hai $\rightarrow$ velocity par depend nahi.

Key takeaway: $T = 2\pi m/qB$ — m , q , B par; v par nahi.

REAL PYQ EXAMPLE — A long solenoid has n turns per metre and current I . The field inside is:

What this tests: Solenoid field formula.

Solution: $B_0 = \mu_0 nI$.

Key takeaway: Solenoid $B = \mu_0 nI$; wire $B = \mu_0 I/2\pi r$ — dono alag.

REAL PYQ EXAMPLE — When viewed from above, current in a loop flows clockwise. The magnetic dipole moment points:

What this tests: Right-hand rule.

Solution: Clockwise (from above) $\rightarrow$ moment DOWNWARD.

Key takeaway: Clockwise = down; anticlockwise = up.

REAL PYQ EXAMPLE — The area enclosed by a complete hysteresis loop denotes:

What this tests: Meaning of hysteresis area.

Solution: Energy dissipated per magnetisation cycle.

Key takeaway: Hysteresis area = energy loss per cycle.

REAL PYQ EXAMPLE — Magnetic field at the centre of a circular coil of radius R carrying current I is B . The field on the axis at distance R is:

What this tests: $B_{\text{axis}} = \mu_0 I R^2 / 2(R^2 + x^2)^{3/2}$ at $x = R$.

Solution: $B_{\text{axis}} = \mu_0 I R^2 / 2(2R^2)^{3/2} = B/(2\sqrt{2}) = B/\sqrt{8}$.

Key takeaway: Axis $x = R \rightarrow B/\sqrt{8}$ (centre ka $1/(2\sqrt{2})$).

REAL PYQ EXAMPLE — If the radius of a current-carrying coil is halved keeping current same, field at centre:

What this tests: $B = \mu_0 I / 2R \rightarrow B \propto 1/R$.

Solution: $R/2 \rightarrow B \times 2$.

Key takeaway: Coil centre $B \propto 1/R$ (radius half $\rightarrow$ field double).

REAL PYQ EXAMPLE — The angular frequency of a charged particle in a cyclotron is:

What this tests: Cyclotron frequency.

Solution: $\omega = qB/m$ ($f = qB/2\pi m$) — velocity se independent.

Key takeaway: Cyclotron $f = qB/2\pi m$ (sirf q , B , m).

REAL PYQ EXAMPLE — For two parallel wires carrying currents I_a and I_b , force per length is:

What this tests: Force between parallel currents.

Solution: $F/L = \mu_0 I_a I_b / 2\pi d$ (same direction $\rightarrow$ attract).

Key takeaway: Parallel wires: $\mu_0 I_a I_b / 2\pi d$ — attraction for same direction.

REAL PYQ EXAMPLE — The relationship between magnetic susceptibility χ and permeability μ is:

What this tests: $\chi = \mu_r - 1$, $\mu = \mu_0 \mu_r$.

Solution: $\chi = (\mu/\mu_0) - 1 \rightarrow \mu = \mu_0(1 + \chi)$.

Key takeaway: $\mu_r = 1 + \chi$; $\mu = \mu_0 \mu_r$.

9. 10-SECOND TRICKS

- Clockwise (upar se) $\rightarrow$ moment down.
- $T = 2\pi m/qB$ mein v nahi $\rightarrow$ "velocity par depend nahi".
- Hysteresis area = energy loss/cycle.
- Coil: radius half $\rightarrow B \times 2$; axis $x = R \rightarrow B/\sqrt{8}$.
- Cyclotron $f = qB/2\pi m$ (no v); same-direction wires attract.

10. IMPORTANT FACTS

- Angle of dip: poles 90° , equator 0° . $\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$.
- Curie temperature ke upar ferromagnetic $\rightarrow$ paramagnetic.
- Ferromagnetic: Fe, Ni, Co. Al paramagnetic, Cu diamagnetic.
- Freely suspended magnet: N-S align (earth ka field).

11. COMMON TRAPS

TRAP: B badhao $\rightarrow$ radius badhe.

CORRECT: $r = mv/qB \rightarrow$ radius GHATTA hai.

TRAP: Angle of dip poles par 0° .

CORRECT: Poles par 90° , equator par 0° .

TRAP: Helical path sirf $\perp$ velocity se banta hai.

CORRECT: Helical path tab banta hai jab velocity ka $\parallel$ component bhi ho.

TRAP: Cyclotron frequency velocity par depend karti hai.

CORRECT: $f = qB/2\pi m$ — velocity par independent.

TRAP: Parallel wires same direction mein repel karti hain.

CORRECT: Same direction $\rightarrow$ ATTRACT; opposite $\rightarrow$ repel.

12. GRAPHS / DIAGRAMS

- Hysteresis (B-H) loop: area = energy loss; retentivity & coercivity loop se milte hain.
- Right-hand rule: wire $\rightarrow$ concentric circles; solenoid $\rightarrow$ andar uniform field.

13. LAST-MINUTE REVISION**LAST-MINUTE REVISION**

- $B(\text{wire}) = \mu_0 I / 2\pi r$; $B(\text{solenoid}) = \mu_0 nI$
- $r = mv/qB$; $T = 2\pi m/qB$ (no v); helical if v has $\parallel$ part
- $m = NIA$; $\chi = \mu_r - 1$; $W = MH(1 - \cos\alpha)$; dip: poles 90°

CHAPTER 14 — ELECTROMAGNETIC INDUCTION AND ALTERNATING CURRENT**1. INTRODUCTION**

Ye chapter batata hai ki changing magnetic field kaise current (emf) banati hai, aur AC circuits kaise kaam karte hain. Fixed patterns: $\varepsilon = -d\Phi/dt$, transformer, aur AC power ($\cos\phi$).

2. CORE CONCEPTS

What is Faraday's law? Flux change hone par induced emf: $\varepsilon = -d\Phi/dt$. Jitni tez flux badle, utni badi emf.

What is Lenz's law? Induced emf ki DIRECTION batata hai — wo change ko OPPOSE karti hai. Ye energy conservation se judi hai.

What does a generator work on? ELECTROMAGNETIC INDUCTION (flux change se emf).

What is mutual inductance? Ek coil ke current change se dusri coil mein emf. Unit henry (H), dimension $[ML^2T^{-2}A^{-2}]$.

What is AC power? $V_{\text{rms}} = V_0/\sqrt{2}$; $P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos\phi$; $\cos\phi$ = power factor. LCR resonance par $\cos\phi = 1$.

What is a transformer? Voltage badalne wala device: $V_p/V_s = N_p/N_s$.

What is LCR resonance? Series LCR mein jab $X_L = X_C$ hota hai, circuit pure resistive ban jaata hai: $Z = R$, current MAX, $\cos\phi = 1$. Resonant frequency $f_0 = 1/(2\pi\sqrt{LC})$ — R par depend NAHI karti (R badle to f_0 same).

What is half-cycle average of AC? $i = I_0 \sin \omega t$ ka half-cycle average $\bar{I} = 2I_0/\pi$ (full-cycle average = 0).

What is mutual inductance emf? $\varepsilon = -M di/dt$ (M = coefficient of mutual induction, henry). Current jitni tez badle, emf utna bada.

What are eddy currents? Bulk conductor mein CHANGING magnetic flux se bante hain — circular induced currents (isliye transformer core laminated hota hai).

Energy in an inductor. $U = (1/2)LI^2$. 50 mH, 4 A $\rightarrow 0.5 \times 0.05 \times 16 = 0.4$ J.

3. KEY RELATIONS

- $\epsilon = -d\Phi/dt = -A dB/dt$ (A fixed): $A = 2 \text{ m}^2$, $\Delta B = 4$ in 2 s $\rightarrow \epsilon = 4 \text{ V}$.
- $\Phi(t)$ diya ho $\rightarrow$ differentiate: $\Phi = 6t^2 + 3t + 2 \rightarrow \epsilon = 12t + 3$; $t = 3 \rightarrow 39 \text{ V}$.
- Transformer: $V_p/V_s = N_p/N_s$; $\eta = V_s I_s / (V_p I_p)$.
- 220 $\rightarrow$ 11 V, 5 A $\rightarrow$ 90 A $\rightarrow \eta = (11 \times 90) / (220 \times 5) = 90\%$.
- Magnet fast $\rightarrow$ coil ki taraf $\rightarrow$ LARGER emf ($\epsilon \propto \text{speed}$).
- $f_0 = 1/(2\pi\sqrt{LC})$; $I_{\text{half}} = 2I_0/\pi$.
- $\epsilon = M dI/dt$: $M = 0.5 \text{ H}$, $\Delta I = 4 \text{ A}$ in 0.5 s $\rightarrow \epsilon = 0.5 \times 8 = 4 \text{ V}$.
- N-turn coil: $\epsilon = N d\Phi/dt$. Half-wave rectifier ka DC output = V_0/π (full sine ka half-cycle average $2V_0/\pi$).
- $U_{\text{inductor}} = (1/2)LI^2$; eddy currents = changing flux in bulk conductor.

4. FORMULAS

$\epsilon = -d\Phi/dt$; $\Phi = BA \cos\theta$; $V_{\text{rms}} = V_0/\sqrt{2}$
 $P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos\phi$; $\eta_{\text{transformer}} = V_s I_s / V_p I_p$
 $f_0 = 1/(2\pi\sqrt{LC})$; $I_{\text{half}} = 2I_0/\pi$; $\epsilon = M dI/dt$
 $U = (1/2)LI^2$; ideal transformer: $P_{\text{in}} = P_{\text{out}}$
 Symbol meanings: ϵ = induced emf, Φ = magnetic flux, B = field, A = area, V_{rms} = rms voltage, ϕ = phase difference, V_p/V_s = primary/secondary voltage, N_p/N_s = turns, I_p/I_s = currents.

5. CONDITIONS / SPECIAL CASES

- LCR resonance: $\cos\phi = 1$ (power factor 1), current max.
- High impedance $\rightarrow$ kam current (same source).
- Generator = EMI principle (force on conductor nahi).
- Pure inductive circuit: V aur I mein 90° phase difference.
- Resonance: $X_L = X_C \rightarrow Z = R \rightarrow \cos\phi = 1$ (current max).
- R decrease karo $\rightarrow f_0$ SAME (sirf L , C decide karte hain).
- $i = 50 \sin 100t \rightarrow I_{\text{half}} = 2 \times 50/\pi = 100/\pi \text{ A}$.
- Ideal step-down transformer mein POWER constant rehti hai.

6. HOW TO RECOGNIZE THE QUESTION

- "emf from flux / $\Phi(t)$ " $\rightarrow \epsilon = -d\Phi/dt$.
- "Transformer" $\rightarrow V_p/V_s = N_p/N_s$, $\eta = V_s I_s / V_p I_p$.
- "AC power / resonance" $\rightarrow \cos\phi = 1$.
- "Resonant frequency" $\rightarrow f_0 = 1/(2\pi\sqrt{LC})$.
- "Half-cycle average" $\rightarrow 2I_0/\pi$.
- "Mutual inductance emf" $\rightarrow \epsilon = M dI/dt$.

7. SOLVING METHODS

- emf = $A \times (\Delta B/\Delta t)$.
- Transformer efficiency: output power/input power.
- Resonance: $\cos\phi = 1$.

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — The magnetic flux linked with a coil is $\Phi = 6t^2 + 3t + 2$ (weber). The induced emf at $t = 3$ s is:

What this tests: $\epsilon = d\Phi/dt$.

Solution: $\epsilon = 12t + 3$; at $t = 3 \rightarrow 39 \text{ V}$.

Key takeaway: Pehle differentiate, phir substitute.

REAL PYQ EXAMPLE — A coil of area 2 m^2 is placed in a field changing from 4 to 8 Wb/m² in 2 s. Induced emf:

What this tests: $\epsilon = A(\Delta B/\Delta t)$.

Solution: $\epsilon = 2 \times (4/2) = 4 \text{ V}$.

Key takeaway: $\epsilon = A \times (\Delta B/\Delta t)$ — area times rate.

REAL PYQ EXAMPLE — A transformer reduces 220 V to 11 V. Primary draws 5 A, secondary 90 A.

Efficiency is:

What this tests: $\eta = \text{output power}/\text{input power}$.

Solution: $\eta = (11 \times 90) / (220 \times 5) = 990/1100 = 0.9 = 90\%$.

Key takeaway: Transformer $\eta = V_s I_s / V_p I_p$ (power ratio, voltage ratio nahi).

REAL PYQ EXAMPLE — The value of power factor $\cos\phi$ in a series LCR circuit at resonance is:

What this tests: At resonance $X_L = X_C$, circuit pure resistive.

Solution: $\cos\phi = 1$.

Key takeaway: Resonance $\rightarrow \cos\phi = 1 \rightarrow P = E_{\text{rms}} I_{\text{rms}}$.

REAL PYQ EXAMPLE — A magnet is moved rapidly toward one coil and slowly toward another identical coil. Induced emf:

What this tests: $\epsilon \propto \text{rate of flux change}$.

Solution: Fast magnet $\rightarrow$ larger emf.

Key takeaway: $\epsilon \propto \text{speed of flux change}$ — faster = bigger emf.

REAL PYQ EXAMPLE — The basic principle on which the AC generator works is:

What this tests: Generator principle.

Solution: Electromagnetic induction.

Key takeaway: Generator = EMI (flux change $\rightarrow$ emf).

REAL PYQ EXAMPLE — The mathematical form of the resonant frequency of an LCR circuit is:

What this tests: Resonance condition.

Solution: $f_0 = 1/(2\pi\sqrt{LC})$.

Key takeaway: $f_0 = 1/(2\pi\sqrt{LC})$ — R ka koi role nahi.

REAL PYQ EXAMPLE — In a series RLC circuit ($R = 1000 \Omega$, $L = 4 \text{ H}$, $C = 10^{-6} \text{ F}$), R is decreased by 20 Ω . Resonant frequency:

What this tests: f_0 sirf L , C par depend.

Solution: R badle to f_0 unchanged.

Key takeaway: Resonant frequency R se independent.

REAL PYQ EXAMPLE — An AC current $i = 50 \sin 100t$ A. Its half-cycle average is:

What this tests: $I_{\text{half}} = 2I_0/\pi$.

Solution: $I_{\text{half}} = 2 \times 50/\pi = 100/\pi \text{ A}$.

Key takeaway: Half-cycle average = $2I_0/\pi$; full-cycle = 0.

REAL PYQ EXAMPLE — Mutual inductance is 0.5 H. Current changes from +2 A to -2 A in 0.5 s. Induced emf:

What this tests: $\epsilon = M dI/dt$.

Solution: $\epsilon = 0.5 \times (4/0.5) = 4 \text{ V}$.

Key takeaway: $\Delta I = 4 \text{ A}$ (sign ka dhyan: $+2 \rightarrow -2$).

REAL PYQ EXAMPLE — The energy stored in a 50 mH inductor carrying 4 A is:

What this tests: $U = (1/2)LI^2$.

Solution: $U = 0.5 \times 50 \times 10^{-3} \times 16 = 0.4 \text{ J}$.

Key takeaway: Inductor energy = $(1/2)LI^2$ — capacitor ke $(1/2)CV^2$ jaisa.

9. 10-SECOND TRICKS

- $\epsilon = A(\Delta B/\Delta t)$.
- Transformer $\eta = (V_s I_s) / (V_p I_p)$.

- Resonance $\rightarrow \cos\phi = 1$.
- $f_0 = 1/(2\pi\sqrt{LC})$; $I_{\text{half}} = 2I_0/\pi$.
- $\varepsilon = M \, dI/dt$; $U = (1/2)LI^2$.

10. IMPORTANT FACTS

- Rectifier AC $\rightarrow$ DC; order: Rectifier $\rightarrow$ Filter $\rightarrow$ Regulator.
- Weber = flux ki unit; henry = inductance ki unit.

11. COMMON TRAPS

TRAP: Rectifier-filter-regulator order galat.

CORRECT: Filter smooth karta hai, regulator stable — yehi order.

TRAP: Transformer efficiency voltage ratio.

CORRECT: Power ratio $V_s I_s / V_p I_p$ hota hai.

TRAP: Generator force-on-conductor principle par chalta hai.

CORRECT: Generator = electromagnetic induction.

TRAP: R badhane se resonant frequency badhti hai.

CORRECT: $f_0 = 1/(2\pi\sqrt{LC})$ — sirf L aur C; R se independent.

TRAP: Full-cycle average = $2I_0/\pi$.

CORRECT: Full-cycle average = 0; HALF-cycle = $2I_0/\pi$.

12. GRAPHS / DIAGRAMS

- AC waveforms: V aur I sine; ϕ decide karta hai power factor.
- Phasor: VL aur VC opposite; resonance par cancel ($\cos\phi = 1$).
- Transformer: core par 2 coils; $V_p/V_s = N_p/N_s$.

13. LAST-MINUTE REVISION

LAST-MINUTE REVISION

- $\varepsilon = -d\Phi/dt = -A \, dB/dt$; $\Phi = BA \cos\theta$
- $P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos\phi$; resonance $\cos\phi = 1$
- Transformer: $V_p/V_s = N_p/N_s$; $\eta = V_s I_s / V_p I_p$
- AC $\rightarrow$ DC: rectifier $\rightarrow$ filter $\rightarrow$ regulator

CHAPTER 15 — ELECTROMAGNETIC WAVES

1. INTRODUCTION

EM waves = light, radio, X-rays, etc. — oscillating electric aur magnetic fields jo vacuum mein bhi chalti hain. Ye chapter chhota hai: spectrum order, $E_0/B_0 = c$, aur "kaun EM wave nahi hai" type facts.

2. CORE CONCEPTS

What is an EM wave? Oscillating E aur B fields, mutually PERPENDICULAR aur propagation direction ke bhi $\perp$ — isliye TRANSVERSE. Vacuum mein speed $c = 3 \times 10^8 \text{ m/s} \approx 1/\sqrt{(\mu_0 \varepsilon_0)}$.

Kya EM waves deflect hoti hain? NAHI. Inka koi charge nahi hota, isliye electric/magnetic field inhe deflect nahi kar sakte.

What does an EM wave carry? ENERGY (charge ya mass nahi).

What is the spectrum order? Frequency badhne ke order: Radio < Micro < IR < Visible < UV < X < Gamma. (Wavelength iska ulta.)

3. KEY RELATIONS

- $c = 1/\sqrt{(\mu_0 \varepsilon_0)} = v\lambda$; $E_0 = cB_0$ ($E_0/B_0 = c$).
- B vector: $B = (1/\omega)(k \times E)$.
- $E = h\nu$ (photon energy bridge).

4. FORMULAS

$$c = v\lambda \text{ ; } E_0 = cB_0 \text{ ; } c = 1/\sqrt{(\mu_0 \varepsilon_0)}$$

$$E = hc/\lambda$$

Symbol meanings: c = speed of light, ν = frequency, λ = wavelength, E_0/B_0 = field amplitudes, h = Planck's constant, ω = angular frequency, k = propagation vector.

5. CONDITIONS / SPECIAL CASES

- Radio, IR, UV EM hain; SOUND nahi.
- EM waves energy carry karti hain (charge/mass nahi).
- Max frequency = gamma; max wavelength = radio; UV/X/micro/gamma mein micro ki wavelength sabse badi.
- Wavelength ranges: microwave $\approx 1 \text{ mm} - 1 \text{ m}$; visible 400–800 nm ; UV < 400 nm ; IR > 800 nm.
- Point source se distance r par intensity $I = P/4\pi r^2$ (P = power). 100 W, 10% $\rightarrow 10 \text{ W}$; $r = 5 \text{ m} \rightarrow I = 10/(4\pi \times 25) \approx 0.032 \text{ W/m}^2$.
- Galaxies ke beech distance BADHTI ja rahi hai (redshift $\rightarrow$ expanding universe, Hubble).

6. HOW TO RECOGNIZE THE QUESTION

- "Not an EM wave" $\rightarrow$ sound.
- " E_0, B_0 relation" $\rightarrow E_0/B_0 = c$.
- "Spectrum order" $\rightarrow$ radio se gamma.

7. SOLVING METHODS

- "Not EM" $\rightarrow$ sound (odd one out).
- $E_0/B_0 = c$ (c^2 nahi).

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — Which of the following is NOT an electromagnetic wave?

What this tests: EM transverse hoti hai, medium nahi chahiye.

Solution: Sound mechanical hai (medium chahiye) — radio/IR/UV EM hain.

Key takeaway: Sound = list mein odd one out.

REAL PYQ EXAMPLE — In an EM wave, the ratio of electric field amplitude to magnetic field amplitude is:

What this tests: EM wave relation.

Solution: $E_0/B_0 = c$.

Key takeaway: $E_0/B_0 = c$ (c^2 nahi).

REAL PYQ EXAMPLE — Which EM wave has the maximum wavelength?

What this tests: Spectrum order.

Solution: Radio sabse lambi wavelength (gamma sabse chhoti).

Key takeaway: Wavelength: radio > micro > IR > visible > UV > X > gamma.

REAL PYQ EXAMPLE — Which statement is INCORRECT about electromagnetic waves?

What this tests: EM wave properties.

Solution: "Deflected by electric/magnetic field" galat hai — EM waves neutral hain.

Key takeaway: EM waves fields se deflect nahi hoti.

REAL PYQ EXAMPLE — The wavelength range of microwaves is:

What this tests: Spectrum ranges.

Solution: Microwave $\approx 1 \text{ mm}$ to 1 m (radio se chhota, IR se bada).

Key takeaway: Microwave 1 mm–1 m ; visible 400–800 nm.

REAL PYQ EXAMPLE — A 100 W bulb converts 10% power to visible radiation. Average intensity 5 m away is:

What this tests: $I = P/4\pi r^2$.

Solution: $I = 10/(4\pi \times 25) \approx 0.032 \text{ W/m}^2$.

Key takeaway: $I = P/4\pi r^2$ — radius ka square neeche.

9. 10-SECOND TRICKS

- $E_0/B_0 = c$.
- Sound = list mein odd one out.

10. IMPORTANT FACTS

- EM waves transverse hain, vacuum mein chalti hain, energy carry karti hain.
- Phase difference between E and B in EM wave = 0 (same phase).

11. COMMON TRAPS

TRAP: EM waves E/B field se deflect hoti hain.

CORRECT: Ye neutral hain — deflect nahi hoti.

TRAP: $E_0/B_0 = c^2$.

CORRECT: $E_0/B_0 = c$.

12. GRAPHS / DIAGRAMS

- EM spectrum: radio < micro < IR < visible < UV < X < gamma.
- E aur B perpendicular sinusoids, sath propagation axis par chalte hain.

13. LAST-MINUTE REVISION**LAST-MINUTE REVISION**

- $c = v\lambda = 1/\sqrt{\mu_0\epsilon_0} = 3 \times 10^8 \text{ m/s}$; $E_0 = cB_0$
- $E \perp B \perp \text{direction}$; transverse; fields se deflect nahi

CHAPTER 16 — OPTICS**1. INTRODUCTION**

Optics = light ka reflection, refraction, mirrors, lenses, prisms aur wave optics (YDSE). Sabse zyada PYQs wala chapter hai. Formulas fixed hain: mirror/lens formula, prism, fringe width, critical angle.

2. CORE CONCEPTS

What is reflection? Plane mirror same-size virtual image deta hai; object distance = image distance.

What is a spherical mirror? $f = R/2$ (concave f +ve, convex f -ve). Mirror formula $1/f = 1/v + 1/u$.

What is refraction? Light medium badalti hai to speed/wavelength badalte hain, FREQUENCY nahi. Snell's law: $\mu_1 \sin i = \mu_2 \sin r$.

What is a lens? $1/f = 1/v - 1/u$; magnification $m = v/u$. m negative → real inverted; m positive → virtual erect.

What is TIR? Total internal reflection: denser → rarer, $i >$ critical angle. $\sin C = \mu_{\text{rarer}}/\mu_{\text{denser}}$.

What does the eye see? Retina par image REAL, INVERTED, DIMINISHED.

What is dispersion? White light prism se guzarti hai to alag-alag colours mein toot jaati hai (alag $\lambda \rightarrow$ alag $\mu \rightarrow$ alag deviation). VIOLET sabse zyada deviate hota hai (μ sabse zyada), RED sabse kam. Isliye glass ka refractive index violet ke liye MAXIMUM.

What is a rainbow? Barish ki boondo mein refraction + TIR + dispersion — teeno milkar rainbow banate hain (red bahar, violet andar).

What is lens power? $P = 1/f$ (f metre mein), unit DIOPTR (D). Convex lens f +ve → P +ve; concave → P -ve. +2.5 D → $f = 1/2.5 = 0.4 \text{ m} = 40 \text{ cm}$ convex.

What is the lens-maker formula? $1/f = (\mu - 1)(1/R_1 - 1/R_2)$. Sign convention: convex surface R +ve, concave R -ve.

Lens ko liquid (μ_l) mein duba do → $\mu \rightarrow \mu_g/\mu_l$.

Mirrors inclined. Do plane mirrors θ angle par → number of images $N = 360/\theta - 1$ (jab $360/\theta$ even integer ho).

Deviation at a mirror. Plane mirror par light i angle par gire to deviation $\delta = 180^\circ - 2i$ ($i = 30^\circ \rightarrow \delta = 120^\circ$).

What is Huygens' principle? Wavefront ka HAR point naya secondary wavelet banata hai; unka envelope agla wavefront hai. Isse reflection AUR refraction dono explain hote hain.

What is polarisation? Sirf TRANSVERSE waves polarise hoti hain (light haan, sound nahi). Reflection se light partially polarise hoti hai; Brewster angle par puri: $\tan \theta_B = \mu$.

Diffraction condition. Jab slit/obstacle ka size wavelength ke barabar ho tab diffraction clearly dikhta hai (corona around moon = diffraction).

Colour of objects. Object apna colour usi wavelength ko reflect karke dikhata hai. Red paper yellow light mein → BLACK (yellow absorb ho jaata hai, kuch reflect nahi hota).

3. KEY RELATIONS

- Mirror: $1/f = 1/v + 1/u$; quick $v = uf/(u-f)$. $f = 20$, $u = 30 \rightarrow v = 60 \text{ cm}$.
- Thin prism: $\delta = (\mu - 1)A \rightarrow \mu = 1 + \delta/A = 1.25$.
- YDSE: $\beta = \lambda D/d$; $y_n = n\lambda D/d$.
- Critical angle: $\sin C = \mu_{\text{rarer}}/\mu_{\text{denser}}$. Water 4/3, glass 5/3 → $C = \sin^{-1}(4/5)$.
- Simple microscope ($D = 25 \text{ cm}$): $M = 1 + D/f$; $f = 4 \text{ cm} \rightarrow 7.25$.
- Dispersion order (deviation): violet > indigo > ... > red; μ_{violet} sabse bada.
- $P = 1/f$ (m); +2.5 D → 40 cm convex.
- $1/f = (\mu - 1)(1/R_1 - 1/R_2)$; $N(\text{mirrors}) = 360/\theta - 1$.
- Plane mirror deviation $\delta = 180^\circ - 2i$; Brewster: $\tan \theta_B = \mu$.
- Telescope: $M = f_o/f_e$; YDSE amplitude ratio = $\sqrt{(I_1/I_2)} = \sqrt{(1/4)} = 1/2$.
- Destructive interference: path difference = $(2n+1)\lambda/2$ (phase 180°).
- $v = c/\mu \rightarrow v_1/v_2 = \mu_2/\mu_1$ (glass 3/2, water 4/3 → $v_w/v_g = 9/8$).
- λ in medium = λ_0/μ (6000 Å in glass $\mu = 1.5 \rightarrow 4000 \text{ Å}$).

4. FORMULAS

$f = R/2$; $1/f = 1/v + 1/u$ (mirror); $1/f = 1/v - 1/u$ (lens)

$m = v/u$; $\delta = (\mu - 1)A$; $\beta = \lambda D/d$; $y_n = n\lambda D/d$

$\sin C = \mu_{\text{rarer}}/\mu_{\text{denser}}$; $M = 1 + D/f$;

resolving power $\propto 1/\lambda$

$P = 1/f$ (dioptr); $1/f = (\mu - 1)(1/R_1 - 1/R_2)$

$N = 360/\theta - 1$; $\delta(\text{mirror}) = 180^\circ - 2i$; $\tan \theta_B = \mu$

$v = c/\mu$; $\lambda_{\text{medium}} = \lambda_0/\mu$; $M_{\text{telescope}} = f_o/f_e$

Symbol meanings: f = focal length, R = radius of curvature, u = object distance, v = image distance, m = magnification, μ = refractive index, δ = deviation, A = prism angle, β = fringe width, D = screen distance, d = slit separation, C = critical angle.

5. CONDITIONS / SPECIAL CASES

- Concave mirror $R = 25$ ($f = 12.5$), $u = 10 \rightarrow v = uf/(u-f) = -50 \text{ cm}$ (virtual, mirror ke peeche).
- YDSE: $d = 0.2 \text{ mm}$, $D = 1 \text{ m}$, 3rd bright $7.5 \text{ mm} \rightarrow \lambda = 500 \text{ nm}$.
- Fringe count vs λ : $n \propto 1/\lambda \rightarrow 92 \times (5898/5461) \approx 99$.

- Red $\rightarrow$ blue microscope: λ ghata $\rightarrow$ resolving power BADHTI hai.
- Glass ($\mu = \sqrt{3}$), $i = 60^\circ \rightarrow r = 30^\circ \rightarrow$ angle between refracted and reflected $= 90^\circ$.
- 30° mirrors asymmetric $\rightarrow N = 360/30 - 1 = 11$.
- Plane mirror $i = 30^\circ \rightarrow \delta = 180 - 60 = 120^\circ$.
- Double convex lens $f = 25$ cm, $\mu = 1.5$, $R_1 = 2R_2 \rightarrow 1/25 = 0.5(1/R_1 + 1/R_2) \rightarrow R_1 = 37.5$ cm, $R_2 = 18.75$ cm.
- Concave lens $\mu_g = 1.5$ in liquid $\mu_l = 2 \rightarrow$ relative $\mu = 0.75 < 1 \rightarrow$ lens CONVEX ban jaati hai ($f = 80$ cm).
- Red paper in yellow light $\rightarrow$ black.

6. HOW TO RECOGNIZE THE QUESTION

- "Mirror/lens image position" $\rightarrow v = uf/(u-f)$; $m = v/u$.
- "Prism μ " $\rightarrow \mu = 1 + \delta/A$.
- "YDSE fringe" $\rightarrow \beta = \lambda D/d$.
- "Critical angle" $\rightarrow \sin C = \mu_r/\mu_d$.
- "Microscope" $\rightarrow M = 1 + D/f$.
- "Power of lens (diopetre)" $\rightarrow P = 1/f$.
- "Two mirrors / number of images" $\rightarrow 360/\theta - 1$.
- "White light / colours / rainbow" $\rightarrow$ dispersion.
- "Polarisation / Brewster" $\rightarrow$ transverse + $\tan \theta_B = \mu$.
- "Frequency in medium / λ change" $\rightarrow \lambda/\mu$, $v = c/\mu$.

7. SOLVING METHODS

- $f = R/2$ (hamesha spherical mirror).
- Prism: $\mu = 1 + \delta/A$ (thin prism).
- Fringe count: wavelength ka inverse ratio.

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — An object is placed 30 cm before a concave mirror of focal length 20 cm.

Image distance is:

What this tests: Mirror formula / quick form.

Solution: $v = uf/(u-f) = 30 \times 20/(30-20) = 600/10 = 60$ cm.

Key takeaway: $v = uf/(u-f)$ concave mirror ke liye fastest.

REAL PYQ EXAMPLE — The radius of curvature of a spherical mirror is 15. Its focal length is:

What this tests: $f = R/2$.

Solution: $f = 15/2 = 7.5$ cm.

Key takeaway: Spherical mirror: $f = R/2$ (hamesha).

REAL PYQ EXAMPLE — A light ray passes through a thin prism of angle 8° with deviation 2° . Refractive index is:

What this tests: $\delta = (\mu-1)A$.

Solution: $\mu = 1 + \delta/A = 1 + 2/8 = 1.25$.

Key takeaway: Thin prism: $\mu = 1 + \delta/A$.

REAL PYQ EXAMPLE — With two slits 0.2 mm apart and screen 1 m away, the third bright fringe is 7.5 mm from centre. Wavelength is:

What this tests: $y_n = n\lambda D/d$.

Solution: $\lambda = y_n d/(nD) = 7.5 \times 10^{-3} \times 0.2 \times 10^{-3} / 3 = 500$ nm.

Key takeaway: $\lambda = y d/(nD)$ — units mm aur nm dhyan se.

REAL PYQ EXAMPLE — If the refractive index of water is $4/3$ and glass is $5/3$, critical angle (glass $\rightarrow$ water) is:

What this tests: $\sin C = \mu_{\text{rarer}}/\mu_{\text{denser}}$.

Solution: $\sin C = (4/3)/(5/3) = 4/5 \rightarrow C = \sin^{-1}(4/5)$.

Key takeaway: $\sin C = \mu_r/\mu_d$ — denser neeche.

REAL PYQ EXAMPLE — A simple microscope has focal length 4 cm. Magnifying power at least distance of distinct vision (25 cm):

What this tests: $M = 1 + D/f$.

Solution: $M = 1 + 25/4 = 7.25$.

Key takeaway: Simple microscope ($D = 25$ cm): $M = 1 + D/f$.

REAL PYQ EXAMPLE — The power of a lens is +2.5 D. The lens is:

What this tests: $P = 1/f$.

Solution: $f = 1/2.5 = 0.4$ m = 40 cm; P positive $\rightarrow$ convex.

Key takeaway: $P = 1/f$ (metre) — +ve = convex, -ve = concave.

REAL PYQ EXAMPLE — Two plane mirrors are at 30° . Number of images (object placed asymmetrically):

What this tests: $N = 360/\theta - 1$.

Solution: $N = 360/30 - 1 = 12 - 1 = 11$.

Key takeaway: $N = 360/\theta - 1$ (θ se divide karke 1 ghatao).

REAL PYQ EXAMPLE — The angle of incidence on a plane mirror is 30° . Angle of deviation is:

What this tests: $\delta = 180^\circ - 2i$.

Solution: $\delta = 180^\circ - 60^\circ = 120^\circ$.

Key takeaway: Mirror deviation $= 180^\circ - 2i$.

REAL PYQ EXAMPLE — A double convex lens has $f = 25$ cm, $\mu = 1.5$, one radius double the other. The radii are:

What this tests: Lens-maker: $1/f = (\mu-1)(1/R_1 + 1/R_2)$, $R_1 = 2R_2$.

Solution: $1/25 = 0.5(1/R_1 + 1/R_2) \rightarrow R_2 = 18.75$ cm, $R_1 = 37.5$ cm.

Key takeaway: Biconvex: $1/f = (\mu-1)(1/R_1 + 1/R_2)$ — signs dhyan se.

REAL PYQ EXAMPLE — Which law can be explained by Huygens' principle?

What this tests: Huygens secondary wavelets.

Solution: Reflection AND refraction dono.

Key takeaway: Huygens $\rightarrow$ reflection + refraction dono.

REAL PYQ EXAMPLE — Refractive index of glass is 1.5 for light of wavelength 6000 Å in vacuum.

Wavelength in glass:

What this tests: $\lambda_{\text{medium}} = \lambda_0/\mu$.

Solution: $\lambda = 6000/1.5 = 4000$ Å.

Key takeaway: Medium mein wavelength GHTATI hai: λ/μ .

REAL PYQ EXAMPLE — In YDSE the slit-width ratio is 1 : 4. The amplitude ratio is:

What this tests: Intensity $\propto$ width, amplitude $\propto \sqrt{I}$.

Solution: $I_1:I_2 = 1:4 \rightarrow A_1:A_2 = \sqrt{1}:\sqrt{4} = 1:2$.

Key takeaway: Amplitude ratio $= \sqrt{(\text{intensity ratio})}$.

9. 10-SECOND TRICKS

- $f = R/2$.
- m negative = real inverted; positive = virtual erect.
- Plane mirror: $v = u$.
- Mirror/image velocity: tum plane mirror ki taraf speed u se bhaago $\rightarrow$ image tumhari taraf $2u$ se aati hai (object ke relative image ki speed $= 2u$).
- Vehicle ka rear-view mirror CONVEX hota hai: wide field of view + hamesha erect, diminished image.
- $P = 1/f$; $N = 360/\theta - 1$; $\delta = 180^\circ - 2i$.
- Violet sabse zyada deviate; μ_{violet} maximum.
- λ in medium $= \lambda_0/\mu$ (ghat-ta hai); frequency same.

10. IMPORTANT FACTS

- TIR: denser→rarer. Corona around moon = diffraction.
- Visible light: 400–800 nm. Near point ≈ 25 cm.
- Dispersion: VIBGYOR (violet sabse zyada deviate).
- Rainbow: refraction + TIR + dispersion.
- Telescope: $M = f_o/f_e$ ($f_o \gg f_e$).

11. COMMON TRAPS

TRAP: Medium badle to frequency badalti hai.

CORRECT: Frequency constant rehti hai;
speed/wavelength badalte hain.

TRAP: TIR rarer→denser mein hota hai.

CORRECT: TIR sirf denser→rarer mein hota hai.

TRAP: Concave mirror object inside focus → real image.

CORRECT: Inside focus → image VIRTUAL, mirror ke peeche (v negative).

TRAP: Convex lens ki power negative hoti hai.

CORRECT: Convex = +ve power (converging); concave = -ve.

TRAP: N mirrors = $360/\theta$.

CORRECT: $N = 360/\theta - 1$ (jab $360/\theta$ even integer ho).

TRAP: Medium change par frequency badalti hai.

CORRECT: Frequency constant; $\lambda = \lambda_o/\mu$ ghat-ta hai.

12. GRAPHS / DIAGRAMS

- Ray diagrams: concave mirror & convex lens real inverted; convex mirror & concave lens hamesha virtual erect.
- YDSE: alternate bright/dark fringes, $\beta = \lambda D/d$.

13. LAST-MINUTE REVISION**LAST-MINUTE REVISION**

- $f = R/2$; mirror $1/f = 1/v + 1/u$; lens $1/f = 1/v - 1/u$
- $m = v/u$ (negative = real inverted); $\delta = (\mu - 1)A$
- $\beta = \lambda D/d$; $\sin C = \mu_r/\mu_d$; $M = 1 + D/f$; TIR denser→rarer

CHAPTER 17 — DUAL NATURE OF MATTER AND RADIATION**1. INTRODUCTION**

Ye chapter batata hai ki light wave BHI hai aur particle BHI, aur moving particle ki apni ek wave bhi hoti hai. Fixed patterns: photon energy ($12400/\lambda$ shortcut), work function, aur de Broglie wavelength.

2. CORE CONCEPTS

What is dual nature? Light wave ki tarah behave karti hai (interference, diffraction) aur particle ki tarah bhi (photon). Photon energy $E = hv = hc/\lambda$.

What is the photoelectric effect? Light metal par girti hai to electrons nikalte hain — lekin TABHI jab $\nu >$ threshold frequency ν_o (minimum FREQUENCY, intensity nahi).

What is work function? $\phi = h\nu_o = hc/\lambda_o$ — electron nikaalne ke liye minimum energy.

What is Einstein's equation? $h\nu = \phi + K_{\max}$. Photon ki energy = work function + max kinetic energy.

What are matter waves? Moving particle ki de Broglie wavelength $\lambda = h/p = h/\sqrt{2mK}$.

3. KEY RELATIONS

- $E = hc/\lambda$; $E(\text{eV}) = 12400/\lambda(\text{\AA}) = 1240/\lambda(\text{nm})$.
- $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ (joules mein convert karne ke liye eV ko 1.6×10^{-19} se multiply karo).
- $\lambda = 6000 \text{ \AA} \rightarrow 2.06 \text{ eV}$; $\lambda_o = 5000 \text{ \AA}$ (sodium) $\rightarrow \phi \approx 4 \times 10^{-19} \text{ J}$.

- de Broglie: $\lambda = h/p = h/\sqrt{2mK}$; $\lambda = 0.2 \text{ \AA} \rightarrow p = 3.3 \times 10^{-23} \text{ kg m/s}$.

- Same KE: $\lambda \propto 1/\sqrt{m} \rightarrow \lambda_p/\lambda_e = \sqrt{(m_e/m_p)} \approx 1/43$.

4. FORMULAS

$$E = h\nu = hc/\lambda ; \lambda = h/p = h/\sqrt{2mK}$$

$$\phi = h\nu_o = hc/\lambda_o ; h\nu = \phi + K_{\max}$$

$$E(\text{eV}) = 1240/\lambda(\text{nm}) = 12400/\lambda(\text{\AA})$$

Symbol meanings: h = Planck's constant, ν = frequency, ν_o = threshold frequency, λ = wavelength, λ_o = threshold wavelength, ϕ = work function, $K_{\max}$ = max kinetic energy, p = momentum, m = mass.

5. CONDITIONS / SPECIAL CASES

- Intensity $\rightarrow$ electrons ki NUMBER (current); frequency $\rightarrow$ emission + KE.
- Photoelectric emission instantaneous hoti hai.
- Threshold = minimum FREQUENCY (not intensity, not power).

6. HOW TO RECOGNIZE THE QUESTION

- "Photon energy from λ " $\rightarrow 12400/\lambda(\text{\AA})$.
- "Work function from λ_o " $\rightarrow hc/\lambda_o$.
- "Threshold" $\rightarrow$ frequency.
- "de Broglie" $\rightarrow \lambda = h/p = h/\sqrt{2mK}$.

7. SOLVING METHODS

- $E(\text{eV}) = 12400/\lambda(\text{\AA})$ one step.
- Same-KE de Broglie: $\lambda \propto 1/\sqrt{m}$.

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — If the wavelength of a photon is 6000 Å, its energy will be:

What this tests: $E(\text{eV}) = 12400/\lambda(\text{\AA})$.

Solution: $E = 12400/6000 = 2.06 \text{ eV}$.

Key takeaway: $12400/\lambda(\text{\AA})$ — eV nikalne ka one-step shortcut.

REAL PYQ EXAMPLE — An electron and a proton have the same kinetic energy. The ratio of their de Broglie wavelengths is:

What this tests: $\lambda \propto 1/\sqrt{m}$ (same KE).

Solution: $\lambda_p/\lambda_e = \sqrt{(m_e/m_p)} \approx 1/43$.

Key takeaway: Halka particle $\rightarrow$ lambi λ . Same KE $\rightarrow \lambda \propto 1/\sqrt{m}$.

REAL PYQ EXAMPLE — If the de Broglie wavelength of a particle is 0.2 Å, its momentum is:

What this tests: $\lambda = h/p$.

Solution: $p = h/\lambda = 6.6 \times 10^{-34} / 0.2 \times 10^{-10} = 3.3 \times 10^{-23} \text{ kg m/s}$.

Key takeaway: $p = h/\lambda$ — direct division.

REAL PYQ EXAMPLE — According to the photoelectric effect, a photoelectron is emitted when the incident light is more than a certain minimum:

What this tests: Threshold condition.

Solution: Frequency (not intensity).

Key takeaway: Emission ka gate FREQUENCY hai; intensity sirf current badhati hai.

REAL PYQ EXAMPLE — If the threshold wavelength for sodium is 5000 Å, its work function is:

What this tests: $\phi = hc/\lambda_o$.

Solution: $\phi = 6.6 \times 10^{-34} \times 3 \times 10^8 / 5 \times 10^{-7} \approx 4 \times 10^{-19} \text{ J}$.

Key takeaway: $\phi = hc/\lambda_o$ — threshold wavelength se work function.

9. 10-SECOND TRICKS

- $12400/\lambda(\text{\AA}) = \text{eV}$.
- "Minimum frequency" $\rightarrow$ threshold.

10. IMPORTANT FACTS

- Photon: $E = h\nu$. Matter wave: $\lambda = h/p$.

11. COMMON TRAPS

TRAP: Emission intensity par depend karta hai.

CORRECT: Frequency threshold se upar honi chahiye; intensity sirf current badhati hai.

TRAP: Work function eV aur J mix karna.

CORRECT: Units consistent rakho — $12400/\lambda(\text{\AA})$ eV deta hai.

12. GRAPHS / DIAGRAMS

- K_{max} vs v : straight line, slope = h , x-intercept = v_0 .
- Stopping potential vs v : same intercept wali line.

13. LAST-MINUTE REVISION**LAST-MINUTE REVISION**

- $E = h\nu = hc/\lambda$; $E(\text{eV}) = 12400/\lambda(\text{\AA})$; $\lambda = h/p = h/\sqrt{2mK}$
- $\phi = hc/\lambda_0$; $h\nu = \phi + K_{\text{max}}$; same KE $\rightarrow \lambda \propto 1/\sqrt{m}$
- $6000 \text{\AA} \rightarrow 2.06 \text{ eV}$; $0.2 \text{\AA} \rightarrow 3.3 \times 10^{-23} \text{ kg m/s}$

CHAPTER 18 — ATOMS AND NUCLEI**1. INTRODUCTION**

Ye chapter nucleus ki structure aur radioactivity cover karta hai. Fixed patterns: neutrons = $A - Z$, spectral lines formula, half-life, aur fusion condition.

2. CORE CONCEPTS

What is a nucleus made of? Atomic number Z = protons; mass number A = protons + neutrons. Isliye neutrons = $A - Z$.

What are spectral lines? Electron jab higher level n se lower level m par girta hai to spectral lines banti hain. Count = $(n-m)(n-m+1)/2$.

What is radioactivity? Nucleus khud decay karta hai: $N = N_0 e^{-\lambda t}$. Half-life $T_{1/2} = 0.693/\lambda = 0.693 \times \tau$ (mean life).

What is fusion? Halke nuclei jud kar bade nucleus banate hain. Iske liye bahut HIGH temperature chahiye — coulomb repulsion todne ke liye.

What is fission? Bhaari nucleus do parts mein tootata hai. Spectral series names. Hydrogen lines: Lyman ($n = 1$, UV), Balmer ($n = 2$, visible), Paschen ($n = 3$, IR). Balmer hi visible series hai. Formula: $1/\lambda = R(1/n_1^2 - 1/n_2^2)$. Nuclear reactions balance. Har nuclear reaction mein mass number (A) aur charge (Z) dono CONSERVE hote hain — unknown particle nikalne ke liye A aur Z ka sum match karo.

3. KEY RELATIONS

- Neutrons = $A - Z$.
- Spectral lines $n \rightarrow m$: $(n-m)(n-m+1)/2$. 8th $\rightarrow$ 2nd: $6 \times 7/2 = 21$.
- $T_{1/2} = 0.693/\lambda = 0.693\tau$; $\tau = 1/\lambda$. $\lambda = 0.00231/\text{day} \rightarrow 300 \text{ days}$.
- $N = N_0(1/2)^{t/T_{1/2}}$.

4. FORMULAS

$$N = N_0 e^{-\lambda t}; T_{1/2} = 0.693/\lambda = 0.693\tau; \tau = 1/\lambda$$

$$\text{Lines } n \rightarrow m = (n-m)(n-m+1)/2$$

Symbol meanings: Z = atomic number, A = mass number, N = nuclei remaining, N_0 = initial, λ = decay constant, $T_{1/2}$ = half-life, τ = mean life.

5. CONDITIONS / SPECIAL CASES

- 8th shell $\rightarrow$ 2nd shell $\rightarrow 21$ spectral lines.
- $\lambda = 0.00231/\text{day} \rightarrow T_{1/2} = 300 \text{ days}$.
- Mean life $\tau = 50 \text{ days} \rightarrow T_{1/2} = 0.693 \times 50 = 34.65 \text{ days}$.

- α, β, γ penetrating power: $\gamma > \beta > \alpha$; γ ka koi mass/charge nahi.

6. HOW TO RECOGNIZE THE QUESTION

- "Neutrons" $\rightarrow A - Z$.
- "Spectral lines $n \rightarrow m$ " $\rightarrow (n-m)(n-m+1)/2$.
- "Half-life / mean life" $\rightarrow 0.693/\lambda = 0.693\tau$.
- "Fusion condition" $\rightarrow$ high temperature.

7. SOLVING METHODS

- Half-life: $0.693/\lambda$, phir options match.
- Line count: $(n-m)$ formula.

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — Atomic number of a nucleus is Z and atomic mass is M . The number of neutrons is:

What this tests: Neutrons = mass number – atomic number.

Solution: Neutrons = $A - Z$ (yahan $M = A$).

Key takeaway: Neutrons = $A - Z$.

REAL PYQ EXAMPLE — The number of spectral lines when electrons in the 8th shell return to the 2nd shell is:

What this tests: Lines formula with $n = 8, m = 2$.

Solution: $(8-2)(8-2+1)/2 = 6 \times 7/2 = 21$.

Key takeaway: Lines = $(n-m)(n-m+1)/2$, NOT $n(n-1)/2$.

REAL PYQ EXAMPLE — The disintegration constant of an element is 0.00231 per day. Its half-life is:

What this tests: $T_{1/2} = 0.693/\lambda$.

Solution: $T_{1/2} = 0.693/0.00231 = 300 \text{ days}$.

Key takeaway: $T_{1/2} = 0.693/\lambda$ — hamesha.

REAL PYQ EXAMPLE — If the mean life of a radioactive element is 50 days, its half-life is:

What this tests: $T_{1/2} = 0.693 \times \tau$.

Solution: $T_{1/2} = 0.693 \times 50 = 34.65 \text{ days}$.

Key takeaway: $T_{1/2} = 0.693\tau$ (τ nahi).

REAL PYQ EXAMPLE — Nuclear fusion cannot occur on Earth because:

What this tests: Fusion condition.

Solution: Temperature itni high nahi ki coulomb repulsion overcome ho.

Key takeaway: Fusion $\rightarrow$ extreme temperature needed for coulomb barrier.

REAL PYQ EXAMPLE — Lyman and Balmer are the names of:

What this tests: Spectral series.

Solution: Hydrogen spectral series (Lyman UV $n = 1$, Balmer visible $n = 2$).

Key takeaway: Balmer = visible ($n = 2$); Lyman = UV ($n = 1$).

REAL PYQ EXAMPLE — In a nuclear reaction, balancing is done using:

What this tests: Conservation in reactions.

Solution: Mass number (A) aur charge (Z) dono conserve.

Key takeaway: Nuclear reactions: A aur Z dono conserve karo.

9. 10-SECOND TRICKS

- Half-life = $0.693/\lambda$ (hamesha).
- Fusion: temperature too low = coulomb repulsion.

10. IMPORTANT FACTS

- γ highest penetration, no mass/charge.
- α = helium nucleus, β = electron.
- Lyman UV, Balmer visible, Paschen IR.

11. COMMON TRAPS

TRAP: Spectral lines = $n(n-1)/2$.

CORRECT: $n \rightarrow m$ ke liye $(n-m)(n-m+1)/2$ hota hai.

TRAP: Mean life = half-life.

CORRECT: $T_{1/2} = 0.693\tau$ (τ nahi).

TRAP: $\lambda = 0.00231 \rightarrow 500$ days.

CORRECT: $0.693/0.00231 = 300$ days.

12. GRAPHS / DIAGRAMS

- Energy-level diagram: Lyman ($n=1$, UV), Balmer ($n=2$, visible).
- Decay curve: N vs t exponential; half-life constant.

13. LAST-MINUTE REVISION**LAST-MINUTE REVISION**

- Neutrons = $A - Z$; $T_{1/2} = 0.693/\lambda = 0.693\tau$
- Lines $n \rightarrow m = (n-m)(n-m+1)/2$; fusion needs extreme temperature

CHAPTER 19 — ELECTRONIC DEVICES**1. INTRODUCTION**

Ye chapter semiconductors, diodes, transistors aur logic gates cover karta hai. Fixed facts: doping (N vs P type), Zener diode, rectifier chain, aur depletion region.

2. CORE CONCEPTS

What is doping? Pure semiconductor mein impurity milana. Pentavalent (P, As, Sb) $\rightarrow$ N-TYPE (extra electrons).

Trivalent (B, Ga, In) $\rightarrow$ P-TYPE (holes).

What is a p-n junction? P aur N ke beech depletion region banta hai. Ye REVERSE bias mein WIDEN hota hai, forward bias mein narrow.

What is a Zener diode? HEAVILY doped diode jo REVERSE bias mein operate karta hai — VOLTAGE REGULATOR ke roop mein.

Why npn over pnp? Electron mobility $>$ hole mobility, isliye npn faster.

What is rectification? AC $\rightarrow$ DC. Chain: Rectifier $\rightarrow$ Filter (smooth) $\rightarrow$ Regulator (constant).

What are logic gates? Digital circuits ke basic blocks. AND (output 1 sirf jab dono inputs 1), OR (koi ek 1 to output 1), NOT (invert), NAND (AND ka ulta), NOR (OR ka ulta), XOR (inputs alag to 1). Truth table se koi bhi combination nikal jaata hai.

What are energy bands? Conduction band = wo band jisme electrons freely conduct karte hain. Valence band = atoms se bandhe electrons. Conductor mein dono overlap; insulator mein bada gap.

Ge at 0 K. Absolute zero par germanium INSULATOR jaisa behave karta hai (koi free electron nahi).

Emitter. Transistor ka emitter HEAVILY doped aur THIN hota hai (zyada carriers inject karne ke liye).

3. KEY RELATIONS

- N-type: pentavalent impurity; P-type: trivalent impurity.
- Zener: heavily doped, reverse bias, voltage regulator.
- npn $>$ pnp (electron mobility).
- AND: $Y = A \cdot B$; OR: $Y = A + B$; NOT: $Y = \bar{A}$; NAND = AND + NOT; NOR = OR + NOT.
- AND + NOT ka combination = NAND gate.
- Conduction band electrons hi conduction mein hissa lete hain.
- Ge at 0 K = insulator; emitter heavily doped + thin.

4. FORMULAS

Diode: forward bias mein conduct; Zener reverse breakdown mein conduct

β (current gain) = I_c/I_b (common emitter)

AND: $Y = A \cdot B$; OR: $Y = A + B$; NOT: $Y = \bar{A}$; NAND = NOT($A \cdot B$); NOR = NOT($A + B$)

5. CONDITIONS / SPECIAL CASES

- Ga doped germanium $\rightarrow$ P-type; P/Sb/Bi doped $\rightarrow$ N-type.
- Reverse bias only $\rightarrow$ depletion width badhti hai.
- AC $\rightarrow$ DC order: rectifier $\rightarrow$ filter $\rightarrow$ regulator.

6. HOW TO RECOGNIZE THE QUESTION

- "N vs P type" $\rightarrow$ pentavalent vs trivalent.
- "Zener" $\rightarrow$ heavily doped, reverse, regulator.
- "Depletion width" $\rightarrow$ reverse bias.
- "AC $\rightarrow$ DC chain" $\rightarrow$ rectifier $\rightarrow$ filter $\rightarrow$ regulator.

7. SOLVING METHODS

- Doping type: periodic group (III vs V).
- AC $\rightarrow$ DC order yaad rakho.

8. REAL PYQ EXAMPLES

REAL PYQ EXAMPLE — Germanium is doped with gallium. The semiconductor obtained is:

What this tests: Doping type by impurity group.

Solution: Gallium (trivalent, group III) $\rightarrow$ p-type.

Key takeaway: Group III $\rightarrow$ p-type; group V $\rightarrow$ n-type.

REAL PYQ EXAMPLE — Which component is used to create an N-type semiconductor?

What this tests: N-type impurity.

Solution: Pentavalent impurity (phosphorus/arsenic).

Key takeaway: Pentavalent $\rightarrow$ N-type; trivalent $\rightarrow$ P-type.

REAL PYQ EXAMPLE — Zener diode is ____ doped and connected in ____ bias, used as a ____:

What this tests: Zener diode properties.

Solution: Heavily doped, reverse bias, voltage regulator.

Key takeaway: Zener = heavily doped + reverse + regulator.

REAL PYQ EXAMPLE — The increase in the width of the depletion region in a p-n junction is due to:

What this tests: Depletion vs bias.

Solution: Reverse bias only.

Key takeaway: Reverse bias widens depletion; forward narrows it.

REAL PYQ EXAMPLE — Constant DC from variable AC. Correct order of operations:

What this tests: AC $\rightarrow$ DC chain.

Solution: Rectifier $\rightarrow$ filter $\rightarrow$ regulator.

Key takeaway: Rectifier (AC $\rightarrow$ DC) $\rightarrow$ filter (smooth) $\rightarrow$ regulator (constant).

REAL PYQ EXAMPLE — n-p-n transistors are preferred over p-n-p because:

What this tests: Transistor mobility.

Solution: Electron mobility $>$ hole mobility.

Key takeaway: npn $>$ pnp — electrons faster.

REAL PYQ EXAMPLE — Which logic gate gives output Y = 1 only when both inputs are 1?

What this tests: AND gate definition.

Solution: AND gate ($Y = A \cdot B$).

Key takeaway: AND = dono 1 tab 1; OR = koi ek 1 tab 1.

REAL PYQ EXAMPLE — AND gate followed by NOT gate gives:

What this tests: Gate combination.

Solution: AND + NOT = NAND gate.

Key takeaway: AND + NOT = NAND; OR + NOT = NOR.

REAL PYQ EXAMPLE — Which band in a solid has electrons that participate in conduction?

What this tests: Energy bands.

Solution: Conduction band (valence band ke electrons bandhe hote hain).

Key takeaway: Conduction = conduction band electrons.

REAL PYQ EXAMPLE — Pure germanium at absolute zero behaves as:

What this tests: Semiconductor at 0 K.

Solution: Insulator (koi free electron nahi).

Key takeaway: 0 K par semiconductor = insulator.

9. 10-SECOND TRICKS

- Pentavalent $\rightarrow$ N; trivalent $\rightarrow$ P.
- npn $>$ pnp (electrons fast).
- Zener = reverse-biased regulator.
- AND+NOT = NAND ; OR+NOT = NOR.
- Conduction band $\rightarrow$ current; Ge at 0 K $\rightarrow$ insulator.

10. IMPORTANT FACTS

- Zener heavily doped, reverse bias, voltage regulator.
- Si diode knee ~ 0.7 V; Ge ~ 0.3 V.
- Emitter: heavily doped + thin; collector: largest area.

11. COMMON TRAPS

TRAP: Depletion width forward bias mein badhti hai.

CORRECT: REVERSE bias mein badhti hai.

TRAP: Gallium $\rightarrow$ n-type.

CORRECT: Gallium (III) $\rightarrow$ p-type; phosphorus (V) $\rightarrow$ n-type.

12. GRAPHS / DIAGRAMS

- Diode V-I: forward mein conduct, reverse mein block (Si ~ 0.7 V, Ge ~ 0.3 V).
- Logic symbols: AND, OR, NOT, NAND, NOR (bubble = inversion).
- Transistor: npn (arrow out), pnp (arrow in).

13. LAST-MINUTE REVISION**LAST-MINUTE REVISION**

- Pentavalent $\rightarrow$ N-type ; trivalent $\rightarrow$ P-type
- Reverse bias widens depletion ; Zener = heavily doped, reverse, regulator
- Rectifier $\rightarrow$ filter $\rightarrow$ regulator ; npn $>$ pnp (electron mobility)

CHAPTER 20 — COMMUNICATION SYSTEMS**1. INTRODUCTION**

Ye chapter batata hai ki signals kaise transmit hote hain. Fixed facts: modulation/demodulation aur propagation modes (ground/sky/space wave).

2. CORE CONCEPTS

What is modulation? Low-frequency baseband signal ko high-frequency carrier par chadhana — effective transmission ke liye ZAROORI hai.

What is demodulation? Receiver par original signal wapaa nikalna — modulation ka reverse.

What are propagation modes? Ground wave (surface ke saath), sky wave (ionosphere reflection), space wave (line-of-sight). Radio waves TEENO use karti hain.

Why does a man's voice sound like a woman's after transmission? Amplifiers ki WRONG bandwidth selection.

3. KEY RELATIONS

- Radio waves = ground + sky + space (all of the above).

- Sky waves ionosphere se reflect; microwave/LOS space wave.

- Modulation index $\mu = A_m/A_c$.

4. FORMULAS

$$c = v\lambda ; \text{modulation index } \mu = A_m/A_c$$

5. CONDITIONS / SPECIAL CASES

- Voice high-pitched (man $\rightarrow$ woman) sunai de = amplifiers ki WRONG bandwidth selection.
- Microwaves/TV space wave use karte hain.

6. HOW TO RECOGNIZE THE QUESTION

- "Propagation modes" $\rightarrow$ ground + sky + space.
- "Modulation definition/necessity" $\rightarrow$ baseband on carrier.
- "Pitch change" $\rightarrow$ wrong bandwidth.

7. SOLVING METHODS

- "Radio waves by?" $\rightarrow$ all three modes.

8. REAL PYQ EXAMPLES**REAL PYQ EXAMPLE — Through which mode of wave propagation are radio waves sent from one place to another?**

What this tests: Radio propagation modes.

Solution: Ground + sky + space — all of the above.

Key takeaway: Radio = teeno modes.

REAL PYQ EXAMPLE — The process in which a low-frequency message signal is superimposed on a high-frequency wave is called:

What this tests: Definition of modulation.

Solution: Modulation (demodulation iska reverse).

Key takeaway: Modulation = baseband on carrier; necessary for transmission.

REAL PYQ EXAMPLE — The speech signal of a man, after modulation, sounds like that of a woman. This is due to:

What this tests: Transmission discrepancy.

Solution: Wrong bandwidth selection of amplifiers.

Key takeaway: Pitch change = wrong bandwidth selection.

9. 10-SECOND TRICKS

- Radio propagation = all three modes.

10. IMPORTANT FACTS

- Ionosphere sky waves reflect karti hai; modulation necessary hai.

11. COMMON TRAPS

TRAP: Radio waves sirf ek mode use karti hain.

CORRECT: Teeso modes: ground, sky, space.

12. GRAPHS / DIAGRAMS

- Communication block: transmitter $\rightarrow$ channel $\rightarrow$ receiver.
- Ground wave (surface), sky wave (ionosphere), space wave (line-of-sight).

13. LAST-MINUTE REVISION**LAST-MINUTE REVISION**

- Modulation (baseband on carrier) $\rightarrow$ transmit $\rightarrow$ demodulation
- Propagation: ground, sky (ionosphere), space (LOS)

FINAL FORMULA REVISION

- $v = u + at$; $s = ut + (1/2)at^2$; $v^2 = u^2 + 2as$; $S_n = u + (a/2)(2n-1)$
- Free-fall per-second = 1:3:5:7 ; $a = v^2/r = \omega^2 r$; $H = u^2 \sin^2 \theta / 2g$; $R = u^2 \sin^2 \theta / g$; $T = 2u \sin \theta / g$

- $F = ma = dp/dt$; $p = mv$; $J = F\Delta t = \Delta p$; $f_{\text{max}} = \mu_s N$; $a = \mu_s g$; $K = p^2/2m$
- $W = Fs \cos\theta$; $U = (1/2)kx^2$; $P = Fv$; $\tau = I\alpha$; $L = I\omega$; $I = I_{\text{cm}} + Md^2$
- Ring: MR^2 , $MR^2/2$, $3MR^2/2$; Disc: $MR^2/2$, $MR^2/4$; Sphere: $2MR^2/5$; Rod: $ML^2/12$, $ML^2/3$
- $F = GMm/r^2$; $g = GM/R^2$; $v_e = \sqrt{2gR}$; $T^2 \propto r^3$; satellite $PE = 2E_{\text{total}}$
- $Y = FL/(A\Delta L)$; $\Delta P = 2T/r$ (drop), $4T/r$ (bubble); $v_t \propto r^2$; $P = \rho gh$
- $\Delta Q = \Delta U + W$; $\eta = 1 - T_2/T_1$; $PV = nRT$; $E = (3/2)Nk_B T$; $\gamma = 1 + 2/f$
- $Q = mc\Delta T$; $\Delta L = \alpha L\Delta T$; $\alpha:\beta:\gamma = 1:2:3$; $h = 2T \cos\theta/(r\rho g)$
- $v_{\text{rms}} = \sqrt{3RT/M}$; $f_0 = 1/(2\pi\sqrt{LC})$; $I_{\text{half}} = 2I_0/\pi$; $\epsilon = M dl/dt$; $U = (1/2)LI^2$
- $I = E/(R+r)$; $I = neAv_d$; $V = W/q$; $\sigma = Q/A$; $E = \sigma/\epsilon_0$
- $B = \mu_0 I/2R$; $B_{\text{axis}} = \mu_0 I R^2/2(R^2+x^2)^{3/2}$; $f_{\text{cyc}} = qB/2\pi m$; $F/L = \mu_0 I_1 I_2/2\pi d$; $\tau = MB \sin\theta$
- $P = 1/f$; $1/f = (\mu-1)(1/R_1 - 1/R_2)$; $N = 360/\theta - 1$; $\delta = 180^\circ - 2i$; $\tan \theta_B = \mu$
- $k = 2\pi/\lambda$; $I \propto A^2$; $R = \sqrt{A^2+B^2+2AB \cos\theta}$; $COP = T_2/(T_1-T_2)$
- $v = \omega\sqrt{A^2-x^2}$; $v_{\text{max}} = \omega A$; $a_{\text{max}} = \omega^2 A$; $a_{\text{max}}/v_{\text{max}} = 2\pi f$
- $F = kq_1 q_2/r^2$; $E = kq/r^2 = V/d$; $C = \epsilon_0 A/d$; $U = Q^2/2C$; $\tau = pE \sin\theta$
- $R = \rho L/A$; $P = VI = I^2 R$; $P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos\phi$
- $B = \mu_0 I/2\pi r = \mu_0 nI$; $r = mv/qB$; $T = 2\pi m/qB$; $m = NIA$; $\chi = \mu_r - 1$
- $\epsilon = -d\Phi/dt$; $V_p/V_s = N_p/N_s$; $\eta = V_{\text{sls}}/V_{\text{plp}}$; $E_0/B_0 = c$
- $E = hc/\lambda$; $\lambda = h/p$; $E(\text{eV}) = 12400/\lambda(\text{\AA})$
- $1/f = 1/v + 1/u$; $m = v/u$; $\delta = (\mu-1)A$; $\beta = \lambda D/d$; $\sin C = \mu_r/\mu_d$; $M = 1 + D/f$
- $T_{1/2} = 0.693/\lambda = 0.693\tau$; lines $n \rightarrow m = (n-m)(n-m+1)/2$

FINAL RELATIONS REVISION

- $g \propto M/R^2$; $v_e \propto \sqrt{M/R}$; $T^2 \propto r^3$; satellite $PE = 2 \times \text{total energy}$
- $v_t \propto r^2$; $\Delta P \propto 1/r$; $\eta = 1 - T_2/T_1$ (kelvin); $\Delta T(^{\circ}\text{C}) = \Delta T(\text{K})$
- $v = \omega\sqrt{A^2-x^2}$; $a_{\text{max}}/v_{\text{max}} = \omega = 2\pi f$; $E \propto A^2$
- $E \propto q/r^2$; $C \propto A/d$; $U = Q^2/2C$; force in medium = F/K
- $R \propto L/A$; $B \propto I/r$; $r \propto 1/B$; $T \propto m/qB$ (no v); $\epsilon \propto d\Phi/dt$
- $K = p^2/2m$; $I = I_{\text{cm}} + Md^2$; $\lambda \propto 1/p$; $\lambda \propto 1/v$; $T_{1/2} = 0.693\tau$
- Resonance: $\cos\phi = 1$; power factor = $\cos\phi$; $P = V_{\text{rms}} I_{\text{rms}} \cos\phi$

FINAL TRICKS REVISION

- Dimensionless: strain, refractive index, specific gravity, angle
- Planck's $h \equiv$ angular momentum; stress $\equiv$ pressure $\equiv$ Young's modulus
- Free fall per-second = odd numbers 1:3:5:7; v doubled $\rightarrow a \times 4$
- Momentum $\propto v$; $K \propto p^2$ ($4 \times K \rightarrow 2 \times p$); airbag $\rightarrow$ smaller force
- Recoil = $(m v)/M$; spring $2 \times \rightarrow$ energy $4 \times$; dot product: matching components
- Kepler equal areas = L conservation; I double $\rightarrow \omega$ half
- $4M$ & $R/2 \rightarrow 16g$; $2M$ & $R/2 \rightarrow v_e \times 2$; satellite $PE = 2E_{\text{total}}$
- Bubble $4T/r$, drop $2T/r$; radius $\times 2 \rightarrow v_t \times 4$; convex meniscus = mercury

- Isothermal $\Delta U = 0$; adiabatic $Q = 0$, $\Delta U = -W$; Carnot: +273 first
- Diatomic $\gamma = 1.4$; sound = NOT EM; mean position $\rightarrow v_{\text{max}}$
- Rest charge $\rightarrow E$ only; E inside conductor = 0; dielectric $K \rightarrow F/K$
- L & A both doubled $\rightarrow R$ unchanged; KCL = charge, KVL = energy
- B up $\rightarrow r$ down; $T = 2\pi m/qB$ (no v); resonance $\cos\phi = 1$
- $E_0/B_0 = c$; $E(\text{eV}) = 12400/\lambda(\text{\AA})$; $T_{1/2} = 0.693/\lambda$
- m negative $\rightarrow$ real inverted; $f = R/2$; TIR denser $\rightarrow$ rarer
- Pentavalent $\rightarrow$ N-type; Zener = reverse regulator; Curie $\rightarrow$ paramagnetic

FINAL TRAPS REVISION

- kg unit hai, quantity nahi; angle ka unit hai, dimension nahi
- Average velocity = NET displacement (closed path $\rightarrow 0$)
- $4 \times KE \rightarrow 2 \times$ momentum ($4 \times$ nahi); bus start = inertia (3rd law nahi)
- Retarding force $\rightarrow$ negative work; $\theta = 90^\circ \rightarrow$ zero work
- Spring: $W = (1/2)k(x_2^2 - x_1^2)$, $(\Delta x)^2$ nahi; torque = force $\times$ moment arm
- Ring tangent = $3MR^2/2$; equal $v_e \rightarrow M/R$ ($M \times R$ nahi); weightless = free fall
- Gauge = $P - P_a$; terminal $v_t \propto r^2$; Carnot kelvin mein
- Adiabatic expansion $\rightarrow \Delta U$ negative; diatomic 1.4 (1.66 = monatomic)
- Sound speed pressure se independent (constant T); mean position v_{max}
- $E_2:E_1$ mein $(r_1/r_2)^2$; charge const + $d \uparrow \rightarrow U \uparrow$; KCL charge / KVL energy
- L & A ; double $\rightarrow R$ same; wire neutral; $r \propto 1/B$; dip poles 90°
- EM fields se deflect nahi; $E_0/B_0 = c$ (c^2 nahi)
- Concave: inside focus $\rightarrow$ virtual behind; TIR sirf denser $\rightarrow$ rarer
- Photoelectric = frequency; lines = $(n-m)(n-m+1)/2$; $T_{1/2} = 0.693\tau$
- $p \times 1.25 \rightarrow K \times 1.5625$; resultant 3-4-5; min zero-resultant = 2 forces
- Isochoric $W = 0$; $COP \neq$ engine η ; rms sirf $\sqrt{T}$ (pressure nahi)
- $f_0 R$ se independent; full-cycle average = 0 (half = $2I_0/\pi$)
- Violet sabse zyada deviate; $\mu_{\text{violet max}}$; $N = 360/\theta - 1$
- Convex = +ve power; Ge at 0 K = insulator; reverse bias widens depletion
- Ga $\rightarrow$ p-type, P/Sb $\rightarrow$ n-type; reverse bias widens depletion

FINAL FACTS REVISION

- 1 kWh = 3.6×10^6 J; 1 N = 10^5 dyne; 1 T = 10^4 gauss; 1 $\mu\text{m} = 10^{-6}$ micron
- Light-year = distance; nit = luminance; second same in SI & CGS
- Bats ultrasonic; sound in air ≈ 332 m/s; Moon $g \approx 1.63$ m/s²
- Kepler 2nd = angular momentum; water surface tension $\downarrow$ with T
- Venturi-meter = flow speed; Bernoulli = energy; continuity = mass
- Second law = entropy; generator = EMI; $\mu_0 = 4\pi \times 10^{-7}$ T m/A

- Charge at rest $\rightarrow$ E only ; Gauss magnetism: $\oint \mathbf{B} \cdot d\mathbf{A} = 0$
- Hysteresis area = energy loss ; dip: poles 90° , equator 0°
- Visible 400–800 nm ; corona = diffraction ; eye: real inverted diminished, near point 25 cm
- Fusion = high temperature ; npn > pnp (electron mobility) ; radio = all modes
- EM waves carry energy ; Poisson = lateral/longitudinal strain
- $27^\circ\text{C} = 27\text{ K}$; henry = inductance ; weber = flux ; Zener = regulator
- Water $c = 1\text{ cal/g}^\circ\text{C}$; $1\text{ cal} = 4.18\text{ J}$; $\sigma_{\text{Stefan}} = 5.67 \times 10^{-8}$
- Ferromagnetic: Fe, Ni, Co ; microwave 1 mm–1 m ; visible 400–800 nm
- Lyman UV / Balmer visible ; emitter heavily doped + thin ; galaxies recede (Hubble)

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END OF PHYSICS MASTER BOOK

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